Clifford and Geometric Algebra

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Abstract

Contents

I. CANONICAL ANTI COMMUTATION RELATIONSHIP (CAR)

The Canonical Anticommutation Relationships are for a given conb $\{|\varphi_j\rangle|\, 1 \leq j \leq r\}$ of $\mathcal{H}^1$

$$\left\{a_j, a_k^\dagger\right\} = \delta_{jk} I_{\mathcal{F}} \tag{1a}$$

$$\left\{a_j^\dagger, a_k^\dagger\right\} = \{a_j, a_k\} = 0 \tag{1b}$$

$$1 \leq j \leq r, \tag{1c}$$

where

$$\{A, B\} = AB + BA, \tag{2}$$

the creation operators $\left\{a_j^\dagger \middle|\, 1 \leq j \leq r\right\}$ are defined by their action on *normalized* vectors $|\Psi\rangle \in \mathcal{H}_{\mathcal{F}}$ by

$$a_j^\dagger |\Psi\rangle = |\varphi_j \wedge \Psi\rangle \tag{3}$$

and the annihilation operators $\{a_j|\, 1 \leq j \leq r\}$ are defined as their adjoints i.e.

$$a_j = \left(a_j^\dagger\right)^\dagger, \, 1 \leq j \leq r. \tag{4}$$

1. *Comments*

- The operators $\left\{a_j^\dagger \middle|\, 1 \leq j \leq r\right\}$ are bounded so have bounded adjoints $\{a_j|\, 1 \leq j \leq r\}$ and hence

$$a_j^\dagger = (a_j)^\dagger\,. \tag{5}$$

- In particular it follows from the definition eq. (3) and eq. $(1a)$ that

$$
\begin{aligned}
a_j^{\dagger}\left|\phi\right\rangle &= \left|\varphi_j\right\rangle \\
a_j\left|\phi\right\rangle &= 0 \\
&\quad 1 \leq j \leq r.
\end{aligned}
\tag{6}
$$

  This follows from

$$\left\langle \phi \right| a_j^{\dagger} a_j \phi \right\rangle = \left\langle \phi \middle| \left(I_{\mathcal{F}} - a_j a_j^{\dagger}\right) \phi \right\rangle = 0, \tag{7}$$

  which implies that

$$a_j\left|\phi\right\rangle = 0. \tag{8}$$

- $a_j^{\dagger}$ and a_j are partial isometries, (and generate the Fermion Fock algebra), as

$$a_j^{\dagger} a_j \text{ and } a_j a_j^{\dagger} \tag{9}$$

  are projectors and equivalently

$$a_j a_j^{\dagger} a_j = a_j\left(1 - a_j a_j^{\dagger}\right) = a_j. \tag{10}$$

  In this particular case

$$a_j^{\dagger} a_j + a_j a_j^{\dagger} = I_F, \tag{11}$$

$$a_j^{\dagger} a_j a_j a_j^{\dagger} = 0 \tag{12}$$

$$a_j a_j^{\dagger} a_j^{\dagger} a_j = 0, \tag{13}$$

  the projectors a $a_j^{\dagger} a_j$ and $a_j a_j^{\dagger}$ are orthogonal as

$$a_j^{\dagger} a_j a_j a_j^{\dagger} = a_j^{\dagger} a_j \left(1 - a_j^{\dagger} a_j\right) = a_j^{\dagger} a_j - a_j^{\dagger} a_j = 0$$

  and are also infinite dimensional.

One can set the operators $\left\{ a_j^{\dagger} \middle| 1 \leq j \leq r \right\}$ as the values of the map

$$\Phi^{\dagger} : \mathcal{H}^1 \rightarrow \mathcal{B}\left(\mathcal{H}_{\mathcal{F}}\right) \tag{14}$$

i.e.

$$a_j^{\dagger} = \Phi^{\dagger}\left(\left|\varphi_j\right\rangle\right), \tag{15}$$

which leads to a more general (basis free) form of the CAR

$$\left\{\Phi\left(|u\rangle\right), \Phi^{\dagger}\left(|v\rangle\right)\right\} = \langle u|\, v\rangle\, I_{\mathcal{F}} \tag{16a}$$

$$\left\{\Phi^{\dagger}\left(|u\rangle\right), \Phi^{\dagger}\left(|v\rangle\right)\right\} = \left\{\Phi\left(|u\rangle\right), \Phi\left(|v\rangle\right)\right\} = 0 \tag{16b}$$

$$\forall\, |u\rangle\,, |v\rangle \in \mathcal{H}^{1}. \tag{16c}$$

The c^{*}-algebra $\mathcal{B}\left(\mathcal{H}_{\mathcal{F}}\right)$, (which is a w^{*}-algebra), is generated by the basis $\left\{a_j, a_j^{\dagger}\middle|\, 1 \leq j \leq r\right\}$. A map T is a c^{*}-algebra isomorphism if

$$T \in \mathcal{B}\left(\mathcal{B}\left(\mathcal{H}_{\mathcal{F}}\right)\right) \tag{17}$$

$$T\left(XY\right) = T\left(X\right) T\left(Y\right) \tag{18}$$

$$T\left(X^{\dagger}\right) = T\left(X\right)^{\dagger} \tag{19}$$

$$T^{-1}T = Id, \tag{20}$$

where the norm of T is defined by

$$\|T\| = \sup_{\|X\|_{\mathcal{B}\left(\mathcal{H}_{\mathcal{F}}\right)}=1} \|T\left(X\right)\|_{\mathcal{B}\left(\mathcal{H}_{\mathcal{F}}\right)}\,. \tag{21}$$

In particular one can see that

$$T\left(I_{\mathcal{F}}\right) = I_{\mathcal{F}} \tag{22}$$

and

$$\left\{T\left(a_j\right), T\left(a_k^{\dagger}\right)\right\} = T\left(\left\{a_j, a_k^{\dagger}\right\}\right) = \delta_{jk} T\left(I_{\mathcal{F}}\right) \tag{23a}$$

$$T\left(\left\{a_j^{\dagger}, a_k^{\dagger}\right\}\right) = T\left(\left\{a_j, a_k\right\}\right) = 0 \tag{23b}$$

$$1 \leq j, k \leq r. \tag{23c}$$

One can note that

$$\|a_j\| = \left\|a_j^{\dagger}\right\| = 1. \tag{24}$$

A. Canonical Transformations

A c^{*}-algebra isomorphism

$$T : \mathcal{F} \rightarrow \mathcal{F} \tag{25}$$

that is given by

$$T\begin{pmatrix}\mathbf{a}^\dagger & \mathbf{a}\end{pmatrix} = \begin{pmatrix}\mathbf{a}^\dagger & \mathbf{a}\end{pmatrix}\mathbb{T} = \begin{pmatrix}\mathbf{a}^\dagger & \mathbf{a}\end{pmatrix}\begin{pmatrix}\mathbb{U} & \overline{\mathbb{V}} \\ \mathbb{V} & \overline{\mathbb{U}}\end{pmatrix} = \begin{pmatrix}\mathbf{b}^\dagger & \mathbf{b}\end{pmatrix} \tag{26}$$

and preserves the CAR's is called a *canonical transformation*. Thus

$$b_k^\dagger = \sum_{1\le j\le r}\left(a_j^\dagger U_{jk} + a_j V_{jk}\right) \tag{27}$$

$$b_k = \sum_{1\le j\le r}\left(a_j^\dagger \bar{V}_{jk} + a_j \bar{U}_{jk}\right) \tag{28}$$

and

$$\begin{aligned}
\delta_{kl} &= \left\{b_k^\dagger, b_l\right\} = \left\{\sum_{1\le j\le r}\left(a_j^\dagger U_{jk} + a_j V_{jk}\right), \sum_{1\le j\le r}\left(a_m^\dagger \bar{V}_{ml} + a_m \bar{U}_{ml}\right)\right\} \\
&= \sum_{1\le j,m\le r}\left(\left\{a_j^\dagger U_{jk}, a_m^\dagger \bar{V}_{ml}\right\} + \left\{a_j^\dagger U_{jk}, a_m \bar{U}_{ml}\right\} + \left\{a_j V_{jk}, a_m^\dagger \bar{V}_{ml}\right\} + \left\{a_j V_{jk}, a_m \bar{U}_{mk}\right\}\right) \\
&= \sum_{1\le j,m\le r}\left(U_{jk}\left\{a_j^\dagger, a_m^\dagger\right\}\bar{V}_{ml} + U_{jk}\left\{a_j^\dagger, a_m\right\}\bar{U}_{ml} + V_{jk}\left\{a_j, a_m^\dagger\right\}\bar{V}_{ml} + V_{jk}\left\{a_j, a_m\right\}\bar{U}_{mk}\right) \\
&= \sum_{1\le j,m\le r}\left(U_{jk}\delta_{jm}\bar{U}_{ml} + V_{jk}\delta_{jm}\bar{V}_{ml}\right) = \sum_{1\le j\le r}\left(U_{jk}\bar{U}_{jl} + V_{jk}\bar{V}_{jl}\right) = \sum_{1\le j\le r}\left(U_{lj}^\dagger U_{jk} + V_{lj}^\dagger V_{jk}\right) \\
&= \left(U^\dagger U\right)_{lk} + \left(V^\dagger V\right)_{lk},
\end{aligned} \tag{29}$$

i.e.

$$\delta_{kl} = \left(U^\dagger U\right)_{lk} + \left(V^\dagger V\right)_{lk}. \tag{30}$$

In the same way

$$\begin{aligned}
0 &= \left\{b_k^\dagger, b_l^\dagger\right\} = \left\{\sum_{1\le j\le r}\left(a_j^\dagger U_{jk} + a_j V_{jk}\right), \sum_{1\le j\le r}\left(a_m^\dagger U_{ml} + a_m V_{ml}\right)\right\} \\
&= \sum_{1\le j,m\le r}\left(U_{jk}\delta_{jm}V_{ml} + V_{jk}\delta_{jm}U_{ml}\right) = \sum_{1\le j\le r}\left(V_{lj}^t U_{jk} + U_{lj}^t V_{jk}\right) \\
&= \left(V^t U\right)_{lk} + \left(U^t V\right)_{lk}
\end{aligned} \tag{31}$$

and

$$0 = \left\{b_k, b_l\right\} = \left(V^\dagger \bar{U}\right)_{lk} + \left(U^\dagger \bar{V}\right)_{lk}, \tag{32}$$

i.e.

$$\left(V^t U\right)_{lk} = -\left(U^t V\right)_{lk}. \tag{33}$$

Thus

$$\begin{pmatrix} \mathbb{U} & \overline{\mathbb{V}} \\ \mathbb{V} & \overline{\mathbb{U}} \end{pmatrix}^{\dagger} \begin{pmatrix} \mathbb{U} & \overline{\mathbb{V}} \\ \mathbb{V} & \overline{\mathbb{U}} \end{pmatrix} = \begin{pmatrix} \mathbb{U}^{\dagger} & \mathbb{V}^{\dagger} \\ \mathbb{V}^{t} & \mathbb{U}^{t} \end{pmatrix} \begin{pmatrix} \mathbb{U} & \overline{\mathbb{V}} \\ \mathbb{V} & \overline{\mathbb{U}} \end{pmatrix} \tag{34}$$

$$= \begin{pmatrix} \mathbb{U}^{\dagger}\mathbb{U} + \mathbb{V}^{\dagger}\mathbb{V} & \mathbb{U}^{\dagger}\overline{\mathbb{V}} + \mathbb{V}^{\dagger}\overline{\mathbb{U}} \\ \mathbb{V}^{t}\mathbb{U} + \mathbb{U}^{t}\mathbb{V} & \mathbb{V}^{t}\overline{\mathbb{V}} + \mathbb{U}^{t}\overline{\mathbb{U}} \end{pmatrix} = \begin{pmatrix} \mathbb{I}_r & \mathbb{O}_r \\ \mathbb{O}_r & \mathbb{I}_r \end{pmatrix}. \tag{35}$$

Hence $\begin{pmatrix} \mathbb{U} & \overline{\mathbb{V}} \\ \mathbb{V} & \overline{\mathbb{U}} \end{pmatrix} \in U\left(2r, \mathbb{R}\right)$ and

$$\mathbb{U}^{\dagger}\mathbb{U} + \mathbb{V}^{\dagger}\mathbb{V} = \mathbb{I}_r \tag{36}$$

$$\mathbb{U}^{\dagger}\overline{\mathbb{V}} + \mathbb{V}^{\dagger}\overline{\mathbb{U}} = \mathbb{O}_r, \tag{37}$$

which we show below are the conditions that $T \in O\left(2r, \mathbb{R}\right)$. We show below that

$$\left\{ \begin{pmatrix} \mathbf{b} \\ \mathbf{b}^{\dagger} \end{pmatrix}, \begin{pmatrix} \mathbf{b}^{\dagger} & \mathbf{b} \end{pmatrix} \right\} = \begin{pmatrix} \mathbb{I}_r & \mathbb{O}_r \\ \mathbb{O}_r & \mathbb{I}_r \end{pmatrix} =$$

$$= \begin{pmatrix} \mathbb{U} & \overline{\mathbb{V}} \\ \mathbb{V} & \overline{\mathbb{U}} \end{pmatrix}^{\dagger} \left\{ \begin{pmatrix} \mathbf{a} \\ \mathbf{a}^{\dagger} \end{pmatrix}, \begin{pmatrix} \mathbf{a}^{\dagger} & \mathbf{a} \end{pmatrix} \right\} \begin{pmatrix} \mathbb{U} & \overline{\mathbb{V}} \\ \mathbb{V} & \overline{\mathbb{U}} \end{pmatrix}$$

$$= \begin{pmatrix} \mathbb{U} & \overline{\mathbb{V}} \\ \mathbb{V} & \overline{\mathbb{U}} \end{pmatrix}^{\dagger} \begin{pmatrix} \left\{\mathbf{a}, \mathbf{a}^{\dagger}\right\} & \left\{\mathbf{a}, \mathbf{a}\right\} \\ \left\{\mathbf{a}^{\dagger}, \mathbf{a}^{\dagger}\right\} & \left\{\mathbf{a}^{\dagger}, \mathbf{a}\right\} \end{pmatrix} \begin{pmatrix} \mathbb{U} & \overline{\mathbb{V}} \\ \mathbb{V} & \overline{\mathbb{U}} \end{pmatrix}$$

$$= \begin{pmatrix} \mathbb{U} & \overline{\mathbb{V}} \\ \mathbb{V} & \overline{\mathbb{U}} \end{pmatrix}^{\dagger} \begin{pmatrix} \mathbb{I}_r & \mathbb{O}_r \\ \mathbb{O}_r & \mathbb{I}_r \end{pmatrix} \begin{pmatrix} \mathbb{U} & \overline{\mathbb{V}} \\ \mathbb{V} & \overline{\mathbb{U}} \end{pmatrix}$$

$$= \begin{pmatrix} \mathbb{U} & \overline{\mathbb{V}} \\ \mathbb{V} & \overline{\mathbb{U}} \end{pmatrix}^{\dagger} \begin{pmatrix} \mathbb{U} & \overline{\mathbb{V}} \\ \mathbb{V} & \overline{\mathbb{U}} \end{pmatrix} = \begin{pmatrix} \mathbb{U}^{\dagger} & \mathbb{V}^{\dagger} \\ \overline{\mathbb{V}}^{\dagger} & \overline{\mathbb{U}}^{\dagger} \end{pmatrix} \begin{pmatrix} \mathbb{U} & \overline{\mathbb{V}} \\ \mathbb{V} & \overline{\mathbb{U}} \end{pmatrix} = \begin{pmatrix} \mathbb{I}_r & \mathbb{O}_r \\ \mathbb{O}_r & \mathbb{I}_r \end{pmatrix}. \tag{38}$$

B. CAR Preserving Transformations

The CAR for $r = 2$ can be expressed as

$$\begin{pmatrix} a_1 a_1^{\dagger} & a_1 a_2^{\dagger} & a_1 a_1 & a_1 a_2 \\ a_2 a_1^{\dagger} & a_2 a_2^{\dagger} & a_2 a_1 & a_2 a_2 \\ a_1^{\dagger} a_1^{\dagger} & a_1^{\dagger} a_2^{\dagger} & a_1^{\dagger} a_1 & a_1^{\dagger} a_2 \\ a_2^{\dagger} a_1^{\dagger} & a_2^{\dagger} a_2^{\dagger} & a_2^{\dagger} a_1 & a_2^{\dagger} a_2 \end{pmatrix} + \begin{pmatrix} a_1^{\dagger} a_1 & a_2^{\dagger} a_1 & a_1 a_1 & a_2 a_1 \\ a_1^{\dagger} a_2 & a_2^{\dagger} a_2 & a_1 a_2 & a_2 a_2 \\ a_1^{\dagger} a_1^{\dagger} & a_2^{\dagger} a_1^{\dagger} & a_1 a_1^{\dagger} & a_2 a_1^{\dagger} \\ a_1^{\dagger} a_2^{\dagger} & a_2^{\dagger} a_2^{\dagger} & a_1 a_2^{\dagger} & a_2 a_2^{\dagger} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{39}$$

where

$$
\begin{pmatrix} a_1a_1^\dagger & a_1a_2^\dagger & a_1a_1 & a_1a_2 \\ a_2a_1^\dagger & a_2a_2^\dagger & a_2a_1 & a_2a_2 \\ a_1^\dagger a_1^\dagger & a_1^\dagger a_2^\dagger & a_1^\dagger a_1 & a_1^\dagger a_2 \\ a_2^\dagger a_1^\dagger & a_2^\dagger a_2^\dagger & a_2^\dagger a_1 & a_2^\dagger a_2 \end{pmatrix} = \begin{pmatrix} a_1 \\ a_2 \\ a_1^\dagger \\ a_2^\dagger \end{pmatrix} \begin{pmatrix} a_1^\dagger & a_2^\dagger & a_1 & a_2 \end{pmatrix} \tag{40}
$$

and noting that

$$
\begin{aligned}
&\left(\begin{pmatrix} \mathbb{O}_2 & \mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O}_2 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_1^\dagger \\ a_2^\dagger \end{pmatrix} \begin{pmatrix} a_1^\dagger & a_2^\dagger & a_1 & a_2 \end{pmatrix} \begin{pmatrix} \mathbb{O}_2 & \mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O}_2 \end{pmatrix} \right)^t = \left(\begin{pmatrix} \mathbb{O}_2 & \mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O}_2 \end{pmatrix} \begin{pmatrix} a_1a_1^\dagger & a_1a_2^\dagger & a_1a_1 & a_1a_2 \\ a_2a_1^\dagger & a_2a_2^\dagger & a_2a_1 & a_2a_2 \\ a_1^\dagger a_1^\dagger & a_1^\dagger a_2^\dagger & a_1^\dagger a_1 & a_1^\dagger a_2 \\ a_2^\dagger a_1^\dagger & a_2^\dagger a_2^\dagger & a_2^\dagger a_1 & a_2^\dagger a_2 \end{pmatrix} \begin{pmatrix} \mathbb{O}_2 & \mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O}_2 \end{pmatrix} \right)^t \\
&= \left(\begin{pmatrix} \mathbb{O}_2 & \mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O}_2 \end{pmatrix} \begin{pmatrix} a_1a_1 & a_1a_2 & a_1a_1^\dagger & a_1a_2^\dagger \\ a_2a_1 & a_2a_2 & a_2a_1^\dagger & a_2a_2^\dagger \\ a_1^\dagger a_1 & a_1^\dagger a_2 & a_1^\dagger a_1^\dagger & a_1^\dagger a_2^\dagger \\ a_2^\dagger a_1 & a_2^\dagger a_2 & a_2^\dagger a_1^\dagger & a_2^\dagger a_2^\dagger \end{pmatrix} \right)^t = \begin{pmatrix} a_1^\dagger a_1 & a_1^\dagger a_2 & a_1^\dagger a_1^\dagger & a_1^\dagger a_2^\dagger \\ a_2^\dagger a_1 & a_2^\dagger a_2 & a_2^\dagger a_1^\dagger & a_2^\dagger a_2^\dagger \\ a_1a_1 & a_1a_2 & a_1a_1^\dagger & a_1a_2^\dagger \\ a_2a_1 & a_2a_2 & a_2a_1^\dagger & a_2a_2^\dagger \end{pmatrix}^t = \begin{pmatrix} a_1^\dagger a_1 & a_2^\dagger a_1 & a_1a_1 & a_2a_1 \\ a_1^\dagger a_2 & a_2^\dagger a_2 & a_1a_2 & a_2a_2 \\ a_1^\dagger a_1^\dagger & a_2^\dagger a_1^\dagger & a_1a_1^\dagger & a_2a_1^\dagger \\ a_1^\dagger a_2^\dagger & a_2^\dagger a_2^\dagger & a_1a_2^\dagger & a_2a_2^\dagger \end{pmatrix}
\end{aligned} \tag{41}
$$

one can see that

$$
\begin{pmatrix} a_1 \\ a_2 \\ a_1^\dagger \\ a_2^\dagger \end{pmatrix} \begin{pmatrix} a_1^\dagger & a_2^\dagger & a_1 & a_2 \end{pmatrix} + \left(\begin{pmatrix} \mathbb{O}_2 & \mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O}_2 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_1^\dagger \\ a_2^\dagger \end{pmatrix} \begin{pmatrix} a_1^\dagger & a_2^\dagger & a_1 & a_2 \end{pmatrix} \begin{pmatrix} \mathbb{O}_2 & \mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O}_2 \end{pmatrix} \right)^t = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{42}
$$

For a canonical transformation T

$$\begin{pmatrix} \bar u_{11} & \bar u_{21} & \bar v_{11} & \bar v_{21} \\ \bar u_{12} & \bar u_{22} & \bar v_{12} & \bar v_{22} \\ v_{11} & v_{21} & u_{11} & u_{21} \\ v_{12} & v_{22} & u_{12} & u_{22} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_1^\dagger \\ a_2^\dagger \end{pmatrix} \begin{pmatrix} a_1^\dagger & a_2^\dagger & a_1 & a_2 \end{pmatrix} \begin{pmatrix} u_{11} & u_{12} & \bar v_{11} & \bar v_{12} \\ u_{21} & u_{22} & \bar v_{21} & \bar v_{22} \\ v_{11} & v_{12} & \bar u_{11} & \bar u_{12} \\ v_{21} & v_{22} & \bar u_{21} & \bar u_{22} \end{pmatrix} +$$

$$\begin{pmatrix} \mathbb{O}_2 & \mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O}_2 \end{pmatrix} \left(\begin{pmatrix} \bar u_{11} & \bar u_{21} & \bar v_{11} & \bar v_{21} \\ \bar u_{12} & \bar u_{22} & \bar v_{12} & \bar v_{22} \\ v_{11} & v_{21} & u_{11} & u_{21} \\ v_{12} & v_{22} & u_{12} & u_{22} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_1^\dagger \\ a_2^\dagger \end{pmatrix} \begin{pmatrix} a_1^\dagger & a_2^\dagger & a_1 & a_2 \end{pmatrix} \begin{pmatrix} u_{11} & u_{12} & \bar v_{11} & \bar v_{12} \\ u_{21} & u_{22} & \bar v_{21} & \bar v_{22} \\ v_{11} & v_{12} & \bar u_{11} & \bar u_{12} \\ v_{21} & v_{22} & \bar u_{21} & \bar u_{22} \end{pmatrix} \right)^t \begin{pmatrix} \mathbb{O}_2 & \mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O}_2 \end{pmatrix} \tag{43}$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{44}$$

This gives

$$\begin{pmatrix} \bar u_{11} & \bar u_{21} & \bar v_{11} & \bar v_{21} \\ \bar u_{12} & \bar u_{22} & \bar v_{12} & \bar v_{22} \\ v_{11} & v_{21} & u_{11} & u_{21} \\ v_{12} & v_{22} & u_{12} & u_{22} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_1^\dagger \\ a_2^\dagger \end{pmatrix} \begin{pmatrix} a_1^\dagger & a_2^\dagger & a_1 & a_2 \end{pmatrix} \begin{pmatrix} u_{11} & u_{12} & \bar v_{11} & \bar v_{12} \\ u_{21} & u_{22} & \bar v_{21} & \bar v_{22} \\ v_{11} & v_{12} & \bar u_{11} & \bar u_{12} \\ v_{21} & v_{22} & \bar u_{21} & \bar u_{22} \end{pmatrix} +$$

$$\begin{pmatrix} \mathbb{O}_2 & \mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O}_2 \end{pmatrix} \begin{pmatrix} u_{11} & u_{12} & \bar v_{11} & \bar v_{12} \\ u_{21} & u_{22} & \bar v_{21} & \bar v_{22} \\ v_{11} & v_{12} & \bar u_{11} & \bar u_{12} \\ v_{21} & v_{22} & \bar u_{21} & \bar u_{22} \end{pmatrix}^t \left(\begin{pmatrix} a_1 \\ a_2 \\ a_1^\dagger \\ a_2^\dagger \end{pmatrix} \begin{pmatrix} a_1^\dagger & a_2^\dagger & a_1 & a_2 \end{pmatrix} \right)^t \begin{pmatrix} \bar u_{11} & \bar u_{21} & \bar v_{11} & \bar v_{21} \\ \bar u_{12} & \bar u_{22} & \bar v_{12} & \bar v_{22} \\ v_{11} & v_{21} & u_{11} & u_{21} \\ v_{12} & v_{22} & u_{12} & u_{22} \end{pmatrix}^t \begin{pmatrix} \mathbb{O}_2 & \mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O}_2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{45}$$

and

$$
\begin{pmatrix} \bar{u}_{11} & \bar{u}_{21} & \bar{v}_{11} & \bar{v}_{21} \\ \bar{u}_{12} & \bar{u}_{22} & \bar{v}_{12} & \bar{v}_{22} \\ v_{11} & v_{21} & u_{11} & u_{21} \\ v_{12} & v_{22} & u_{12} & u_{22} \end{pmatrix}
\begin{pmatrix} a_1 \\ a_2 \\ a_1^\dagger \\ a_2^\dagger \end{pmatrix}
\begin{pmatrix} a_1^\dagger & a_2^\dagger & a_1 & a_2 \end{pmatrix}
\begin{pmatrix} u_{11} & u_{12} & \bar{v}_{11} & \bar{v}_{12} \\ u_{21} & u_{22} & \bar{v}_{21} & \bar{v}_{22} \\ v_{11} & v_{12} & \bar{u}_{11} & \bar{u}_{12} \\ v_{21} & v_{22} & \bar{u}_{21} & \bar{u}_{22} \end{pmatrix} +
$$

$$
= \begin{pmatrix} \mathbb{O}_2 & \mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O}_2 \end{pmatrix}
\begin{pmatrix} u_{11} & u_{21} & v_{11} & v_{12} \\ u_{12} & u_{22} & v_{21} & v_{22} \\ \bar{v}_{11} & \bar{v}_{21} & \bar{u}_{11} & \bar{u}_{21} \\ \bar{v}_{12} & \bar{v}_{22} & \bar{u}_{12} & \bar{u}_{22} \end{pmatrix}
\left(\begin{pmatrix} a_1 \\ a_2 \\ a_1^\dagger \\ a_2^\dagger \end{pmatrix} \begin{pmatrix} a_1^\dagger & a_2^\dagger & a_1 & a_2 \end{pmatrix} \right)^t
\begin{pmatrix} \bar{u}_{11} & \bar{u}_{21} & \bar{v}_{11} & \bar{v}_{21} \\ \bar{u}_{12} & \bar{u}_{22} & \bar{v}_{12} & \bar{v}_{22} \\ v_{11} & v_{12} & u_{11} & u_{21} \\ v_{21} & v_{22} & u_{12} & u_{22} \end{pmatrix}
\begin{pmatrix} \mathbb{O}_2 & \mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O}_2 \end{pmatrix}
$$

$$
= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{46}
$$

Now

$$
\begin{pmatrix} \mathbb{O}_2 & \mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O}_2 \end{pmatrix}
\begin{pmatrix} u_{11} & u_{21} & v_{11} & v_{12} \\ u_{12} & u_{22} & v_{21} & v_{22} \\ \bar{v}_{11} & \bar{v}_{21} & \bar{u}_{11} & \bar{u}_{21} \\ \bar{v}_{12} & \bar{v}_{22} & \bar{u}_{12} & \bar{u}_{22} \end{pmatrix}
= \begin{pmatrix} \bar{v}_{11} & \bar{v}_{21} & \bar{u}_{11} & \bar{u}_{21} \\ \bar{v}_{12} & \bar{v}_{22} & \bar{u}_{12} & \bar{u}_{22} \\ u_{11} & u_{21} & v_{11} & v_{12} \\ u_{12} & u_{22} & v_{21} & v_{22} \end{pmatrix}
\begin{pmatrix} \mathbb{O}_2 & \mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O}_2 \end{pmatrix}
\begin{pmatrix} \mathbb{O}_2 & \mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O}_2 \end{pmatrix} \tag{47}
$$

$$
= \begin{pmatrix} \bar{u}_{11} & \bar{u}_{21} & \bar{v}_{11} & \bar{v}_{21} \\ \bar{u}_{12} & \bar{u}_{22} & \bar{v}_{12} & \bar{v}_{22} \\ v_{11} & v_{12} & u_{11} & u_{21} \\ v_{21} & v_{22} & u_{12} & u_{22} \end{pmatrix}
\begin{pmatrix} \mathbb{O}_2 & \mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O}_2 \end{pmatrix} \tag{48}
$$

i.e.

$$
\begin{pmatrix} \mathbb{O}_2 & \mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O}_2 \end{pmatrix}
\begin{pmatrix} u_{11} & u_{21} & v_{11} & v_{12} \\ u_{12} & u_{22} & v_{21} & v_{22} \\ \bar{v}_{11} & \bar{v}_{21} & \bar{u}_{11} & \bar{u}_{21} \\ \bar{v}_{12} & \bar{v}_{22} & \bar{u}_{12} & \bar{u}_{22} \end{pmatrix}
= \begin{pmatrix} \bar{u}_{11} & \bar{u}_{21} & \bar{v}_{11} & \bar{v}_{21} \\ \bar{u}_{12} & \bar{u}_{22} & \bar{v}_{12} & \bar{v}_{22} \\ v_{11} & v_{12} & u_{11} & u_{21} \\ v_{21} & v_{22} & u_{12} & u_{22} \end{pmatrix}
\begin{pmatrix} \mathbb{O}_2 & \mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O}_2 \end{pmatrix} \tag{49}
$$

Hence

$$
\begin{pmatrix} \bar{u}_{11} & \bar{u}_{21} & \bar{v}_{11} & \bar{v}_{21} \\ \bar{u}_{12} & \bar{u}_{22} & \bar{v}_{12} & \bar{v}_{22} \\ v_{11} & v_{21} & u_{11} & u_{21} \\ v_{12} & v_{22} & u_{12} & u_{22} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_1^\dagger \\ a_2^\dagger \end{pmatrix} \begin{pmatrix} a_1^\dagger & a_2^\dagger & a_1 & a_2 \end{pmatrix} \begin{pmatrix} u_{11} & u_{12} & \bar{v}_{11} & \bar{v}_{12} \\ u_{21} & u_{22} & \bar{v}_{21} & \bar{v}_{22} \\ v_{11} & v_{12} & \bar{u}_{11} & \bar{u}_{12} \\ v_{21} & v_{22} & \bar{u}_{21} & \bar{u}_{22} \end{pmatrix} +
$$
$$
\begin{pmatrix} \bar{u}_{11} & \bar{u}_{21} & \bar{v}_{11} & \bar{v}_{21} \\ \bar{u}_{12} & \bar{u}_{22} & \bar{v}_{12} & \bar{v}_{22} \\ v_{11} & v_{12} & u_{11} & u_{21} \\ v_{21} & v_{22} & u_{12} & u_{22} \end{pmatrix} \begin{pmatrix} \mathbb{O}_2 & \mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O}_2 \end{pmatrix} \left(\begin{pmatrix} a_1 \\ a_2 \\ a_1^\dagger \\ a_2^\dagger \end{pmatrix} \begin{pmatrix} a_1^\dagger & a_2^\dagger & a_1 & a_2 \end{pmatrix} \right)^t \begin{pmatrix} \mathbb{O}_2 & \mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O}_2 \end{pmatrix} \begin{pmatrix} u_{11} & u_{21} & v_{11} & v_{12} \\ u_{12} & u_{22} & v_{21} & v_{22} \\ \bar{v}_{11} & \bar{v}_{21} & \bar{u}_{11} & \bar{u}_{21} \\ \bar{v}_{12} & \bar{v}_{22} & \bar{u}_{12} & \bar{u}_{22} \end{pmatrix}
$$
$$
= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{50}
$$

giving that

$$
\begin{pmatrix} \bar{u}_{11} & \bar{u}_{21} & \bar{v}_{11} & \bar{v}_{21} \\ \bar{u}_{12} & \bar{u}_{22} & \bar{v}_{12} & \bar{v}_{22} \\ v_{11} & v_{21} & u_{11} & u_{21} \\ v_{12} & v_{22} & u_{12} & u_{22} \end{pmatrix} \begin{pmatrix} u_{11} & u_{12} & \bar{v}_{11} & \bar{v}_{12} \\ u_{21} & u_{22} & \bar{v}_{21} & \bar{v}_{22} \\ v_{11} & v_{12} & \bar{u}_{11} & \bar{u}_{12} \\ v_{21} & v_{22} & \bar{u}_{21} & \bar{u}_{22} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{51}
$$

In matrix notation the CAR can thus be expressed as

$$
\begin{pmatrix} \mathbf{a}\mathbf{a}^\dagger & \mathbf{a}\mathbf{a} \\ \mathbf{a}^\dagger\mathbf{a}^\dagger & \mathbf{a}^\dagger\mathbf{a} \end{pmatrix} + \begin{pmatrix} (\mathbf{a}^\dagger\mathbf{a})^t & (\mathbf{a}\mathbf{a})^t \\ (\mathbf{a}^\dagger\mathbf{a}^\dagger)^t & (\mathbf{a}\mathbf{a}^\dagger)^t \end{pmatrix} = \begin{pmatrix} \mathbf{a} \\ \mathbf{a}^\dagger \end{pmatrix} \begin{pmatrix} \mathbf{a}^\dagger & \mathbf{a} \end{pmatrix} + \begin{pmatrix} \mathbb{O}_r & \mathbb{I}_r \\ \mathbb{I}_r & \mathbb{O}_r \end{pmatrix} \left(\begin{pmatrix} \mathbf{a} \\ \mathbf{a}^\dagger \end{pmatrix} \begin{pmatrix} \mathbf{a}^\dagger & \mathbf{a} \end{pmatrix} \right)^t \begin{pmatrix} \mathbb{O}_r & \mathbb{I}_r \\ \mathbb{I}_r & \mathbb{O}_r \end{pmatrix}
$$
$$
= \begin{pmatrix} \mathbf{a}\mathbf{a}^\dagger & \mathbf{a}\mathbf{a} \\ \mathbf{a}^\dagger\mathbf{a}^\dagger & \mathbf{a}^\dagger\mathbf{a} \end{pmatrix} + \begin{pmatrix} \mathbb{O}_r & \mathbb{I}_r \\ \mathbb{I}_r & \mathbb{O}_r \end{pmatrix} \begin{pmatrix} \mathbf{a}\mathbf{a}^\dagger & \mathbf{a}\mathbf{a} \\ \mathbf{a}^\dagger\mathbf{a}^\dagger & \mathbf{a}^\dagger\mathbf{a} \end{pmatrix}^t \begin{pmatrix} \mathbb{O}_r & \mathbb{I}_r \\ \mathbb{I}_r & \mathbb{O}_r \end{pmatrix}
$$
$$
= \begin{pmatrix} \{\mathbf{a}, \mathbf{a}^\dagger\} & \{\mathbf{a}, \mathbf{a}\} \\ \{\mathbf{a}^\dagger, \mathbf{a}^\dagger\} & \{\mathbf{a}^\dagger, \mathbf{a}\} \end{pmatrix} = \begin{pmatrix} \mathbb{I}_r & \mathbb{O}_r \\ \mathbb{O}_r & \mathbb{I}_r \end{pmatrix}, \tag{52}
$$

where

$$
\begin{pmatrix} \mathbf{a}\mathbf{a}^\dagger & \mathbf{a}\mathbf{a} \\ \mathbf{a}^\dagger\mathbf{a}^\dagger & \mathbf{a}^\dagger\mathbf{a} \end{pmatrix} = \begin{pmatrix} \mathbf{a} \\ \mathbf{a}^\dagger \end{pmatrix} \begin{pmatrix} \mathbf{a}^\dagger & \mathbf{a} \end{pmatrix}. \tag{53}
$$

One can test this by

$$\begin{pmatrix}\mathbb{U} & \overline{\mathbb{V}}\\ \mathbb{V} & \overline{\mathbb{U}}\end{pmatrix}^{\dagger}\begin{pmatrix}\mathbf{a}\\ \mathbf{a}^{\dagger}\end{pmatrix}\begin{pmatrix}\mathbf{a}^{\dagger} & \mathbf{a}\end{pmatrix}\begin{pmatrix}\mathbb{U} & \overline{\mathbb{V}}\\ \mathbb{V} & \overline{\mathbb{U}}\end{pmatrix}+\begin{pmatrix}\mathbb{O}_r & \mathbb{I}_r\\ \mathbb{I}_r & \mathbb{O}_r\end{pmatrix}\left(\begin{pmatrix}\mathbb{U} & \overline{\mathbb{V}}\\ \mathbb{V} & \overline{\mathbb{U}}\end{pmatrix}^{\dagger}\begin{pmatrix}\mathbf{a}\\ \mathbf{a}^{\dagger}\end{pmatrix}\begin{pmatrix}\mathbf{a}^{\dagger} & \mathbf{a}\end{pmatrix}\begin{pmatrix}\mathbb{U} & \overline{\mathbb{V}}\\ \mathbb{V} & \overline{\mathbb{U}}\end{pmatrix}\right)^{t}\left(\begin{matrix}\mathbb{O}_r & \mathbb{I}_r\\ \mathbb{I}_r & \mathbb{O}_r\end{matrix}\right. \tag{54}$$

$$=\begin{pmatrix}\mathbb{U}^{\dagger} & \mathbb{V}^{\dagger}\\ \mathbb{V}^{t} & \mathbb{U}^{t}\end{pmatrix}\begin{pmatrix}\mathbf{a}\\ \mathbf{a}^{\dagger}\end{pmatrix}\begin{pmatrix}\mathbf{a}^{\dagger} & \mathbf{a}\end{pmatrix}\begin{pmatrix}\mathbb{U} & \overline{\mathbb{V}}\\ \mathbb{V} & \overline{\mathbb{U}}\end{pmatrix}+\begin{pmatrix}\mathbb{O}_r & \mathbb{I}_r\\ \mathbb{I}_r & \mathbb{O}_r\end{pmatrix}\begin{pmatrix}\mathbb{U}^{t} & \mathbb{V}^{t}\\ \mathbb{V}^{\dagger} & \mathbb{U}^{\dagger}\end{pmatrix}\left(\begin{pmatrix}\mathbf{a}\\ \mathbf{a}^{\dagger}\end{pmatrix}\begin{pmatrix}\mathbf{a}^{\dagger} & \mathbf{a}\end{pmatrix}\right)^{t}\begin{pmatrix}\overline{\mathbb{U}} & \mathbb{V}\\ \overline{\mathbb{V}} & \mathbb{U}\end{pmatrix}\left(\begin{matrix}\mathbb{O}_r\\ \mathbb{I}_r\end{matrix}\right. \tag{55}$$

$$=\begin{pmatrix}\mathbb{U}^{\dagger} & \mathbb{V}^{\dagger}\\ \mathbb{V}^{t} & \mathbb{U}^{t}\end{pmatrix}\begin{pmatrix}\mathbf{a}\\ \mathbf{a}^{\dagger}\end{pmatrix}\begin{pmatrix}\mathbf{a}^{\dagger} & \mathbf{a}\end{pmatrix}\begin{pmatrix}\mathbb{U} & \overline{\mathbb{V}}\\ \mathbb{V} & \overline{\mathbb{U}}\end{pmatrix}+\begin{pmatrix}\mathbb{V}^{\dagger} & \mathbb{U}^{\dagger}\\ \mathbb{U}^{t} & \mathbb{V}^{t}\end{pmatrix}\begin{pmatrix}\mathbb{O}_r & \mathbb{I}_r\\ \mathbb{I}_r & \mathbb{O}_r\end{pmatrix}\left(\begin{pmatrix}\mathbf{a}\\ \mathbf{a}^{\dagger}\end{pmatrix}\begin{pmatrix}\mathbf{a}^{\dagger} & \mathbf{a}\end{pmatrix}\right)^{t}\begin{pmatrix}\mathbb{O}_r & \mathbb{I}_r\\ \mathbb{I}_r & \mathbb{O}_r\end{pmatrix}\left(\begin{matrix}\mathbb{V}\\ \mathbb{U}\end{matrix}\right. \tag{56}$$

$$=\begin{pmatrix}\mathbb{U}^{\dagger} & \mathbb{V}^{\dagger}\\ \mathbb{V}^{t} & \mathbb{U}^{t}\end{pmatrix}\begin{pmatrix}\mathbf{a}\\ \mathbf{a}^{\dagger}\end{pmatrix}\begin{pmatrix}\mathbf{a}^{\dagger} & \mathbf{a}\end{pmatrix}\begin{pmatrix}\mathbb{U} & \overline{\mathbb{V}}\\ \mathbb{V} & \overline{\mathbb{U}}\end{pmatrix}+\begin{pmatrix}\mathbb{V}^{\dagger} & \mathbb{U}^{\dagger}\\ \mathbb{U}^{t} & \mathbb{V}^{t}\end{pmatrix}\left(\begin{pmatrix}\mathbf{a}\\ \mathbf{a}^{\dagger}\end{pmatrix}\begin{pmatrix}\mathbf{a}^{\dagger} & \mathbf{a}\end{pmatrix}\right)^{t}\begin{pmatrix}\mathbb{V} & \overline{\mathbb{U}}\\ \mathbb{U} & \overline{\mathbb{V}}\end{pmatrix} \tag{57}$$

Consider a general isomorphism

$$T:\mathcal{F}_1\!\rightarrow\mathcal{F}_1 \tag{58}$$

given by

$$T\begin{pmatrix}\mathbf{a}^{\dagger} & \mathbf{a}\end{pmatrix}=\begin{pmatrix}\mathbf{a}^{\dagger} & \mathbf{a}\end{pmatrix}\begin{pmatrix}\mathbb{T}_{12} & \mathbb{T}_{11}\\ \mathbb{T}_{22} & \mathbb{T}_{21}\end{pmatrix}=\begin{pmatrix}\beta & \alpha\end{pmatrix} \tag{59}$$

Hence T preserves the CAR if

$$\begin{pmatrix}\mathbb{T}_{11}^{\dagger} & \mathbb{T}_{12}^{\dagger}\\ \mathbb{T}_{21}^{\dagger} & \mathbb{T}_{22}^{\dagger}\end{pmatrix}\begin{pmatrix}\mathbf{a}\\ \mathbf{a}^{\dagger}\end{pmatrix}\begin{pmatrix}\mathbf{a}^{\dagger} & \mathbf{a}\end{pmatrix}\begin{pmatrix}\mathbb{T}_{12} & \mathbb{T}_{11}\\ \mathbb{T}_{22} & \mathbb{T}_{21}\end{pmatrix} \tag{60}$$

$$+\begin{pmatrix}\mathbb{O}_r & \mathbb{I}_r\\ \mathbb{I}_r & \mathbb{O}_r\end{pmatrix}\begin{pmatrix}\mathbb{T}_{11}^{\dagger} & \mathbb{T}_{12}^{\dagger}\\ \mathbb{T}_{21}^{\dagger} & \mathbb{T}_{22}^{\dagger}\end{pmatrix}\begin{pmatrix}\mathbf{a}\\ \mathbf{a}^{\dagger}\end{pmatrix}\begin{pmatrix}\mathbf{a}^{\dagger} & \mathbf{a}\end{pmatrix}\begin{pmatrix}\mathbb{T}_{12} & \mathbb{T}_{11}\\ \mathbb{T}_{22} & \mathbb{T}_{21}\end{pmatrix}\begin{pmatrix}\mathbb{O}_r & \mathbb{I}_r\\ \mathbb{I}_r & \mathbb{O}_r\end{pmatrix} \tag{61}$$

$$=\begin{pmatrix}\mathbb{I}_r & \mathbb{O}_r\\ \mathbb{O}_r & \mathbb{I}_r\end{pmatrix}. \tag{62}$$

$$
\mathbb{T}\mathbb{T}^\dagger \begin{pmatrix} \mathbf{a} \\ \mathbf{a}^\dagger \end{pmatrix} \begin{pmatrix} \mathbf{a}^\dagger & \mathbf{a} \end{pmatrix} \mathbb{T}\mathbb{T}^\dagger \tag{63}
$$

$$
+ \mathbb{T} \begin{pmatrix} \mathbb{O}_r & \mathbb{I}_r \\ \mathbb{I}_r & \mathbb{O}_r \end{pmatrix} \mathbb{T}^\dagger \begin{pmatrix} \mathbf{a} \\ \mathbf{a}^\dagger \end{pmatrix} \begin{pmatrix} \mathbf{a}^\dagger & \mathbf{a} \end{pmatrix} \mathbb{T} \begin{pmatrix} \mathbb{O}_r & \mathbb{I}_r \\ \mathbb{I}_r & \mathbb{O}_r \end{pmatrix} \mathbb{T}^\dagger \tag{64}
$$

$$
= \begin{pmatrix} \mathbb{I}_r & \mathbb{O}_r \\ \mathbb{O}_r & \mathbb{I}_r \end{pmatrix}. \tag{65}
$$

$$
\begin{pmatrix} \mathbf{a} \\ \mathbf{a}^\dagger \end{pmatrix} \begin{pmatrix} \mathbf{a}^\dagger & \mathbf{a} \end{pmatrix} \tag{66}
$$

$$
+ \mathbb{T} \begin{pmatrix} \mathbb{O}_r & \mathbb{I}_r \\ \mathbb{I}_r & \mathbb{O}_r \end{pmatrix} \mathbb{T}^\dagger \begin{pmatrix} \mathbf{a} \\ \mathbf{a}^\dagger \end{pmatrix} \begin{pmatrix} \mathbf{a}^\dagger & \mathbf{a} \end{pmatrix} \mathbb{T} \begin{pmatrix} \mathbb{O}_r & \mathbb{I}_r \\ \mathbb{I}_r & \mathbb{O}_r \end{pmatrix} \mathbb{T}^\dagger \tag{67}
$$

$$
= \begin{pmatrix} \mathbb{I}_r & \mathbb{O}_r \\ \mathbb{O}_r & \mathbb{I}_r \end{pmatrix}. \tag{68}
$$

If

$$
\mathbb{T} \begin{pmatrix} \mathbb{O}_r & \mathbb{I}_r \\ \mathbb{I}_r & \mathbb{O}_r \end{pmatrix} \mathbb{T}^\dagger = \begin{pmatrix} \mathbb{I}_r & \mathbb{O}_r \\ \mathbb{O}_r & \mathbb{I}_r \end{pmatrix} \Leftrightarrow \mathbb{T} \begin{pmatrix} \mathbb{O}_r & \mathbb{I}_r \\ \mathbb{I}_r & \mathbb{O}_r \end{pmatrix} = \begin{pmatrix} \mathbb{I}_r & \mathbb{O}_r \\ \mathbb{O}_r & \mathbb{I}_r \end{pmatrix} \mathbb{T} \tag{69}
$$

then

$$
\left\{ \sum_{1\le j\le r} \left(a_j^\dagger T_{1jk} + a_j T_{3jk} \right), \sum_{1\le j\le r} \left(a_j^\dagger T_{2jl} + a_j T_{4jl} \right) \right\} \tag{70}
$$

$$
= \left\{ \sum_{1\le i\le r} a_i^\dagger T_{1ik}, \sum_{1\le j\le r} a_j^\dagger T_{2jl} \right\} + \left\{ \sum_{1\le i\le r} a_i^\dagger T_{1ik}, \sum_{1\le j\le r} a_j T_{4jl} \right\} \tag{71}
$$

$$
+ \left\{ \sum_{1\le i\le r} a_i T_{3ik}, \sum_{1\le j\le r} a_j^\dagger T_{2jl} \right\} + \left\{ \sum_{1\le i\le r} a_i T_{3ik}, \sum_{1\le j\le r} a_j^\dagger T_{4jl} \right\} \tag{72}
$$

$$
= \sum_{1\le i,j\le r} \left\{ T_{1ik} T_{2jl} \left\{ a_i^\dagger, a_j^\dagger \right\} + T_{1ik} T_{4jl} \left\{ a_i^\dagger, a_j \right\} + T_{3ik} T_{2jl} \left\{ a_i, a_j^\dagger \right\} + T_{3ik} T_{4jl} \left\{ a_i, a_j \right\} \right\} \tag{73}
$$

$$
= \sum_{1\le i,j\le r} \left\{ T_{1ik} T_{4jl} \delta_{ij} + T_{3ik} T_{2jl} \delta_{ij} \right\} \tag{74}
$$

C. Dressed Vacuum Vectors

One can define dressed vacuum states as

$$|\phi_g\rangle = \mathcal{N}(\phi_g) \exp\left\{ \sum_{1\leq j\leq s} c_j a_j^\dagger a_{j+s}^\dagger \right\} |\phi\rangle = \mathcal{N}(\phi_g) \prod_{1\leq j\leq s} \exp\left\{ c_j a_j^\dagger a_{j+s}^\dagger \right\} |\phi\rangle = \prod_{1\leq j\leq s} \left(d_j I_{\mathcal{F}} + c_j a_j^\dagger a_{j+s}^\dagger \right) |\phi\rangle \,. \tag{75}$$

These states have the properties

$$\begin{aligned} a_k^\dagger \prod_{1\leq j\leq s} \left(d_j I_{\mathcal{F}} + c_j a_j^\dagger a_{j+s}^\dagger \right) |\phi\rangle &= \prod_{1\leq j\neq k\leq s} \left(d_j I_{\mathcal{F}} + c_j a_j^\dagger a_{j+s}^\dagger \right) a_k^\dagger \left(d_k I_{\mathcal{F}} + c_k a_k^\dagger a_{k+s}^\dagger \right) |\phi\rangle \\ &= \prod_{1\leq j\neq k\leq s} \left(d_j I_{\mathcal{F}} + c_j a_j^\dagger a_{j+s}^\dagger \right) a_k^\dagger |\phi\rangle = d_k a_k^\dagger \prod_{1\leq j\neq k\leq s} \left(d_j I_{\mathcal{F}} + c_j a_j^\dagger a_{j+s}^\dagger \right) |\phi\rangle \,, \end{aligned} \tag{76}$$

i.e.

$$a_k^\dagger |\phi_g\rangle = d_k a_k^\dagger \left| \phi_g \left(\hat{k} \right) \right\rangle$$

$$\begin{aligned} & a_k \prod_{1\leq j\leq s} \left(d_j I_{\mathcal{F}} + c_j a_j^\dagger a_{j+s}^\dagger \right) |\phi\rangle = \prod_{1\leq j\neq k\leq s} \left(d_j I_{\mathcal{F}} + c_j a_j^\dagger a_{j+s}^\dagger \right) a_k \left(d_k I_{\mathcal{F}} + c_k a_k^\dagger a_{k+s}^\dagger \right) |\phi\rangle \\ & = \prod_{1\leq j\neq k\leq s} \left(d_j I_{\mathcal{F}} + c_j a_j^\dagger a_{j+s}^\dagger \right) \left(d_k a_k + c_k a_k a_k^\dagger a_{k+s}^\dagger \right) |\phi\rangle = \prod_{1\leq j\neq k\leq s} \left(d_j I_{\mathcal{F}} + c_j a_j^\dagger a_{j+s}^\dagger \right) \left(d_k a_k + c_k \left(1 - a_k^\dagger a_k \right) a \right. \\ & = \prod_{1\leq j\neq k\leq s} \left(d_j I_{\mathcal{F}} + c_j a_j^\dagger a_{j+s}^\dagger \right) \left(d_k a_{k+s} - c_k a_{k+s}^\dagger \right) |\phi\rangle = c_k a_{k+s}^\dagger \prod_{1\leq j\neq k\leq s} \left(d_j I_{\mathcal{F}} + c_j a_j^\dagger a_{j+s}^\dagger \right) |\phi\rangle \end{aligned} \tag{77}$$

i.e.

$$a_k |\phi_g\rangle = c_k a_{k+s}^\dagger \left| \phi_g \left(\hat{k} \right) \right\rangle$$

$$\begin{aligned} a_{k+s}^\dagger \prod_{1\leq j\leq s} \left(d_j I_{\mathcal{F}} + c_j a_j^\dagger a_{j+s}^\dagger \right) |\phi\rangle &= \prod_{1\leq j\neq k\leq s} \left(d_j I_{\mathcal{F}} + c_j a_j^\dagger a_{j+s}^\dagger \right) a_{k+s}^\dagger \left(d_k I_{\mathcal{F}} + c_k a_k^\dagger a_{k+s}^\dagger \right) |\phi\rangle \\ &= \prod_{1\leq j\neq k\leq s} \left(d_j I_{\mathcal{F}} + c_j a_j^\dagger a_{j+s}^\dagger \right) a_{k+s}^\dagger |\phi\rangle = d_k a_{k+s}^\dagger \prod_{1\leq j\neq k\leq s} \left(d_j I_{\mathcal{F}} + c_j a_j^\dagger a_{j+s}^\dagger \right) |\phi \end{aligned} \tag{78}$$

i.e.

$$a_{k+s}^\dagger |\phi_g\rangle = d_k a_{k+s}^\dagger \left| \phi_g \left(\hat{k} \right) \right\rangle$$

$$a_{k+s}\prod_{1\leq j\leq s}\left(d_jI_{\mathcal{F}}+c_ja_j^\dagger a_{j+s}^\dagger\right)|\phi\rangle=\prod_{1\leq j\neq k\leq s}\left(d_jI_{\mathcal{F}}+c_ja_j^\dagger a_{j+s}^\dagger\right)a_{k+s}\left(d_kI_{\mathcal{F}}+c_ka_k^\dagger a_{k+s}^\dagger\right)|\phi\rangle$$
$$=\prod_{1\leq j\neq k\leq s}\left(d_jI_{\mathcal{F}}+c_ja_j^\dagger a_{j+s}^\dagger\right)\left(d_ka_{k+s}+c_ka_{k+s}a_k^\dagger a_{k+s}^\dagger\right)|\phi\rangle=\prod_{1\leq j\neq k\leq s}\left(d_jI_{\mathcal{F}}+c_ja_j^\dagger a_{j+s}^\dagger\right)\left(d_ka_{k+s}-c_ka_k^\dagger\left(1-\right.\right.$$
$$=\prod_{1\leq j\neq k\leq s}\left(d_jI_{\mathcal{F}}+c_ja_j^\dagger a_{j+s}^\dagger\right)\left(d_ka_{k+s}-c_ka_{k+s}^\dagger\right)|\phi\rangle=-c_ka_k^\dagger\prod_{1\leq j\neq k\leq s}\left(d_jI_{\mathcal{F}}+c_ja_j^\dagger a_{j+s}^\dagger\right)|\phi\rangle \tag{79}$$

i.e.

$$a_{k+s}|\phi_g\rangle=-c_ka_k^\dagger\left|\phi_g\left(\hat{k}\right)\right\rangle.$$

Summarizing

$$\begin{aligned}a_k^\dagger|\phi_g\rangle&=d_ka_k^\dagger\left|\phi_g\left(\hat{k}\right)\right\rangle\\ a_k|\phi_g\rangle&=c_ka_{k+s}^\dagger\left|\phi_g\left(\hat{k}\right)\right\rangle\\ a_{k+s}^\dagger|\phi_g\rangle&=d_ka_{k+s}^\dagger\left|\phi_g\left(\hat{k}\right)\right\rangle\\ a_{k+s}|\phi_g\rangle&=-c_ka_k^\dagger\left|\phi_g\left(\hat{k}\right)\right\rangle,\end{aligned} \tag{80}$$

which leads to dressed creation and annihilation operators

$$\begin{aligned}b_k&=c_ka_k^\dagger+d_ka_{k+s}\\ b_{k+s}&=c_ka_{k+s}^\dagger-d_ka_k\\ b_k^\dagger&=d_ka_{k+s}^\dagger+c_ka_k\\ b_{k+s}^\dagger&=-d_ka_k^\dagger+c_ka_{k+s}\end{aligned} \tag{81}$$

$$\left(a_k^\dagger\ \ a_{k+s}^\dagger\ \ a_k\ \ a_{k+s}\right)\mathbb{T}_k=\left(a_k^\dagger\ \ a_{k+s}^\dagger\ \ a_k\ \ a_{k+s}\right)\begin{pmatrix}\mathbb{U}_k&\overline{\mathbb{V}}_k\\ \mathbb{V}_k&\overline{\mathbb{U}}_k\end{pmatrix}$$
$$=\left(a_k^\dagger\ \ a_{k+s}^\dagger\ \ a_k\ \ a_{k+s}\right)\begin{pmatrix}0&-d_k&c_k&0\\ d_k&0&0&c_k\\ c_k&0&0&-d_k\\ 0&c_k&d_k&0\end{pmatrix}=\left(b_k^\dagger\ \ b_{k+s}^\dagger\ \ b_k\ \ b_{k+s}\right). \tag{82}$$

This gives

$$T\left(\mathbf{a}^\dagger\ \ \mathbf{a}\right)=\left(\mathbf{a}^\dagger\ \ \mathbf{a}\right)\mathbb{T}=\left(\mathbf{a}^\dagger\ \ \mathbf{a}\right)\begin{pmatrix}\mathbb{U}&\overline{\mathbb{V}}\\ \mathbb{V}&\overline{\mathbb{U}}\end{pmatrix}=\left(\mathbf{b}^\dagger\ \ \mathbf{b}\right), \tag{83}$$

where

$$\mathbb{T} = \begin{pmatrix} \mathbb{T}_1 & \mathbb{O} & \mathbb{O} \\ \mathbb{O} & \ddots & \mathbb{O} \\ \mathbb{O} & \mathbb{O} & \mathbb{T}_r \end{pmatrix}.$$

The conditions

$$\mathbb{U}^\dagger \mathbb{U} + \mathbb{V}^\dagger \mathbb{V} = \mathbb{I}_r \tag{84}$$

$$\mathbb{U}^\dagger \overline{\mathbb{V}} + \mathbb{V}^\dagger \overline{\mathbb{U}} = \mathbb{O}_r \tag{85}$$

give

$$\begin{pmatrix} 0 & d_k \\ -d_k & 0 \end{pmatrix} \begin{pmatrix} 0 & -d_k \\ d_k & 0 \end{pmatrix} + \begin{pmatrix} c_k & 0 \\ 0 & c_k \end{pmatrix} \begin{pmatrix} c_k & 0 \\ 0 & c_k \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

i.e.

$$d_k^2 + c_k^2 = 1.$$

By construction

$$b_k \left|\phi_g\right\rangle = \left(c_k a_k^\dagger + d_k a_{k+s} \right) \left|\phi_g\right\rangle = 0,$$

and as the transformation $\mathbb{T}$ is canonical and $\left|\phi_g\right\rangle$ is normalized then

$$\begin{aligned} \left\langle \phi_g \middle| \left\{ b_k^\dagger, b_k \right\} \phi_g \right\rangle &= \left\langle \phi_g \middle| \phi_g \right\rangle = \left\langle \phi_g \middle| \left(b_k^\dagger b_k + b_k b_k^\dagger \right) \phi_g \right\rangle \\ &= \left\langle \phi_g \middle| b_k b_k^\dagger \phi_g \right\rangle = 1, \end{aligned}$$

which implies

$$\left|b_k\right\rangle = b_k^\dagger \left|\phi_g\right\rangle$$

is a normalized state.

If

$$c_k = d_k$$

then

$$\begin{aligned} b_k &= c_k \left(a_k^\dagger + a_{k+s} \right) \\ b_{k+s} &= c_k \left(a_{k+s}^\dagger - a_k \right) \\ b_k^\dagger &= c_k \left(a_{k+s}^\dagger + a_k \right) \\ b_{k+s}^\dagger &= c_k \left(-a_k^\dagger + a_{k+s} \right) \end{aligned} \tag{86}$$

and

$$b_k + b_{k+s} = c_k \left(a_k^\dagger + a_{k+s} + a_{k+s}^\dagger - a_k \right)$$

D. General Self Adjoint Field Operators

In general

$$X(|u\rangle) = \frac{1}{\sqrt{2}}\left(\Phi^\dagger(|u\rangle) + \Phi(|u\rangle)\right) \tag{87}$$

$$Y(|u\rangle) = \frac{i}{\sqrt{2}}\left(\Phi^\dagger(|u\rangle) - \Phi(|u\rangle)\right) \tag{88}$$

$$\Phi^\dagger(|u\rangle) = \frac{1}{\sqrt{2}}\left(X(|u\rangle) - iY(|u\rangle)\right) \tag{89}$$

$$\Phi(|u\rangle) = \frac{1}{\sqrt{2}}\left(X(|u\rangle) + iY(|u\rangle)\right). \tag{90}$$

Hence

$$\begin{aligned}
&\{X(|u\rangle), X(|v\rangle)\} = \frac{1}{2}\left\{\left(\Phi^\dagger(|u\rangle) + \Phi(|u\rangle)\right), \left(\Phi^\dagger(|v\rangle) + \Phi(|v\rangle)\right)\right\} \\
&= \left\{\Phi^\dagger(|u\rangle), \Phi^\dagger(|v\rangle)\right\} + \left\{\Phi^\dagger(|u\rangle), \Phi(|v\rangle)\right\} + \left\{\Phi(|u\rangle), \Phi^\dagger(|v\rangle)\right\} + \{\Phi(|u\rangle), \Phi(|v\rangle)\} \\
&= 0 + \overline{\langle u|\, v\rangle} I_{\mathcal{F}} + \langle u|\, v\rangle\, I_{\mathcal{F}} + 0 = \operatorname{Re}\langle u|\, v\rangle\, I_{\mathcal{F}} && (91) \\
&\{X(|u\rangle), Y(|v\rangle)\} = \left\{\left(\Phi^\dagger(|u\rangle) + \Phi(|u\rangle)\right), i\left(\Phi^\dagger(|v\rangle) - \Phi(|v\rangle)\right)\right\} \\
&= i\left\{\Phi^\dagger(|u\rangle), \Phi^\dagger(|v\rangle)\right\} - i\left\{\Phi^\dagger(|u\rangle), \Phi(|v\rangle)\right\} + i\left\{\Phi(|u\rangle), \Phi^\dagger(|v\rangle)\right\} - i\{\Phi(|u\rangle), \Phi(|v\rangle)\} \\
&= 0 - i\overline{\langle u|\, v\rangle} I_{\mathcal{F}} + i\langle u|\, v\rangle\, I_{\mathcal{F}} = \operatorname{Im}\langle u|\, v\rangle\, I_{\mathcal{F}} && (92) \\
&\{Y(|u\rangle), Y(|v\rangle)\} = \left\{i\left(\Phi^\dagger(|u\rangle) - \Phi(|u\rangle)\right), i\left(\Phi^\dagger(|v\rangle) - \Phi(|v\rangle)\right)\right\} \\
&= -\left\{\Phi^\dagger(|u\rangle), \Phi^\dagger(|v\rangle)\right\} + \left\{\Phi^\dagger(|u\rangle), \Phi(|v\rangle)\right\} + \left\{\Phi(|u\rangle), \Phi^\dagger(|v\rangle)\right\} - \{\Phi(|u\rangle), \Phi(|v\rangle)\} \\
&= 0 + \overline{\langle u|\, v\rangle} I_{\mathcal{F}} + \langle u|\, v\rangle\, I_{\mathcal{F}} + 0 = \operatorname{Re}\langle u|\, v\rangle\, I_{\mathcal{F}} && (93)
\end{aligned}$$

and

$$\{X(|u\rangle), X(|u\rangle)\} = \langle u|\, u\rangle\, I_{\mathcal{F}} \tag{94}$$

$$\{X(|u\rangle), Y(|u\rangle)\} = 0 \tag{95}$$

$$\{Y(|u\rangle), Y(|u\rangle)\} = \langle u|\, u\rangle\, I_{\mathcal{F}}. \tag{96}$$

In particular for a conb $\{|u_i\rangle|\, 1 \le j \le r\}$

$$\{X(|u_i\rangle), X(|u_j\rangle)\} = \delta_{ij} I_{\mathcal{F}} \tag{97}$$

$$\{X(|u_i\rangle), Y(|u_j\rangle)\} = 0 \tag{98}$$

$$\{Y(|u_i\rangle), Y(|u_j\rangle)\} = \delta_{ij} I_{\mathcal{F}}. \tag{99}$$

E. Self Ajoint Basis of the CAR Fock Algebra $\mathcal{F}$

One can define self adjoint operators

$$x_j = \frac{1}{\sqrt{2}}\left(a_j^\dagger + a_j\right) = x_j^\dagger \tag{100}$$

$$x_{j+r} = \frac{i}{\sqrt{2}}\left(a_j^\dagger - a_j\right) = x_{j+r}^\dagger. \tag{101}$$

Then noting that

$$-ix_{j+r} = \frac{1}{\sqrt{2}}\left(a_j^\dagger - a_j\right) \tag{102}$$

gives the inverse relationships

$$a_j^\dagger = \frac{1}{\sqrt{2}}\left(x_j - ix_{j+r}\right) \tag{103}$$

$$a_j = \frac{1}{\sqrt{2}}\left(x_j + ix_{j+r}\right). \tag{104}$$

This can be checked by

$$\begin{aligned}
\frac{1}{\sqrt{2}}\left(a_j^\dagger + a_j\right) &= \frac{1}{2}\left(\left(x_j - ix_{j+r}\right) + \left(x_j + ix_{j+r}\right)\right) = x_j \\
\frac{i}{\sqrt{2}}\left(a_j^\dagger - a_j\right) &= \frac{i}{2}\left(\left(x_j - ix_{j+r}\right) - \left(x_j + ix_{j+r}\right)\right) = x_{j+s}.
\end{aligned} \tag{105}$$

In matrix notation the transformation between the $\left\{a_j, a_j^\dagger \middle| 1 \leq j \leq r\right\}$ and $\left\{x_j \middle| 1 \leq j \leq 2r\right\}$ bases is given by

$$\begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{pmatrix} = J \begin{pmatrix} \mathbf{a}^\dagger & \mathbf{a} \end{pmatrix} = \begin{pmatrix} \mathbf{a}^\dagger & \mathbf{a} \end{pmatrix} \mathbb{J} \tag{106}$$

$$= \begin{pmatrix} \mathbf{a}^\dagger & \mathbf{a} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix}, \tag{107}$$

where

$$\mathbb{J} = \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix} \in U\left(2r, \mathbb{R}\right) \tag{108}$$

i.e.

$$\begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix}^\dagger \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix} = \tag{109}$$

$$\begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{1}{\sqrt{2}}\mathbb{I}_r \\ \frac{-i}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix} = \begin{pmatrix} \mathbb{I}_r & \mathbb{O}_r \\ \mathbb{O}_r & \mathbb{I}_r \end{pmatrix}, \tag{110}$$

but is NOT a canonical transformation as it is not a c^*-algebra map. One should also note that

$$\begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_{p,q} & \frac{i}{\sqrt{2}}\mathbb{I}_{p,q} \\ \frac{1}{\sqrt{2}}\mathbb{I}_{p,q} & -\frac{i}{\sqrt{2}}\mathbb{I}_{p,q} \end{pmatrix}^\dagger \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_{p,q} & \frac{i}{\sqrt{2}}\mathbb{I}_{p,q} \\ \frac{1}{\sqrt{2}}\mathbb{I}_{p,q} & -\frac{i}{\sqrt{2}}\mathbb{I}_{p,q} \end{pmatrix} = \tag{111}$$

$$\begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_{p,q} & \frac{1}{\sqrt{2}}\mathbb{I}_{p,q} \\ \frac{-i}{\sqrt{2}}\mathbb{I}_{p,q} & \frac{i}{\sqrt{2}}\mathbb{I}_{p,q} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_{p,q} & \frac{i}{\sqrt{2}}\mathbb{I}_{p,q} \\ \frac{1}{\sqrt{2}}\mathbb{I}_{p,q} & -\frac{i}{\sqrt{2}}\mathbb{I}_{p,q} \end{pmatrix} = \begin{pmatrix} \mathbb{I}_r & \mathbb{O}_r \\ \mathbb{O}_r & \mathbb{I}_r \end{pmatrix}, \tag{112}$$

where

$$\mathbb{I}_{p,q} = \begin{pmatrix} \mathbb{I}_p & \mathbb{O} \\ \mathbb{O} & -\mathbb{I}_q \end{pmatrix} \text{ and } p+q=r. \tag{113}$$

One can set

$$J\left(a_j^\dagger\right) = x_j \tag{114}$$

$$J\left(a_j\right) = x_{j+r}. \tag{115}$$

The self adjoint basis produced by the canonically transformed $\left\{ a_j^\dagger, a_j \middle| 1 \le j \le r \right\}$ is

$$\begin{aligned}
\begin{pmatrix} \mathbf{y}_1 & \mathbf{y}_2 \end{pmatrix} = J\begin{pmatrix} \mathbf{b}^\dagger & \mathbf{b} \end{pmatrix} &= \begin{pmatrix} \mathbf{b}^\dagger & \mathbf{b} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix} \\
&= \begin{pmatrix} \mathbf{a}^\dagger & \mathbf{a} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix}^\dagger \begin{pmatrix} \mathbb{U} & \overline{\mathbb{V}} \\ \mathbb{V} & \overline{\mathbb{U}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix} \\
&= \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix}^\dagger \begin{pmatrix} \mathbb{U} & \overline{\mathbb{V}} \\ \mathbb{V} & \overline{\mathbb{U}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
&= \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{1}{\sqrt{2}}\mathbb{I}_r \\ \frac{-i}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix} \begin{pmatrix} \mathbb{U} & \overline{\mathbb{V}} \\ \mathbb{V} & \overline{\mathbb{U}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix} \\
&= \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{U}+\frac{1}{\sqrt{2}}\mathbb{V} & \frac{1}{\sqrt{2}}\overline{\mathbb{V}}+\frac{1}{\sqrt{2}}\overline{\mathbb{U}} \\ \frac{-i}{\sqrt{2}}\mathbb{U}+\frac{i}{\sqrt{2}}\mathbb{V} & \frac{-i}{\sqrt{2}}\overline{\mathbb{V}}+\frac{i}{\sqrt{2}}\overline{\mathbb{U}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix} \\
&= \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{pmatrix} \begin{pmatrix} \frac{1}{2}\mathbb{U}+\frac{1}{2}\mathbb{V}+\frac{1}{2}\overline{\mathbb{V}}+\frac{1}{2}\overline{\mathbb{U}} & \frac{i}{2}\mathbb{U}-\frac{i}{2}\mathbb{V}+\frac{i}{2}\overline{\mathbb{V}}-\frac{i}{2}\overline{\mathbb{U}} \\ \frac{-i}{2}\mathbb{U}+\frac{i}{2}\mathbb{V}-\frac{i}{2}\overline{\mathbb{V}}+\frac{i}{2}\overline{\mathbb{U}} & \frac{1}{2}\mathbb{U}+\frac{1}{2}\mathbb{V}-\frac{1}{2}\overline{\mathbb{V}}+\frac{1}{2}\overline{\mathbb{U}} \end{pmatrix} \\
&= \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{pmatrix} \begin{pmatrix} \mathrm{Re}\left(\mathbb{U}+\mathbb{V}\right) & \mathrm{Im}\left(\mathbb{V}+\mathbb{U}\right) \\ \mathrm{Im}\left(\mathbb{V}-\mathbb{U}\right) & \mathrm{Re}\left(\mathbb{U}-\mathbb{V}\right) \end{pmatrix}.
\end{aligned} \tag{116}$$

i.e.

$$\begin{pmatrix} \mathbf{y}_1 & \mathbf{y}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{pmatrix} \begin{pmatrix} \mathrm{Re}\,(\mathbb{U}+\mathbb{V}) & \mathrm{Im}\,(\mathbb{V}+\mathbb{U}) \\ \mathrm{Im}\,(\mathbb{V}-\mathbb{U}) & \mathrm{Re}\,(\mathbb{U}-\mathbb{V}) \end{pmatrix}. \tag{117}$$

Thus as

$$\begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{pmatrix} = J \begin{pmatrix} \mathbf{a}^\dagger & \mathbf{a} \end{pmatrix} = \begin{pmatrix} \mathbf{a}^\dagger & \mathbf{a} \end{pmatrix} \mathbb{J} \tag{118}$$

$$\begin{pmatrix} \mathbf{y}_1 & \mathbf{y}_2 \end{pmatrix} = J \begin{pmatrix} \mathbf{b}^\dagger & \mathbf{b} \end{pmatrix} = \begin{pmatrix} \mathbf{b}^\dagger & \mathbf{b} \end{pmatrix} \mathbb{J} \tag{119}$$

one obtains

$$\begin{pmatrix} \mathbf{y}_1 & \mathbf{y}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{pmatrix} \begin{pmatrix} \mathrm{Re}\,(\mathbb{U}+\mathbb{V}) & \mathrm{Im}\,(\mathbb{V}+\mathbb{U}) \\ \mathrm{Im}\,(\mathbb{V}-\mathbb{U}) & \mathrm{Re}\,(\mathbb{U}-\mathbb{V}) \end{pmatrix} \tag{120}$$

$$= \begin{pmatrix} \mathbf{a}^\dagger & \mathbf{a} \end{pmatrix} \mathbb{J}\mathbb{J}^\dagger \begin{pmatrix} \mathbb{U} & \overline{\mathbb{V}} \\ \mathbb{V} & \overline{\mathbb{U}} \end{pmatrix} \mathbb{J} \tag{121}$$

and hence

$$\begin{pmatrix} \mathrm{Re}\,(\mathbb{U}+\mathbb{V}) & \mathrm{Im}\,(\mathbb{V}+\mathbb{U}) \\ \mathrm{Im}\,(\mathbb{V}-\mathbb{U}) & \mathrm{Re}\,(\mathbb{U}-\mathbb{V}) \end{pmatrix} = \mathbb{J}^\dagger \begin{pmatrix} \mathbb{U} & \overline{\mathbb{V}} \\ \mathbb{V} & \overline{\mathbb{U}} \end{pmatrix} \mathbb{J}. \tag{122}$$

As both $\begin{pmatrix} \mathbb{U} & \overline{\mathbb{V}} \\ \mathbb{V} & \overline{\mathbb{U}} \end{pmatrix}$ and $\begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix}$ are unitary one has that

$$\begin{aligned} &\begin{pmatrix} \mathrm{Re}\,(\mathbb{U}+\mathbb{V}) & \mathrm{Im}\,(\mathbb{V}+\mathbb{U}) \\ \mathrm{Im}\,(\mathbb{V}-\mathbb{U}) & \mathrm{Re}\,(\mathbb{U}-\mathbb{V}) \end{pmatrix} \begin{pmatrix} \mathrm{Re}\,(\mathbb{U}+\mathbb{V}) & \mathrm{Im}\,(\mathbb{V}+\mathbb{U}) \\ \mathrm{Im}\,(\mathbb{V}-\mathbb{U}) & \mathrm{Re}\,(\mathbb{U}-\mathbb{V}) \end{pmatrix}^\dagger \\ &= \begin{pmatrix} \mathrm{Re}\,(\mathbb{U}+\mathbb{V}) & \mathrm{Im}\,(\mathbb{V}+\mathbb{U}) \\ \mathrm{Im}\,(\mathbb{V}-\mathbb{U}) & \mathrm{Re}\,(\mathbb{U}-\mathbb{V}) \end{pmatrix}^\dagger \begin{pmatrix} \mathrm{Re}\,(\mathbb{U}+\mathbb{V}) & \mathrm{Im}\,(\mathbb{V}+\mathbb{U}) \\ \mathrm{Im}\,(\mathbb{V}-\mathbb{U}) & \mathrm{Re}\,(\mathbb{U}-\mathbb{V}) \end{pmatrix} \\ &= \mathbb{I}_{2r} \end{aligned} \tag{123}$$

and as $\begin{pmatrix} \mathrm{Re}\,(\mathbb{U}+\mathbb{V}) & \mathrm{Im}\,(\mathbb{V}+\mathbb{U}) \\ \mathrm{Im}\,(\mathbb{V}-\mathbb{U}) & \mathrm{Re}\,(\mathbb{U}-\mathbb{V}) \end{pmatrix}$ is a real matrix

$$\begin{pmatrix} \mathrm{Re}\,(\mathbb{U}+\mathbb{V}) & \mathrm{Im}\,(\mathbb{V}+\mathbb{U}) \\ \mathrm{Im}\,(\mathbb{V}-\mathbb{U}) & \mathrm{Re}\,(\mathbb{U}-\mathbb{V}) \end{pmatrix}^\dagger = \begin{pmatrix} \mathrm{Re}\,(\mathbb{U}+\mathbb{V}) & \mathrm{Im}\,(\mathbb{V}+\mathbb{U}) \\ \mathrm{Im}\,(\mathbb{V}-\mathbb{U}) & \mathrm{Re}\,(\mathbb{U}-\mathbb{V}) \end{pmatrix}^t \tag{124}$$

and

$$\begin{pmatrix} \mathrm{Re}\,(\mathbb{U}+\mathbb{V}) & \mathrm{Im}\,(\mathbb{V}+\mathbb{U}) \\ \mathrm{Im}\,(\mathbb{V}-\mathbb{U}) & \mathrm{Re}\,(\mathbb{U}-\mathbb{V}) \end{pmatrix} \in O\,(2r, \mathbb{R})\,, \tag{125}$$

i.e. *the group of canonical transformations is* $O(2r,\mathbb{R}) \subset U(2r,\mathbb{R})$. There are other transformation of $\mathcal{F}_1$ that preserve the CAR's that are not canonical, in particular $\begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix}$. In fact one can see that any transformation belonging to $O(2r,\mathbb{C})$ preserve the CAR's, but does not in general preserve the relationships

$$x_j = x_j^{\dagger}. \tag{126}$$

The action of $O(2r,\mathbb{R})$ on $\mathcal{F}$ can be expressed in terms of inner automorphisms i.e.

$$y_j = \hat{O}(x_j) = O(\lambda)\, x_j O(\lambda)^t = \exp\left\{\sum_{1\le k<l\le 2r} \lambda_{kl} x_k x_l\right\} x_j \exp\left\{-\sum_{1\le k<l\le 2r} \lambda_{kl} x_k x_l\right\}, \tag{127}$$

where

$$\lambda = -\lambda^t, \tag{128}$$

thus

$$\begin{pmatrix} \mathbf{y}_1 & \mathbf{y}_2 \end{pmatrix} = \widehat{O(\lambda)} \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{pmatrix} \mathbb{T} = \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{pmatrix} \begin{pmatrix} \mathrm{Re}\,(\mathbb{U}+\mathbb{V}) & \mathrm{Im}\,(\mathbb{V}+\mathbb{U}) \\ \mathrm{Im}\,(\mathbb{V}-\mathbb{U}) & \mathrm{Re}\,(\mathbb{U}-\mathbb{V}) \end{pmatrix} \tag{129}$$

$$\begin{pmatrix} \mathbf{y}_1 & \mathbf{y}_2 \end{pmatrix} = \widehat{O(\lambda)} \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{pmatrix} = \begin{pmatrix} O(\lambda)\,\mathbf{x}_1 O(\lambda)^t & O(\lambda)\,\mathbf{x}_2 O(\lambda)^t \end{pmatrix} = \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{pmatrix} \widehat{\mathbb{O}} = \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{pmatrix} \exp\{\mathrm{adj}\,(\boldsymbol{\lambda})\} \tag{130}$$

F.

G.

H. Clifford Algebra

The squares of the self adjoint operators $\{x_j | 1 \le j \le 2r\}$ are multiples of the identity i.e.

$$x_j^2 = \frac{1}{\sqrt{2}}\left(a_j^{\dagger} + a_j\right) \frac{1}{\sqrt{2}}\left(a_j^{\dagger} + a_j\right) = \frac{1}{2} I_{\mathcal{F}} \tag{131}$$

$$x_{j+r}^2 = \frac{i}{\sqrt{2}}\left(a_j^{\dagger} - a_j\right) \frac{i}{\sqrt{2}}\left(a_j^{\dagger} - a_j\right) = \frac{1}{2} I_{\mathcal{F}} \tag{132}$$

Thus

$$\|x_j\|_{\mathcal{B}(\mathcal{H}_{\mathcal{F}})} = \frac{1}{\sqrt{2}}, 1 \leq j \leq 2r. \tag{133}$$

The products for $1 \leq j \neq k \leq r$ are

$$\begin{aligned} x_j x_k &= \frac{1}{\sqrt{2}}\left(a_j^\dagger + a_j\right)\frac{1}{\sqrt{2}}\left(a_k^\dagger + a_k\right) = \frac{1}{2}\left(a_j^\dagger a_k + a_j a_k + a_j^\dagger a_k^\dagger + a_j a_k^\dagger\right) \\ x_k x_j &= \frac{1}{\sqrt{2}}\left(a_k^\dagger + a_k\right)\frac{1}{\sqrt{2}}\left(a_j^\dagger + a_j\right) = \frac{1}{2}\left(a_k^\dagger a_j + a_k a_j + a_k^\dagger a_j^\dagger + a_k a_j^\dagger\right) \end{aligned} \tag{134}$$

hence

$$x_j x_k = -x_k x_j. \tag{135}$$

The products for $r+1 \leq j+r \neq k+r \leq 2r$ are

$$\begin{aligned} x_{j+r} x_{k+r} &= \frac{i}{\sqrt{2}}\left(a_j^\dagger - a_j\right)\frac{i}{\sqrt{2}}\left(a_k^\dagger - a_k\right) = \frac{1}{2}\left(-a_j^\dagger a_k^\dagger + a_j^\dagger a_k + a_j a_k^\dagger - a_j a_k\right) \\ x_{k+r} x_{j+r} &= \frac{i}{\sqrt{2}}\left(a_k^\dagger - a_k\right)\frac{i}{\sqrt{2}}\left(a_j^\dagger - a_j\right) = \frac{1}{2}\left(-a_k^\dagger a_j^\dagger + a_k^\dagger a_j + a_k a_j^\dagger - a_k a_j\right) \end{aligned} \tag{136}$$

hence

$$x_{j+r} x_{k+r} = -x_{k+r} x_{j+r}. \tag{137}$$

The products for $1 \leq j \leq r$; $r+1 \leq k+r \leq 2r$ are

$$\begin{aligned} x_j x_{k+r} &= \frac{1}{\sqrt{2}}\left(a_j^\dagger + a_j\right)\frac{i}{\sqrt{2}}\left(a_k^\dagger - a_k\right) = \frac{1}{2}\left(i a_j^\dagger a_k^\dagger - i a_j^\dagger a_k + i a_j a_k^\dagger - i a_j a_k\right) \\ x_{k+r} x_j &= \frac{i}{\sqrt{2}}\left(a_k^\dagger - a_k\right)\frac{1}{\sqrt{2}}\left(a_j^\dagger + a_j\right) = (x_j x_{k+s})^\dagger = \frac{1}{2}\left(i a_k^\dagger a_j^\dagger + i a_k^\dagger a_j - i a_k a_j^\dagger - i a_k a_j\right) \end{aligned} \tag{138}$$

hence $x_j x_{k+r} = -x_{k+r} x_j$.

The squares for $1 \leq j \leq r$ are

$$\begin{aligned} x_j^2 &= x_j x_j = \frac{1}{\sqrt{2}}\left(a_j^\dagger + a_j\right)\frac{1}{\sqrt{2}}\left(a_j^\dagger + a_j\right) = \frac{1}{2}\left(a_j a_j^\dagger + a_j a_j + a_j^\dagger a_j^\dagger + a_j^\dagger a_j\right) = \frac{1}{2} I_{\mathcal{F}} \\ x_{j+r}^2 &= x_{j+r} x_{j+r} = \frac{i}{\sqrt{2}}\left(a_j^\dagger - a_j\right)\frac{i}{\sqrt{2}}\left(a_j^\dagger - a_j\right) \\ &= -\frac{1}{2}\left(a_j^\dagger a_j^\dagger - a_j^\dagger a_j - a_j a_j^\dagger + a_j a_j\right) = \frac{1}{2} I_{\mathcal{F}}. \end{aligned} \tag{139}$$

Thus the self adjoint operators $\{x_j |\, 1 \leq j \leq 2r\}$ satisfy the CAR's

$$\{x_j, x_k\} = \delta_{jk} I_{\mathcal{F}};\ 1 \leq i, j \leq 2r. \tag{140}$$

The Fermi Fock algebra $\mathcal{F}$, which is representation of a CAR algebra, is generated by *complex* polynomials of the generators $\{x_j | 1 \leq j \leq 2r\}$ that satisfy the conditions

$$x_j x_k = -x_k x_j,\ 1 \leq j < k \leq 2r \tag{141}$$

$$x_j^2 = \frac{1}{2} I_{\mathcal{F}},\ 1 \leq j \leq 2r \tag{142}$$

or equivalently

$$\{x_j, x_k\} = \delta_{jk} I_{\mathcal{F}};\ 1 \leq i, j \leq 2r, \tag{143}$$

thus is a *Clifford Algebra*. This is a graded algebra and be expressed as the exterior algebra over $\mathcal{F}_1$ i.e. as the vector space direct sum

$$\mathcal{F} = \bigwedge \mathcal{F}_1 = \bigoplus_{0 \leq n \leq 2r} \mathcal{F}_n, \tag{144}$$

where $\mathcal{F}_n$ are multilinear homogeneous polynomials of degree n in $\{x_j\}$ and the scaler part is defined as

$$\mathcal{F}_0 = \{ z I_{\mathcal{F}} | z \in \mathbb{C} \}. \tag{145}$$

One needs to show that

$$I_{\mathcal{F}} \neq \sum_{1 \leq n \leq 2r} \sum_{1 \leq j_1 < \cdots < j_n \leq 2r} A_{j_1 \cdots j_{2r}} x_{j_1} \ldots x_{j_n}$$

in order that the subspaces $\{\mathcal{F}_n\}$ are disjoint.

This is a different decomposition to the ones introduced before i.e. in terms of degrees of creation and annihilation operators

$$\mathcal{F} = \bigoplus_{0 \leq n,m \leq r} \mathcal{B}\left(\mathcal{H}_{\mathcal{F}}\right)_{nm} \tag{146}$$

and mappings between the component Hilbert spaces

$$\mathcal{F} = \bigoplus_{0 \leq n,m \leq r} \mathcal{B}_{nm}\left(\mathcal{H}_{\mathcal{F}}\right) = \bigoplus_{0 \leq n,m \leq r} \mathcal{B}_{nm}\left(\mathcal{H}^n, \mathcal{H}^m\right). \tag{147}$$

If A is a general element of $\mathcal{F}$ then

$$A = \sum_{0 \leq n \leq 2r} \langle A \rangle_n = \sum_{0 \leq n \leq 2r} A_n, \tag{148}$$

where

$$\langle A \rangle_n = A_n \in \mathcal{F}_n. \tag{149}$$

Thus

$$\langle A\rangle_n = \sum_{1\le j_1<\cdots<j_n\le 2r} A_{nj_1\cdots j_n} x_{j_1}\cdots x_{j_n} = \sum_{1\le j_1<\cdots<j_n\le 2r} A_{nj_1\cdots j_n} x_{j_1}\wedge\cdots\wedge x_{j_n} \tag{150}$$

and

$$\langle A\rangle_0 = \alpha I_{\mathcal{F}} = A_0; \alpha \in \mathbb{C}. \tag{151}$$

Thus

$$\langle A_n\rangle_m = \delta_{nm} A_n. \tag{152}$$

II. GEOMETRIC ALGEBRA

A. Scaler Product

The product C_1D_1 of two grade 1 operators (vectors) can be expressed as the sum of a symmetric and antisymmetric part of the operator product in $\mathcal{F}$ i.e.

$$C_1D_1 = \frac{1}{2}\{C_1D_1 + D_1C_1\} + \frac{1}{2}\{C_1D_1 - D_1C_1\} = \frac{1}{2}\{C_1, D_1\} + \frac{1}{2}[C_1, D_1] = C_1\vee D_1 + C_1\wedge D_1, \tag{153}$$

which allows one to define a scalar product $(\cdot|\cdot)$ between vectors as

$$(C_1|D_1) = C_1 \vee D_1 = \frac{1}{2}\{C_1, D_1\}. \tag{154}$$

The basis $\{x_j|\ 1\le j\le 2r\}$ is orthogonal with respect to this scalar product i.e.

$$(x_j|x_k) = x_j \vee x_k = \frac{1}{2}\{x_jx_k + x_kx_j\} = \{x_j, x_k\} = \frac{1}{2}\delta_{jk} I_{\mathcal{F}}; 1\le j\le 2r, \tag{155}$$

but not normalized. One can define normalized vectors by

$$\tilde{x}_j = \sqrt{2}x_j; 1\le j\le 2r, \tag{156}$$

but these vectors do not satisfy the CAR as

$$\{\tilde{x}_j, \tilde{x}_k\} = 2\delta_{jk} I_{\mathcal{F}}.$$

One can expand any two vectors C_1 and D_1 in terms of this orthonormal basis as

$$C_1 = \sum_{1\le j\le 2r} c_j\tilde{x}_j \tag{157}$$

$$D_1 = \sum_{1\le j\le 2r} d_j\tilde{x}_j \tag{158}$$

and obtain

$$(C_1|D_1) = (D_1|C_1) = \sum_{1\leq j\leq 2r} c_j d_j I_{\mathcal{F}}. \tag{159}$$

The CAR's can then be expressed in terms of this scalar product as

$$\{C_1, D_1\} = \sum_{1\leq j,k\leq 2r} c_j d_k \{\tilde{x}_j, \tilde{x}_k\} I_{\mathcal{F}} = \sum_{1\leq j\leq 2r} c_j d_j I_{\mathcal{F}} = 2(C|D). \tag{160}$$

The invariance group of the scalar product $(\cdot|\cdot)$ is $O\left(2r, \mathbb{C}\right)$.

One can define a quadratic function Q in terms of this scalar product by

$$Q\left(C_1\right) = \left(C_1 \left| C_1\right.\right) \tag{161}$$

and then one can recover the scalar product by using the polarization identity

$$\left(C_1 \left| D_1\right.\right) = Q\left(C_1 + D_1\right) - Q\left(C_1\right) - Q\left(D_1\right). \tag{162}$$

B. Contraction

Left and right contraction are defined by

$$v\rfloor D = \frac{1}{2}\left\{vD - \hat{D}v\right\}$$
$$v\lfloor D = \frac{1}{2}\left\{Dv - v\hat{D}\right\},$$

where the *grade involution operation*

$$\widehat{\ }: D \rightarrow \hat{D},$$

is defined by

$$D = \sum_{0\leq n\leq 2r} D_n$$

$$\hat{D}_n = (-1)^n D_n.$$

The scaler product is a special case of contraction i.e.

$$\left(C_1 \left| D_1\right.\right) = \frac{1}{2}\left\{C_1 D_1 - (-1)^1 D_1 C_1\right\}.$$

Contraction can be generalized to act between any elements of the GA by

$$A\rfloor B = \sum_{0\leq p,q\leq 2r} \left\langle A_p B_q\right\rangle_{p-q} \tag{163}$$

$$A\lfloor B = \sum_{0\leq p,q\leq 2r} \left\langle A_p B_q\right\rangle_{q-p}. \tag{164}$$

C. Wedge Product

The wedge product between vectors is defined as the antisymmetric part of the operator product i.e.

$$C_1 \wedge D_1 = \frac{1}{2}\left[C_1, D_1\right]$$

and *might NOT* correspond to the wedge product of operators of degree 1 (vectors in the Geometric Algebra) induced by the underlying exterior algebra, $\mathcal{H}_{\mathcal{F}}$,of $\mathcal{F}$. In order to investigate this we consider the first quantized expansions of the creation and annihilation operators

$$a_1^{\dagger} = \left|\varphi_1\right\rangle\left\langle\phi\right| + \sum_{2\leq n\leq r}\ \sum_{2\leq j_1<\cdots<j_{n-1}<r} \left|\varphi_1\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right\rangle\left\langle\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right|, \tag{165}$$

$$a_1 = \left|\phi\right\rangle\left\langle\varphi_1\right| + \sum_{2\leq n\leq r}\ \sum_{2\leq j_1<\cdots<j_{n-1}<r} \left|\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right\rangle\left\langle\varphi_1\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right|. \tag{166}$$

Then the GA wedge product between a creator and an annihilator gives

$$a_1^{\dagger}\wedge a_1 = \frac{1}{2}\left[a_1^{\dagger}, a_1\right] = \frac{1}{2}\left(a_1^{\dagger}a_1 - a_1a_1^{\dagger}\right) = \frac{1}{2}\left(a_1^{\dagger}a_1 - I_{\mathcal{F}} + a_1^{\dagger}a_1\right) = a_1^{\dagger}a_1 - \frac{1}{2}I_{\mathcal{F}} \tag{167}$$

$\lceil$ which is antisymmetric

$$a_1\wedge a_1^{\dagger} = \frac{1}{2}\left[a_1, a_1^{\dagger}\right] = \frac{1}{2}\left(a_1a_1^{\dagger} - a_1^{\dagger}a_1\right) = \frac{1}{2}\left(I_{\mathcal{F}} - a_1^{\dagger}a_1 - a_1^{\dagger}a_1\right) = \frac{1}{2}I_{\mathcal{F}} - a_1^{\dagger}a_1 \tag{168}$$

$\rfloor$

while the operator wedge product gives

$$\left(\left|\varphi_1\right\rangle\left\langle\phi\right| + \sum_{2\leq n\leq r}\ \sum_{2\leq j_1<\cdots<j_{n-1}<r} \left|\varphi_1\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right\rangle\left\langle\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right|\right)\wedge\left(\left|\phi\right\rangle\left\langle\varphi_1\right| + \sum_{2\leq n\leq r}\ \sum_{2\leq j_1<\cdots<j_{n-1}<r} \left|\varphi_{j_1}\cdots\varphi\right. \tag{169}$$

$$= \left|\varphi_1\right\rangle\left\langle\varphi_1\right| + \sum_{2\leq n\leq r}\ \sum_{2\leq j_1<\cdots<j_{n-1}<r} \left|\varphi_1\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right\rangle\left\langle\varphi_1\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right| + \sum_{2\leq n\leq r}\ \sum_{2\leq j_1<\cdots<j_{n-1}<r} \left|\varphi_1\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right\rangle\left\langle\varphi\right. \tag{170}$$

$$+ \sum_{2\leq n\leq r}\ \sum_{2\leq j_1<\cdots<j_{n-1}<r} \left|\varphi_1\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right\rangle\left\langle\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right| \wedge \sum_{2\leq n\leq r}\ \sum_{2\leq j_1<\cdots<j_{n-1}<r} \left|\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right\rangle\left\langle\varphi_1\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right| \tag{171}$$

The GA wedge product between two creators gives

$$a_1^\dagger \wedge a_2^\dagger = a_1^\dagger a_2^\dagger = \left|\varphi_1\varphi_2\right\rangle\left\langle\phi\right| + \sum_{3\le n\le r}\sum_{3\le j_1<\cdots<j_{n-2}<r}\left|\varphi_1\varphi_2\varphi_{j_1}\cdots\varphi_{j_{n-2}}\right\rangle\left\langle\varphi_{j_1}\cdots\varphi_{j_{n-2}}\right|, \tag{172}$$

and we need to see if $a_1^\dagger \wedge a_2^\dagger = a_1^\dagger a_2^\dagger$

$$a_1^\dagger \wedge a_2^\dagger = \left(\left|\varphi_1\right\rangle\left\langle\phi\right| + \sum_{2\le n\le r}\sum_{2\le j_1<\cdots<j_{n-2}<r}\left|\varphi_1\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right\rangle\left\langle\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right|\right) \wedge \left(\left|\varphi_2\right\rangle\left\langle\phi\right| + \sum_{2\le n\le r}\sum_{1\le j_1<\cdots<j_{n-2}\ne} \right. \tag{173}$$

$$= \left|\varphi_1\right\rangle\left\langle\phi\right| \wedge \left|\varphi_2\right\rangle\left\langle\phi\right| + \left|\varphi_1\right\rangle\left\langle\phi\right| \wedge \left(\sum_{2\le n\le r}\sum_{1\le j_1<\cdots<j_{n-2}\ne 2<r}\left|\varphi_2\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right\rangle\left\langle\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right|\right) \tag{174}$$

$$+ \left(\sum_{2\le n\le r}\sum_{2\le j_1<\cdots<j_{n-2}<r}\left|\varphi_1\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right\rangle\left\langle\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right|\right) \wedge \left|\varphi_2\right\rangle\left\langle\phi\right| \tag{175}$$

$$+ \left(\sum_{2\le n\le r}\sum_{2\le j_1<\cdots<j_{n-2}<r}\left|\varphi_1\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right\rangle\left\langle\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right|\right) \wedge \left(\sum_{2\le n\le r}\sum_{1\le j_1<\cdots<j_{n-2}\ne 2<r}\left|\varphi_2\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right\rangle \right. \tag{176}$$

while

$$a_1^\dagger a_2^\dagger = \left(\left|\varphi_1\right\rangle\left\langle\phi\right| + \sum_{2\le n\le r}\sum_{2\le j_1<\cdots<j_{n-2}<r}\left|\varphi_1\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right\rangle\left\langle\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right|\right)\left(\left|\varphi_2\right\rangle\left\langle\phi\right| + \sum_{2\le n\le r}\sum_{1\le j_1<\cdots<j_{n-2}\ne 2<r} \right. \tag{177}$$

$$= \left(\sum_{2\le n\le r}\sum_{2\le j_1<\cdots<j_{n-2}<r}\left|\varphi_1\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right\rangle\left\langle\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right|\right)\left|\varphi_2\right\rangle\left\langle\phi\right| \tag{178}$$

$$+ \left(\sum_{2\le n\le r}\sum_{2\le j_1<\cdots<j_{n-2}<r}\left|\varphi_1\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right\rangle\left\langle\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right|\right)\left(\sum_{2\le n\le r}\sum_{1\le j_1<\cdots<j_{n-2}\ne 2<r}\left|\varphi_2\varphi_{j_1}\cdots\varphi_{j_{n-1}}\right\rangle\left\langle\varphi_{j_1}\cdots \right. \right. \tag{179}$$

$$a_j^\dagger = \frac{1}{\sqrt{2}}\left(x_j - ix_{j+r}\right) \tag{180}$$

$$a_j = \frac{1}{\sqrt{2}}\left(x_j + ix_{j+r}\right). \tag{181}$$

The wedge product can be generalized to give

$$v \wedge D = \frac{1}{2}\left\{vD + \hat{D}v\right\} \tag{182}$$

so that

$$vD = v\rfloor D + v \wedge D. \tag{183}$$

The wedge product can be further generalized by

$$A \wedge D = \sum_{0 \le p,q \le 2r} \left\langle A_p B_q \right\rangle_{q+p}. \tag{184}$$

D. Products of Basis Vectors

The product of vectors C_1, D_1 can be expressed as

$$C_1 D_1 = (C_1 | D_1) + C_1 \wedge D_1 \tag{185}$$

and in particular the product of two basis vectors as

$$x_i x_j = (x_i | x_j) + x_i \wedge x_j. \tag{186}$$

The third order product decomposes as

$$x_{j_1} x_{j_2} x_{j_3} = x_{j_1} \wedge x_{j_2} \wedge x_{j_3} + (x_{j_2} | x_{j_3}) x_{j_1} - (x_{j_1} | x_{j_3}) x_{j_2} + (x_{j_1} | x_{j_2}) x_{j_3}. \tag{187}$$

The fourth order decompose as

$$\begin{aligned}
x_{j_1} x_{j_2} x_{j_3} x_{j_4} &= x_{j_1} \wedge x_{j_2} \wedge x_{j_3} \wedge x_{j_4} \\
&+ (x_{j_1} | x_{j_2}) x_{j_3} \wedge x_{j_4} - (x_{j_1} | x_{j_3}) x_{j_2} \wedge x_{j_4} + (x_{j_1} | x_{j_4}) x_{j_2} \wedge x_{j_3} \\
&+ (x_{j_2} | x_{j_3}) x_{j_1} \wedge x_{j_4} - (x_{j_2} | x_{j_4}) x_{j_1} \wedge x_{j_3} + (x_{j_3} | x_{j_4}) x_{j_1} \wedge x_{j_2} \\
&+ (x_{j_1} | x_{j_2}) (x_{j_3} | x_{j_4}) - (x_{j_1} | x_{j_3}) (x_{j_2} | x_{j_4}) + (x_{j_1} | x_{j_4}) (x_{j_2} | x_{j_3}).
\end{aligned} \tag{188}$$

⌈To check we derive this again by

$$\begin{aligned}
x_i x_j x_k x_l &= \{(x_i | x_j) + x_i \wedge x_j\} \{(x_k | x_l) + x_k \wedge x_l\} \\
&= (x_i | x_j)(x_k | x_l) + (x_i | x_j) x_k \wedge x_l + (x_k | x_l) x_i \wedge x_j + (x_i \wedge x_j)(x_k \wedge x_l) \\
&= (x_i | x_j)(x_k | x_l) + (x_i | x_j) x_k \wedge x_l + (x_k | x_l) x_i \wedge x_j \\
&- (x_i | x_k) x_j \wedge x_l + (x_i | x_l) x_j \wedge x_k + (x_j | x_k) x_i \wedge x_l - (x_j | x_l) x_i \wedge x_k \\
&+ x_i \wedge x_j \wedge x_k \wedge x_l
\end{aligned} \tag{189}$$

$\rfloor$

The fifth order by

$$\begin{aligned} x_i x_j x_k x_l x_m = \{ & x_i \wedge x_j \wedge x_k \wedge x_l + (x_i | x_j)\, x_k \wedge x_l - (x_i | x_k)\, x_j \wedge x_l + (x_i | x_l)\, x_j \wedge x_k \\ & + (x_j | x_k)\, x_i \wedge x_l - (x_j | x_l)\, x_i \wedge x_k + (x_k | x_l)\, x_i \wedge x_j \\ & + (x_i | x_j)(x_k | x_l) - (x_i | x_k)(x_j | x_l) + (x_i | x_l)(x_j | x_k) \}\, x_m . \end{aligned} \tag{190}$$

$\lceil$

$$\begin{aligned} \{x_i \wedge x_j \wedge x_k \wedge x_l\}\, x_m &= x_i \wedge x_j \wedge x_k \wedge x_l \wedge x_m + x_i \wedge x_j \wedge x_k\, (x_l | x_m) - x_i \wedge x_j \wedge x_l\, (x_k | x_m) \\ &\quad + x_i \wedge x_k \wedge x_l\, (x_j | x_m) - x_j \wedge x_k \wedge x_l\, (x_i | x_m) \\ \{(x_i | x_j)\, x_k \wedge x_l\}\, x_m &= (x_i | x_j) \{x_k \wedge x_l \wedge x_m + (x_l | x_m)\, x_k - (x_k | x_m)\, x_l\} \\ &= (x_i | x_j)\, x_k \wedge x_l \wedge x_m + (x_i | x_j)(x_l | x_m)\, x_k - (x_i | x_j)(x_k | x_m)\, x_l \\ (x_i | x_j)(x_k | x_l)\, x_m &= (x_i | x_j)(x_k | x_l)\, x_m \end{aligned} \tag{191}$$

$\rfloor$

$$\begin{aligned} & x_i x_j x_k x_l x_m = x_i \wedge x_j \wedge x_k \wedge x_l \wedge x_m \\ & + (x_i | x_j)\, x_k \wedge x_l \wedge x_m - (x_i | x_k)\, x_j \wedge x_l \wedge x_m + (x_i | x_l)\, x_j \wedge x_k \wedge x_m \\ & + (x_j | x_k)\, x_i \wedge x_l \wedge x_m - (x_j | x_l)\, x_i \wedge x_k \wedge x_m + (x_k | x_l)\, x_i \wedge x_j \wedge x_m \\ & + (x_l | x_m)\, x_i \wedge x_j \wedge x_k - (x_k | x_m)\, x_i \wedge x_j \wedge x_l + (x_j | x_m)\, x_i \wedge x_k \wedge x_l - (x_i | x_m)\, x_j \wedge x_k \wedge x_l \\ & + (x_i | x_j)(x_l | x_m)\, x_k - (x_i | x_j)(x_k | x_m)\, x_l \\ & - (x_i | x_k)(x_l | x_m)\, x_j + (x_i | x_k)(x_j | x_m)\, x_l \\ & - (x_i | x_l)(x_j | x_m)\, x_k + (x_i | x_l)(x_k | x_m)\, x_j \\ & + (x_j | x_k)(x_l | x_m)\, x_i - (x_j | x_k)(x_i | x_m)\, x_l \\ & - (x_j | x_l)(x_k | x_m)\, x_i + (x_j | x_l)(x_k | x_m)\, x_k \\ & + (x_k | x_l)(x_j | x_m)\, x_i - (x_k | x_l)(x_i | x_m)\, x_j \\ & + \{(x_i | x_j)(x_k | x_l) - (x_i | x_k)(x_j | x_l) + (x_i | x_l)(x_j | x_k)\}\, x_m . \end{aligned} \tag{192}$$

$$
\begin{aligned}
&x_i x_j x_k x_l x_m = x_i \wedge x_j \wedge x_k \wedge x_l \wedge x_m \\
&+ (x_i| x_j)\, x_k \wedge x_l \wedge x_m - (x_i| x_k)\, x_j \wedge x_l \wedge x_m + (x_i| x_l)\, x_j \wedge x_k \wedge x_m \\
&+ (x_j| x_k)\, x_i \wedge x_l \wedge x_m - (x_j| x_l)\, x_i \wedge x_k \wedge x_m + (x_k| x_l)\, x_i \wedge x_j \wedge x_m \\
&+ (x_l| x_m)\, x_i \wedge x_j \wedge x_k - (x_k| x_m)\, x_i \wedge x_j \wedge x_l + (x_j| x_m)\, x_i \wedge x_k \wedge x_l - (x_i| x_m)\, x_j \wedge x_k \wedge x_l \\
&+ \{(x_i| x_j)\,(x_l| x_m) - (x_i| x_l)\,(x_j| x_m) + (x_j| x_l)\,(x_k| x_m)\}\, x_k \\
&+ \{(x_i| x_k)\,(x_j| x_m) - (x_j| x_k)\,(x_i| x_m) - (x_i| x_j)\,(x_k| x_m)\}\, x_l \\
&+ \{(x_i| x_l)\,(x_k| x_m) - (x_k| x_l)\,(x_i| x_m) - (x_i| x_k)\,(x_l| x_m)\}\, x_j \\
&+ \{(x_j| x_k)\,(x_l| x_m) - (x_j| x_l)\,(x_k| x_m) + (x_k| x_l)\,(x_j| x_m)\}\, x_i \\
&+ \{(x_i| x_j)\,(x_k| x_l) - (x_i| x_k)\,(x_j| x_l) + (x_i| x_l)\,(x_j| x_k)\}\, x_m. \qquad (193)
\end{aligned}
$$

Thus

$$
\begin{aligned}
&x_i x_j x_k x_l x_m = x_i \wedge x_j \wedge x_k \wedge x_l \wedge x_m \\
&+ \Delta_{ij} x_k \wedge x_l \wedge x_m - \Delta_{ik} x_j \wedge x_l \wedge x_m + \Delta_{il} x_j \wedge x_k \wedge x_m + \Delta_{jk} x_i \wedge x_l \wedge x_m - \Delta_{jl} x_i \wedge x_k \wedge x_m \\
&+ \Delta_{kl} x_i \wedge x_j \wedge x_m + \Delta_{lm} x_i \wedge x_j \wedge x_k - \Delta_{km} x_i \wedge x_j \wedge x_l + \Delta_{jm} x_i \wedge x_k \wedge x_l - \Delta_{im} x_j \wedge x_k \wedge x_l \\
&+ \{\Delta_{ij}\Delta_{lm} - \Delta_{il}\Delta_{jm} + \Delta_{jl}\Delta_{km}\}\, x_k \\
&+ \{\Delta_{ik}\Delta_{jm} - \Delta_{jk}\Delta_{im} - \Delta_{ij}\Delta_{km}\}\, x_l \\
&+ \{\Delta_{il}\Delta_{km} - \Delta_{kl}\Delta_{im} - \Delta_{ik}\Delta_{lm}\}\, x_j \\
&+ \{\Delta_{lm}\Delta_{jk} - \Delta_{jl}\Delta_{km} + \Delta_{kl}\Delta_{jm}\}\, x_i \\
&+ \{\Delta_{ij}\Delta_{kl} - \Delta_{ik}\Delta_{jl} + \Delta_{il}\Delta_{jk}\}\, x_m. \qquad (194)
\end{aligned}
$$

Hence the fifth order products are

$$
\begin{aligned}
&x_{j_1} x_{j_2} x_{j_3} x_{j_4} x_{j_5} = x_{j_1} \wedge x_{j_2} \wedge x_{j_3} \wedge x_{j_4} \wedge x_{j_5} \\
&+ \Delta_{j_1 j_2} x_{j_3} \wedge x_{j_4} \wedge x_{j_5} - \Delta_{j_1 j_3} x_{j_2} \wedge x_{j_4} \wedge x_{j_5} + \Delta_{j_1 j_4} x_{j_2} \wedge x_{j_3} \wedge x_{j_5} \\
&+ \Delta_{j_2 j_3} x_{j_1} \wedge x_{j_4} \wedge x_{j_5} - \Delta_{j_2 j_4} x_{j_1} \wedge x_{j_3} \wedge x_{j_5} + \Delta_{j_3 j_4} x_{j_1} \wedge x_{j_2} \wedge x_{j_5} \\
&+ \Delta_{j_4 j_5} x_{j_1} \wedge x_{j_2} \wedge x_{j_3} - \Delta_{j_3 j_5} x_{j_1} \wedge x_{j_2} \wedge x_{j_4} + \Delta_{j_2 j_5} x_{j_1} \wedge x_{j_3} \wedge x_{j_4} - \Delta_{j_1 j_5} x_{j_2} \wedge x_{j_3} \wedge x_{j_4} \\
&+ \mathrm{Pf}\,(\boldsymbol{\Delta})_{j_2 j_3 j_4 j_5}\, x_{j_1} - \mathrm{Pf}\,(\boldsymbol{\Delta})_{j_1 j_3 j_4 j_5}\, x_j + \mathrm{Pf}\,(\boldsymbol{\Delta})_{j_1 j_2 j_4 j_5}\, x_{j_3} - \mathrm{Pf}\,(\boldsymbol{\Delta})_{j_1 j_2 j_3 j_5}\, x_{j_4} + \mathrm{Pf}\,(\boldsymbol{\Delta})_{j_1 j_2 j_3 j_4}\, x_{j_5}, \qquad (195)
\end{aligned}
$$

where the extended pfaffian (diagonal terms allowed) is given by

$$
\mathrm{Pf}\,(\boldsymbol{\Delta})_{j_1 j_2 j_3 j_4} = \Delta_{j_1 j_2}\Delta_{j_3 j_4} - \Delta_{j_1 j_3}\Delta_{j_2 j_4} + \Delta_{j_1 j_4}\Delta_{j_2 j_3}. \qquad (196)
$$

The sixth order products decompose as

$$
\begin{aligned}
x_i x_j x_k x_l x_m x_n = \{ & x_i \wedge x_j \wedge x_k \wedge x_l \wedge x_m \\
& + \Delta_{ij} x_k \wedge x_l \wedge x_m - \Delta_{ik} x_j \wedge x_l \wedge x_m + \Delta_{il} x_j \wedge x_k \wedge x_m + \Delta_{jk} x_i \wedge x_l \wedge x_m - \Delta_{jl} x_i \wedge x_k \wedge x_m \\
& + \Delta_{kl} x_i \wedge x_j \wedge x_m + \Delta_{lm} x_i \wedge x_j \wedge x_k - \Delta_{km} x_i \wedge x_j \wedge x_l + \Delta_{jm} x_i \wedge x_k \wedge x_l - \Delta_{im} x_j \wedge x_k \wedge x_l \\
& + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{jklm} x_i - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{iklm} x_j + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{ijlm} x_k - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{ijkm} x_l + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{ijkl} x_m \Big\} x_n
\end{aligned} \tag{197}
$$

$$
\begin{aligned}
x_i x_j x_k x_l x_m x_n = \{ & x_i \wedge x_j \wedge x_k \wedge x_l \wedge x_m \\
& + \Delta_{ij} x_k \wedge x_l \wedge x_m - \Delta_{ik} x_j \wedge x_l \wedge x_m + \Delta_{il} x_j \wedge x_k \wedge x_m + \Delta_{jk} x_i \wedge x_l \wedge x_m - \Delta_{jl} x_i \wedge x_k \wedge x_m \\
& + \Delta_{kl} x_i \wedge x_j \wedge x_m + \Delta_{lm} x_i \wedge x_j \wedge x_k - \Delta_{km} x_i \wedge x_j \wedge x_l + \Delta_{jm} x_i \wedge x_k \wedge x_l - \Delta_{im} x_j \wedge x_k \wedge x_l \\
& + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{jklm} x_i - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{iklm} x_j + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{ijlm} x_k - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{ijkm} x_l + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{ijkl} x_m \Big\} x_n
\end{aligned} \tag{198}
$$

leading to

$$
\begin{aligned}
&x_i x_j x_k x_l x_m x_n = x_i \wedge x_j \wedge x_k \wedge x_l \wedge x_m \wedge x_n \\
&+ \Delta_{mn} x_i \wedge x_j \wedge x_k \wedge x_l - \Delta_{\ln} x_i \wedge x_j \wedge x_k \wedge x_m + \Delta_{kn} x_i \wedge x_j \wedge x_l \wedge x_m \\
&- \Delta_{jn} x_i \wedge x_k \wedge x_l \wedge x_m + \Delta_{in} x_j \wedge x_k \wedge x_l \wedge x_m \\
&+ \Delta_{ij} x_k \wedge x_l \wedge x_m \wedge x_n + \Delta_{ij}\Delta_{mn} x_k \wedge x_l - \Delta_{ij}\Delta_{\ln} x_k \wedge x_m + \Delta_{ij}\Delta_{kn} x_l \wedge x_m \\
&- \Delta_{ik} x_j \wedge x_l \wedge x_m \wedge x_n + \Delta_{ik}\Delta_{mn} x_j \wedge x_l - \Delta_{ik}\Delta_{\ln} x_j \wedge x_m + \Delta_{ik}\Delta_{jn} x_l \wedge x_m \\
&+ \Delta_{il} x_j \wedge x_k \wedge x_m \wedge x_n - \Delta_{il}\Delta_{mn} x_j \wedge x_k + \Delta_{il}\Delta_{kn} x_j \wedge x_m - \Delta_{il}\Delta_{jn} x_k \wedge x_m \\
&+ \Delta_{jk} x_i \wedge x_l \wedge x_m \wedge x_n - \Delta_{jk}\Delta_{mn} x_i \wedge x_l + \Delta_{jk}\Delta_{\ln} x_i \wedge x_m - \Delta_{jk}\Delta_{in} x_l \wedge x_m \\
&- \Delta_{jl} x_i \wedge x_k \wedge x_m \wedge x_n + \Delta_{jl}\Delta_{mn} x_i \wedge x_k - \Delta_{jl}\Delta_{kn} x_i \wedge x_m + \Delta_{jl}\Delta_{in} x_k \wedge x_m \\
&+ \Delta_{kl} x_i \wedge x_j \wedge x_m \wedge x_n - \Delta_{kl}\Delta_{mn} x_i \wedge x_j + \Delta_{kl}\Delta_{jn} x_i \wedge x_m - \Delta_{kl}\Delta_{in} x_j \wedge x_m \\
&- \Delta_{im} x_j \wedge x_k \wedge x_l \wedge x_n - \Delta_{im}\Delta_{\ln} x_j \wedge x_k + \Delta_{im}\Delta_{kn} x_j \wedge x_l - \Delta_{im}\Delta_{jn} x_k \wedge x_l \\
&+ \Delta_{lm} x_i \wedge x_j \wedge x_k \wedge x_n - \Delta_{lm}\Delta_{kn} x_i \wedge x_j + \Delta_{lm}\Delta_{jn} x_i \wedge x_k - \Delta_{lm}\Delta_{in} x_j \wedge x_k \\
&- \Delta_{km} x_i \wedge x_j \wedge x_l \wedge x_n + \Delta_{km}\Delta_{\ln} x_i \wedge x_j - \Delta_{km}\Delta_{jn} x_i \wedge x_l + \Delta_{km}\Delta_{in} x_j \wedge x_l \\
&+ \Delta_{jm} x_i \wedge x_k \wedge x_l \wedge x_n - \Delta_{jm}\Delta_{\ln} x_i \wedge x_k + \Delta_{jm}\Delta_{kn} x_i \wedge x_l - \Delta_{jm}\Delta_{in} x_k \wedge x_l \\
&+ \Delta_{lm}\Delta_{kn} x_i \wedge x_j - \Delta_{lm}\Delta_{jn} x_i \wedge x_k + \Delta_{lm}\Delta_{in} x_j \wedge x_k \\
&- \Delta_{km}\Delta_{\ln} x_i \wedge x_j + \Delta_{km}\Delta_{jn} x_i \wedge x_l - \Delta_{km}\Delta_{in} x_j \wedge x_l \\
&+ \Delta_{jm}\Delta_{\ln} x_i \wedge x_k - \Delta_{jm}\Delta_{kn} x_i \wedge x_l + \Delta_{jm}\Delta_{in} x_k \wedge x_l \\
&- \Delta_{im}\Delta_{\ln} x_j \wedge x_k + \Delta_{im}\Delta_{kn} x_j \wedge x_l - \Delta_{im}\Delta_{jn} x_k \wedge x_l \\
&+ \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{jklm} x_i \wedge x_n + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{jklm} \Delta_{in} - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{iklm} x_j \wedge x_n \\
&- \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{iklm} \Delta_{jn} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{ijlm} x_k \wedge x_n + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{ijlm} \Delta_{kn} \\
&- \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{ijkm} x_l \wedge x_n - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{ijkm} \Delta_{\ln} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{ijkl} x_m \wedge x_n + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{ijkl} \Delta_{mn} \qquad (199)
\end{aligned}
$$

$$
\begin{aligned}
&x_i x_j x_k x_l x_m x_n = x_i \wedge x_j \wedge x_k \wedge x_l \wedge x_m \wedge x_n \\
&+ \Delta_{ij} x_k \wedge x_l \wedge x_m \wedge x_n - \Delta_{ik} x_j \wedge x_l \wedge x_m \wedge x_n + \Delta_{il} x_j \wedge x_k \wedge x_m \wedge x_n - \Delta_{im} x_j \wedge x_k \wedge x_l \wedge x_n \\
&+ \Delta_{in} x_j \wedge x_k \wedge x_l \wedge x_m \\
&+ \Delta_{jk} x_i \wedge x_l \wedge x_m \wedge x_n - \Delta_{jl} x_i \wedge x_k \wedge x_m \wedge x_n + \Delta_{jm} x_i \wedge x_k \wedge x_l \wedge x_n - \Delta_{jn} x_i \wedge x_k \wedge x_l \wedge x_m \\
&+ \Delta_{kl} x_i \wedge x_j \wedge x_m \wedge x_n - \Delta_{km} x_i \wedge x_j \wedge x_l \wedge x_n + \Delta_{kn} x_i \wedge x_j \wedge x_l \wedge x_m \\
&+ \Delta_{lm} x_i \wedge x_j \wedge x_k \wedge x_n - \Delta_{\ln} x_i \wedge x_j \wedge x_k \wedge x_m + \Delta_{mn} x_i \wedge x_j \wedge x_k \wedge x_l \\
&+ \left\{\Delta_{lm}\Delta_{kn} - \Delta_{lm}\Delta_{jn} - \Delta_{km}\Delta_{\ln}\right\} x_i \wedge x_j + \left\{\Delta_{jm}\Delta_{\ln} - \Delta_{lm}\Delta_{jn} + \Delta_{jl}\Delta_{mn}\right\} x_i \wedge x_k \\
&+ \left\{\Delta_{km}\Delta_{jn} - \Delta_{jm}\Delta_{kn} - \Delta_{jk}\Delta_{mn}\right\} x_i \wedge x_l + \left\{\Delta_{jk}\Delta_{\ln} - \Delta_{jl}\Delta_{kn} + \Delta_{kl}\Delta_{jn}\right\} x_i \wedge x_m \\
&- \left\{\Delta_{im}\Delta_{\ln} + \Delta_{lm}\Delta_{in} - \Delta_{il}\Delta_{mn}\right\} x_j \wedge x_k - \left\{\Delta_{km}\Delta_{in} + \Delta_{im}\Delta_{kn} + \Delta_{ik}\Delta_{mn}\right\} x_j \wedge x_l \\
&- \left\{\Delta_{ik}\Delta_{\ln} + \Delta_{il}\Delta_{kn} - \Delta_{kl}\Delta_{in}\right\} x_j \wedge x_m + \left\{\Delta_{jm}\Delta_{in} - \Delta_{im}\Delta_{jn} + \Delta_{ij}\Delta_{mn}\right\} x_k \wedge x_l \\
&- \left\{\Delta_{ij}\Delta_{\ln} + \Delta_{jl}\Delta_{in} - \Delta_{il}\Delta_{jn}\right\} x_k \wedge x_m + \left\{\Delta_{ij}\Delta_{kn} - \Delta_{jk}\Delta_{in} + \Delta_{ik}\Delta_{jn}\right\} x_l \wedge x_m \\
&+ \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{jklm} x_i \wedge x_n - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{iklm} x_j \wedge x_n - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{ijkm} x_l \wedge x_n \\
&+ \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{ijlm} x_k \wedge x_n + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{ijkl} x_m \wedge x_n + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{jklm} \Delta_{in} - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{ijkm} \Delta_{\ln} \\
&- \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{iklm} \Delta_{jn} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{ijkl} \Delta_{mn} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{ijlm} \Delta_{kn}
\end{aligned} \tag{200}
$$

and

$$
\begin{aligned}
&x_{j_1} x_j x_{j_3} x_{j_4} x_{j_5} x_{j_6} = x_{j_1} \wedge x_{j_2} \wedge x_{j_3} \wedge x_{j_4} \wedge x_{j_5} \wedge x_{j_6} \\
&+ \Delta_{j_1j_2} x_{j_3} \wedge x_{j_4} \wedge x_{j_5} \wedge x_{j_6} - \Delta_{j_1j_3} x_{j_2} \wedge x_{j_4} \wedge x_{j_5} \wedge x_{j_6} + \Delta_{j_1j_4} x_{j_2} \wedge x_{j_3} \wedge x_{j_5} \wedge x_{j_6} - \Delta_{j_1j_5} x_{j_2} \wedge x_{j_3} \wedge x_{j_4} \wedge \\
&+ \Delta_{j_1j_6} x_{j_2} \wedge x_{j_3} \wedge x_{j_4} \wedge x_{j_5} + \Delta_{j_2j_3} x_{j_1} \wedge x_{j_4} \wedge x_{j_5} \wedge x_{j_6} - \Delta_{j_2j_4} x_{j_1} \wedge x_{j_3} \wedge x_{j_5} \wedge x_{j_6} + \Delta_{j_2j_5} x_{j_1} \wedge x_{j_3} \wedge x_{j_4} \wedge \\
&- \Delta_{j_2j_6} x_{j_1} \wedge x_{j_3} \wedge x_{j_4} \wedge x_{j_5} + \Delta_{j_3j_4} x_{j_1} \wedge x_{j_2} \wedge x_{j_5} \wedge x_{j_6} - \Delta_{j_3j_5} x_{j_1} \wedge x_{j_2} \wedge x_{j_4} \wedge x_{j_6} + \Delta_{j_3j_6} x_{j_1} \wedge x_{j_2} \wedge x_{j_4} \wedge \\
&+ \Delta_{j_4j_5} x_{j_1} \wedge x_{j_2} \wedge x_{j_3} \wedge x_{j_6} - \Delta_{\ln} x_{j_1} \wedge x_{j_2} \wedge x_{j_3} \wedge x_{j_5} + \Delta_{j_5j_6} x_{j_1} \wedge x_{j_2} \wedge x_{j_3} \wedge x_{j_4} \\
&+ \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_4j_5j_3j_6} x_{j_1} \wedge x_{j_2} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_2j_5\ln} x_{j_1} \wedge x_{j_3} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_3j_5j_2j_6} x_{j_1} \wedge x_{j_4} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_2j_3\ln} x_{j_1} \wedge x_{j_5} - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1} \\
&- \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_3\min} x_{j_2} \wedge x_{j_4} - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_3\ln} x_{j_2} \wedge x_{j_5} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_2\min} x_{j_3} \wedge x_{j_4} - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2\ln} x_{j_3} \wedge x_{j_5} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_3} \\
&+ \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_2j_3j_4j_5} x_{j_1} \wedge x_{j_6} - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_3j_4j_5} x_{j_2} \wedge x_{j_6} - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_3j_5} x_{j_4} \wedge x_{j_6} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_4j_5} x_{j_3} \wedge x_{j_6} + \mathrm{Pf}\left(\boldsymbol{\Delta} \right. \\
&+ \left\{\mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_3j_4j_5j_6} - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_4j_3j_5j_6} - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_3j_2j_4j_5j_6} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_5j_3j_4j_6} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_4j_5j_3j_6}\right\} I_{\mathcal{F}}
\end{aligned} \tag{201}
$$

$$
\begin{aligned}
\mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_3j_4j_5j_6} &= \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_3j_4} \Delta_{j_5j_6} - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_3j_5} \Delta_{j_4j_6} \\
&+ \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_4j_5} \Delta_{j_3j_6} - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_3j_4j_5} \Delta_{j_2j_6} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_2j_3j_4j_5} \Delta_{j_1j_6}.
\end{aligned} \tag{202}
$$

2. The second degree products are given by

$$x_i x_j = x_i \wedge x_j + \Delta_{ij}$$

3. The third degree products are given by

$$x_{j_1} x_{j_2} x_{j_3} = x_{j_1} \wedge x_{j_2} \wedge x_{j_3} + \Delta_{j_2 j_3} x_{j_1} - \Delta_{j_1 j_3} x_{j_2} + \Delta_{j_1 j_2} x_{j_3} \tag{203}$$

4. The fourth degree products are given by

$$\begin{aligned}
&x_{j_1} x_{j_2} x_{j_3} x_{j_4} = x_{j_1} \wedge x_j \wedge x_{j_3} \wedge x_{j_4} \\
&+ \Delta_{j_1 j_2} x_{j_3} \wedge x_{j_4} - \Delta_{j_1 j_3} x_{j_2} \wedge x_{j_4} + \Delta_{j_1 j_4} x_{j_2} \wedge x_{j_3} + \Delta_{j_2 j_3} x_{j_1} \wedge x_{j_4} - \Delta_{j_2 j_4} x_{j_1} \wedge x_{j_3} + \Delta_{j_3 j_4} x_{j_1} \wedge x_{j_2} \\
&+ \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1 j_2 j_3 j_4} I_{\mathcal{F}}
\end{aligned} \tag{204}$$

5. The fifth degree products are given by

$$\begin{aligned}
&x_{j_1} x_{j_2} x_{j_3} x_{j_4} x_{j_5} = x_{j_1} \wedge x_{j_2} \wedge x_{j_3} \wedge x_{j_4} \wedge x_{j_5} \\
&+ \Delta_{j_1 j_2} x_{j_3} \wedge x_{j_4} \wedge x_{j_5} - \Delta_{j_1 j_3} x_{j_2} \wedge x_{j_4} \wedge x_{j_5} + \Delta_{j_1 j_4} x_{j_2} \wedge x_{j_3} \wedge x_{j_5} + \Delta_{j_2 j_3} x_{j_1} \wedge x_{j_4} \wedge x_{j_5} \\
&- \Delta_{j_2 j_4} x_{j_1} \wedge x_{j_3} \wedge x_{j_5} + \Delta_{j_3 j_4} x_{j_1} \wedge x_{j_2} \wedge x_{j_5} + \Delta_{j_4 j_5} x_{j_1} \wedge x_{j_2} \wedge x_{j_3} - \Delta_{j_3 j_5} x_{j_1} \wedge x_{j_2} \wedge x_{j_4} \\
&+ \Delta_{j_2 j_5} x_{j_1} \wedge x_{j_3} \wedge x_{j_4} - \Delta_{j_1 j_5} x_{j_2} \wedge x_{j_3} \wedge x_{j_4} \\
&+ \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_2 j_3 j_4 j_5} x_{j_1} - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1 j_3 j_4 j_5} x_j + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1 j_2 j_4 j_5} x_{j_3} - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1 j_2 j_3 j_5} x_{j_4} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1 j_2 j_3 j_4} x_{j_5}
\end{aligned} \tag{205}$$

$$\begin{aligned}
&= x_{j_1} \wedge x_{j_2} \wedge x_{j_3} \wedge x_{j_4} \wedge x_{j_5} \\
&+ \sum_{\sigma \in S_5/(S_2 \times S_3)} \pi\left(\sigma\right) \Delta_{j_{\sigma(1)} j \sigma(2)} x_{j_{\sigma(3)}} \wedge x_{j_{\sigma(4)}} \wedge x_{j_{\sigma(5)}} + \sum_{\sigma \in S_5/(S_4 \times S_1)} \pi\left(\sigma\right) \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_{\sigma(1)} j_{\sigma(2)} j_{\sigma(3)} j_{\sigma(4)}} x_{j_{\sigma(5)}}.
\end{aligned} \tag{206}$$

6. The sixth degree products are given by

$$
\begin{aligned}
& x_{j_1}x_{j_2}x_{j_3}x_{j_4}x_{j_5}x_{j_6} = x_{j_1}\wedge x_{j_2}\wedge x_{j_3}\wedge x_{j_4}\wedge x_{j_5}\wedge x_{j_6} \\
& + \Delta_{j_1j_2}x_{j_3}\wedge x_{j_4}\wedge x_{j_5}\wedge x_{j_6} - \Delta_{j_1j_3}x_{j_2}\wedge x_{j_4}\wedge x_{j_5}\wedge x_{j_6} + \Delta_{j_1j_4}x_{j_2}\wedge x_{j_3}\wedge x_{j_5}\wedge x_{j_6} \\
& - \Delta_{j_1j_5}x_{j_2}\wedge x_{j_3}\wedge x_{j_4}\wedge x_{j_6} + \Delta_{j_1j_6}x_{j_2}\wedge x_{j_3}\wedge x_{j_4}\wedge x_{j_5} + \Delta_{j_2j_3}x_{j_1}\wedge x_{j_4}\wedge x_{j_5}\wedge x_{j_6} \\
& - \Delta_{j_2j_4}x_{j_1}\wedge x_{j_3}\wedge x_{j_5}\wedge x_{j_6} + \Delta_{j_2j_5}x_{j_1}\wedge x_{j_3}\wedge x_{j_4}\wedge x_{j_6} - \Delta_{j_2j_6}x_{j_1}\wedge x_{j_3}\wedge x_{j_4}\wedge x_{j_5} \\
& + \Delta_{j_3j_4}x_{j_1}\wedge x_{j_2}\wedge x_{j_5}\wedge x_{j_6} - \Delta_{j_3j_5}x_{j_1}\wedge x_{j_2}\wedge x_{j_4}\wedge x_{j_6} + \Delta_{j_3j_6}x_{j_1}\wedge x_{j_2}\wedge x_{j_4}\wedge x_{j_5} \\
& + \Delta_{j_4j_5}x_{j_1}\wedge x_{j_2}\wedge x_{j_3}\wedge x_{j_6} - \Delta_{j_4j_6}x_{j_1}\wedge x_{j_2}\wedge x_{j_3}\wedge x_{j_5} + \Delta_{j_5j_6}x_{j_1}\wedge x_{j_2}\wedge x_{j_3}\wedge x_{j_4} \\
& + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_4j_5j_3j_6}x_{j_1}\wedge x_{j_2} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_2j_5j_4j_6}x_{j_1}\wedge x_{j_3} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_3j_5j_2j_6}x_{j_1}\wedge x_{j_4} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_2j_3j_4j_6}x_{j_1}\wedge x_{j_5} \\
& - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_5j_4j_6}x_{j_2}\wedge x_{j_3} - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_3j_5j_1j_6}x_{j_2}\wedge x_{j_4} - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_3j_4j_6}x_{j_2}\wedge x_{j_5} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_2j_1j_6}x_{j_3}\wedge x_{j_4} \\
& - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_4j_6}x_{j_3}\wedge x_{j_5} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_3j_6}x_{j_4}\wedge x_{j_5} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_3j_4}x_{j_5}\wedge x_{j_6} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_2j_3j_4j_5}x_{j_1}\wedge x_{j_6} \\
& - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_3j_4j_5}x_{j_2}\wedge x_{j_6} - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_3j_5}x_{j_4}\wedge x_{j_6} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_4j_5}x_{j_3}\wedge x_{j_6} \\
& + \left\{\mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_3j_4j_5j_6} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_3j_4j_5j_6} - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_4j_3j_5j_6} - \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_3j_2j_4j_5j_6} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_5j_3j_4j_6}\right\} I_{\mathcal{F}}
\end{aligned} \tag{207}
$$

$$
\begin{aligned}
& = x_{j_1}\wedge x_{j_2}\wedge x_{j_3}\wedge x_{j_4}\wedge x_{j_5}\wedge x_{j_6} \\
& + \sum_{\sigma\in S_6/(S_2\times S_4)} \pi\left(\sigma\right)\Delta_{j_{\sigma(1)}j_{\sigma(2)}}x_{j_{\sigma(3)}}\wedge x_{j_{\sigma(4)}}\wedge x_{j_{\sigma(5)}}\wedge x_{j_{\sigma(6)}} \\
& \qquad + \sum_{\sigma\in S_6/(S_4\times S_2)} \pi\left(\sigma\right)\mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_{\sigma(1)}j_{\sigma(2)}j_{\sigma(3)}j_{\sigma(4)}}x_{j_{\sigma(5)}}\wedge x_{j_{\sigma(6)}} + \mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_3j_4j_5j_6} I_{\mathcal{F}}.
\end{aligned} \tag{208}
$$

7. The seventh degree products are given by

$$
\begin{aligned}
& x_{j_1}x_{j}x_{j_3}x_{j_4}x_{j_5}x_{j_6}x_{j_7} = x_{j_1}\wedge x_{j_2}\wedge x_{j_3}\wedge x_{j_4}\wedge x_{j_5}\wedge x_{j_6}\wedge x_{j_7} \\
& + \sum_{\sigma\in S_7/(S_2\times S_5)} \pi\left(\sigma\right)\Delta_{j_{\sigma(1)}j\sigma(2)}x_{j_{\sigma_{(3)}}}\wedge x_{j_{\sigma_{(4)}}}\wedge x_{j_{\sigma(5)}}\wedge x_{j_{\sigma_{(6)}}}\wedge x_{j_{\sigma(7)}} \\
& + \sum_{\sigma\in S_7/(S_4\times S_3)} \pi\left(\sigma\right)\mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_{\sigma(1)}j_{\sigma_{(2)}}j_{\sigma_{(3)}}j_{\sigma_{(4)}}}x_{j_{\sigma_{(5)}}}\wedge x_{j_{\sigma_{(6)}}}\wedge x_{j_{\sigma(7)}} \\
& + \sum_{\sigma\in S_7/(S_6\times S_1)} \pi\left(\sigma\right)\mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_{\sigma(1)}j_{\sigma(2)}j_{\sigma(3)}j_{\sigma(4)}j_{\sigma(5)}j_{\sigma(6)}}x_{j_{\sigma(7)}}
\end{aligned} \tag{209}
$$

$$
\begin{aligned}
x_{j_1} x_j x_{j_3} x_{j_4} x_{j_5} x_{j_6} x_{j_7} &= x_{j_1} \wedge x_{j_2} \wedge x_{j_3} \wedge x_{j_4} \wedge x_{j_5} \wedge x_{j_6} \wedge x_{j_7} \\
&+ \sum_{\sigma \in S_7/(S_2 \times S_5)} \pi(\sigma)\, \Delta_{j_{\sigma(1)} j \sigma(2)} x_{j_{\sigma(3)}} \wedge x_{j_{\sigma(4)}} \wedge x_{j_{\sigma(5)}} \wedge x_{j_{\sigma(6)}} \wedge x_{j_{\sigma(7)}} \\
&+ \sum_{\sigma \in S_7/(S_4 \times S_3)} \pi(\sigma)\, \mathrm{Pf}(\mathbf{\Delta})_{j_{\sigma(1)} j_{\sigma(2)} j_{\sigma(3)} j_{\sigma(4)}} x_{j_{\sigma(5)}} \wedge x_{j_{\sigma(6)}} \wedge x_{j_{\sigma(7)}} \\
&+ \sum_{\sigma \in S_7/(S_6 \times S_1)} \pi(\sigma)\, \mathrm{Pf}(\mathbf{\Delta})_{j_{\sigma(1)} j_{\sigma(2)} j_{\sigma(3)} j_{\sigma(4)} j_{\sigma(5)} j_{\sigma(6)}} x_{j_{\sigma(7)}}
\end{aligned}
\tag{210}
$$

and the eigth degree products are given by

8.

$$
\begin{aligned}
x_{j_1} x_{j_2} x_{j_3} x_{j_4} x_{j_5} x_{j_6} x_{j_7} x_{j_8} &= x_{j_1} \wedge x_{j_2} \wedge x_{j_3} \wedge x_{j_4} \wedge x_{j_5} \wedge x_{j_6} \wedge x_{j_7} \wedge x_{j_8} \\
&+ \sum_{\sigma \in S_8/(S_2 \times S_6)} \pi(\sigma)\, \Delta_{j_{\sigma(1)} j_{\sigma(2)}} x_{j_{\sigma(3)}} \wedge x_{j_{\sigma(4)}} \wedge x_{j_{\sigma(5)}} \wedge x_{j_{\sigma(6)}} \wedge x_{j_{\sigma(7)}} \wedge x_{j_{\sigma(8)}} \\
&+ \sum_{\sigma \in S_8/(S_4 \times S_4)} \pi(\sigma)\, \mathrm{Pf}(\mathbf{\Delta})_{j_{\sigma(1)} j_{\sigma(2)} j_{\sigma(3)} j_{\sigma(4)}} x_{j_{\sigma(5)}} \wedge x_{j_{\sigma(6)}} \wedge x_{j_{\sigma(7)}} \wedge x_{j_{\sigma(8)}} \\
&+ \sum_{\sigma \in S_8/(S_6 \times S_2)} \pi(\sigma)\, \mathrm{Pf}(\mathbf{\Delta})_{j_{\sigma(1)} j_{\sigma(2)} j_{\sigma(3)} j_{\sigma(4)} j_{\sigma(5)} j_{\sigma(6)}} x_{j_{\sigma(7)}} \wedge x_{j_{\sigma(8)}} + \mathrm{Pf}(\mathbf{\Delta})_{j_1 j_2 j_3 j_4 j_5 j_6 j_7 j_8} I_{\mathcal{F}}.
\end{aligned}
\tag{211}
$$

E. Geometric Products of Operators

First consider the homogeneous second degree operator

$$
A_2 = \sum_{1 \leq j_1 < j_2 \leq 2r} A_{2 j_1 j_2} x_{j_1} x_{j_2} \tag{212}
$$

and its square

$$
(A_2)^2 = \sum_{\substack{1 \leq j_1 < j_2 \leq 2r \\ 1 \leq j_3 < j_4 \leq 2r}} A_{2 j_1 j_2} A_{2 j_3 j_4} x_{j_1} x_{j_2} x_{j_3} x_{j_4}. \tag{213}
$$

Utilizing the relationship

$$
\begin{aligned}
x_{j_1} x_{j_2} x_{j_3} x_{j_4} &= x_{j_1} \wedge x_{j_2} \wedge x_{j_3} \wedge x_{j_4} \\
&+ (x_{j_1} | x_{j_2})\, x_{j_3} \wedge x_{j_4} - (x_{j_1} | x_{j_3})\, x_{j_2} \wedge x_{j_4} + (x_{j_1} | x_{j_4})\, x_{j_2} \wedge x_{j_3} \\
&+ (x_{j_2} | x_{j_3})\, x_{j_1} \wedge x_{j_4} - (x_{j_2} | x_{j_4})\, x_{j_1} \wedge x_{j_3} + (x_{j_3} | x_{j_4})\, x_{j_1} \wedge x_{j_2} \\
&+ (x_{j_1} | x_{j_2}) (x_{j_3} | x_{j_4}) - (x_{j_1} | x_{j_3}) (x_{j_2} | x_{j_4}) + (x_{j_1} | x_{j_4}) (x_{j_2} | x_{j_3}).
\end{aligned}
\tag{214}
$$

one obtains

$$(A_2)^2 = \sum_{\substack{1\le j_1<j_2\le 2r \\ 1\le j_3<j_4\le 2r}} \{A_{2j_1j_2}A_{2j_3j_4}x_{j_1}\wedge x_{j_2}\wedge x_{j_3}\wedge x_{j_4}$$

$$- \Delta_{j_1j_3}A_{2j_1j_2}A_{2j_3j_4}x_{j_2}\wedge x_{j_4} + \Delta_{j_1j_4}A_{2j_1j_2}A_{2j_3j_4}x_{j_2}\wedge x_{j_3}$$

$$+ \Delta_{j_2j_3}A_{2j_1j_2}A_{2j_3j_4}x_{j_1}\wedge x_{j_4} - \Delta_{j_2j_4}A_{2j_1j_2}A_{2j_3j_4}x_{j_1}\wedge x_{j_3}$$

$$+ A_{2j_1j_2}\Delta_{j_1j_4}A_{2j_3j_4}\Delta_{j_2j_3}I_{\mathcal{F}} - A_{2j_1j_2}\Delta_{j_1j_3}A_{2j_3j_4}\Delta_{j_2j_4}I_{\mathcal{F}}\}\,. \tag{215}$$

The "scaler" term is easily seen to be

$$-\sum_{1\le j_1<j_2\le 2r} A_{2j_1j_2}\Delta_{j_1j_1}A_{2j_1j_2}\Delta_{j_2j_2} = -\frac{1}{4}\sum_{1\le j_1<j_2\le 2r}(A_{2j_1j_2})^2\, I_{\mathcal{F}}, \tag{216}$$

while one can examine the two-vector term by considering $r=2$, leading to the terms

$$A_{212}A_{213}x_1x_2x_1x_3 + A_{212}A_{214}x_1x_2x_1x_4 + A_{212}A_{223}x_1x_2x_2x_3$$

$$+ A_{212}A_{224}x_1x_2x_2x_4 + A_{213}A_{212}x_1x_3x_1x_2 + A_{213}A_{214}x_1x_3x_1x_4$$

$$+ A_{213}A_{223}x_1x_3x_2x_3 + A_{213}A_{234}x_1x_3x_3x_4 + A_{214}A_{212}x_1x_4x_1x_2$$

$$+ A_{214}A_{213}x_1x_4x_1x_3 + A_{214}A_{214}x_1x_4x_1x_4 + A_{214}A_{224}x_1x_4x_2x_4$$

$$+ A_{214}A_{234}x_1x_4x_3x_4 + A_{223}A_{212}x_2x_3x_1x_2 + A_{223}A_{213}x_2x_3x_1x_3$$

$$+ A_{223}A_{224}x_2x_3x_2x_4 + A_{223}A_{234}x_2x_3x_3x_4 + A_{224}A_{212}x_2x_4x_1x_2$$

$$+ A_{224}A_{213}x_2x_4x_1x_3 + A_{224}A_{214}x_2x_4x_1x_4 + A_{224}A_{223}x_2x_4x_2x_3$$

$$+ A_{224}A_{224}x_2x_4x_2x_4 + A_{224}A_{234}x_2x_4x_3x_4 + A_{234}A_{213}x_3x_4x_1x_3$$

$$+ A_{234}A_{214}x_3x_4x_1x_4 + A_{234}A_{223}x_3x_4x_2x_3 + A_{234}A_{224}x_3x_4x_2x_4 \tag{217}$$

and see that the term involving e.g. $x_2\wedge x_3$ is given by

$$- \Delta_{11}A_{212}A_{213}x_2\wedge x_3 - \Delta_{11}A_{213}A_{212}x_3\wedge x_2 - \Delta_{44}A_{224}A_{234}x_2\wedge x_3$$

$$- \Delta_{44}A_{234}A_{224}x_3\wedge x_2 = 0, \tag{218}$$

and the term involving e.g. $x_1\wedge x_2$ is given by

$$- \Delta_{33}A_{213}A_{223}x_1\wedge x_2 - \Delta_{44}A_{214}A_{224}x_1\wedge x_2 - \Delta_{33}A_{223}A_{213}x_2\wedge x_1$$

$$- \Delta_{44}A_{224}A_{214}x_2\wedge x_1 = 0, \tag{219}$$

thus the entire two vector term must be zero. The four vector term in this case is

$$
\begin{aligned}
& A_{212}A_{234}x_1x_2x_3x_4 + A_{213}A_{224}x_1x_3x_2x_4 + A_{214}A_{223}x_1x_4x_2x_3 \\
& + A_{223}A_{214}x_2x_3x_1x_4 + A_{224}A_{213}x_2x_4x_1x_3 + A_{234}A_{212}x_3x_4x_1x_2 \\
& = 2\left(A_{212}A_{234} - A_{213}A_{224} + A_{214}A_{223}\right)x_1x_2x_3x_4 \\
& = 2\,\mathrm{Pf}\left\{\mathbb{A}_2\right\}_{1234} x_1x_2x_3x_4 = 2\,\mathrm{Pf}\left\{\mathbb{A}_2\right\}_{1234} x_1 \wedge x_2 \wedge x_3 \wedge x_4 = A_2 \wedge A_2.
\end{aligned} \tag{220}
$$

Hence

$$
(A_2)^2 = \frac{1}{8}\,\mathrm{Tr}\left\{\mathbb{A}_2^t\mathbb{A}_2\right\} I_{\mathcal{F}} + A_2 \wedge A_2. \tag{221}
$$

Thus the square of the operator $A_0 I_{\mathcal{F}} + A_2$ is thus given by

$$
(A_0 I_{\mathcal{F}} + A_2)^2 = A_0^2 I_{\mathcal{F}} + 2A_0A_2 + A_2^2 \tag{222}
$$

and the idempotency condition for such operators is

$$
\begin{aligned}
& A_0^2 I_{\mathcal{F}} + 2A_2 + \frac{1}{8}\,\mathrm{Tr}\left\{\mathbb{A}_2^t\mathbb{A}_2\right\} I_{\mathcal{F}} + A_2 \wedge A_2 - A_0 I_{\mathcal{F}} - A_2 \\
& = \left(A_0^2 + \frac{1}{8}\,\mathrm{Tr}\left\{\mathbb{A}_2^t\mathbb{A}_2\right\} - A_0\right) I_{\mathcal{F}} + A_2 \wedge A_2 + \left(2A_0 - 1\right)A_2 = 0.
\end{aligned} \tag{223}
$$

Hence all three of the following conditions must be satisfied

$$
\left(A_0^2 + \frac{1}{8}\,\mathrm{Tr}\left\{\mathbb{A}_2^t\mathbb{A}_2\right\} - A_0\right) = \left(A_0^2 - \frac{1}{4}\sum_{1\le j_1<j_2\le 2r}\left(A_{2j_1j_2}\right)^2 - A_0\right) = 0 \tag{224}
$$

$$
\left(2A_0 - 1\right)A_2 = 0 \tag{225}
$$

$$
A_2 \wedge A_2 = 0. \tag{226}
$$

This gives that

$$
A_0 = \frac{1}{2} \text{ and/or } A_2 = 0 \tag{227}
$$

$$
A_2 \text{ is a rank one two-vector or } A_2 = 0 \tag{228}
$$

which imply that

$$
\left(A_0^2 + \frac{1}{8}\,\mathrm{Tr}\left\{\mathbb{A}_2^t\mathbb{A}_2\right\} - A_0\right) = \frac{1}{4} + \frac{2}{8} - \frac{1}{2} = 0 \tag{229}
$$

or

$$
0 + 0 - 0 = 0. \tag{230}
$$

Thus

$$A = \frac{1}{2} I_{\mathcal{F}} + a \wedge b. \tag{231}$$

Next we consider the product of the homogeneous fourth degree operator with the homogeneous second degree operator

$$A_4 A_2 = \sum_{\substack{1 \le j_1 < j_2 < j_3 < j_4 \le 2r \\ 1 \le j_5 < j_6 \le 2r}} A_{4 j_1 j_2 j_3 j_4} A_{2 j_5 j_6} x_{j_1} x_{j_2} x_{j_3} x_{j_4} x_{j_5} x_{j_6} \tag{232}$$

Utilising the relationship

$$\begin{aligned} x_{j_1} x_{j_2} x_{j_3} x_{j_4} x_{j_5} x_{j_6} &= x_{j_1} \wedge x_{j_2} \wedge x_{j_3} \wedge x_{j_4} \wedge x_{j_5} \wedge x_{j_6} \\ &+ \sum_{\sigma \in S_6/(S_2 \times S_4)} \pi(\sigma) \, \Delta_{j_{\sigma(1)} j_{\sigma(2)}} x_{j_{\sigma(3)}} \wedge x_{j_{\sigma(4)}} \wedge x_{j_{\sigma(5)}} \wedge x_{j_{\sigma(6)}} \\ &+ \sum_{\sigma \in S_6/(S_4 \times S_2)} \pi(\sigma) \, \mathrm{Pf}\,(\boldsymbol{\Delta})_{j_{\sigma(1)} j_{\sigma(2)} j_{\sigma(3)} j_{\sigma(4)}} x_{j_{\sigma(5)}} \wedge x_{j_{\sigma(6)}} + \mathrm{Pf}\,(\boldsymbol{\Delta})_{j_1 j_2 j_3 j_4 j_5} \end{aligned} \tag{233}$$

one obtains

$$\begin{aligned} A_4 A_2 = \sum_{\substack{1 \le j_1 < j_2 < j_3 < j_4 \le 2r \\ 1 \le j_5 < j_6 \le 2r}} & A_{4 j_1 j_2 j_3 j_4} A_{2 j_5 j_6} \left\{ x_{j_1} \wedge x_{j_2} \wedge x_{j_3} \wedge x_{j_4} \wedge x_{j_5} \wedge x_{j_6} \right. \\ &+ \sum_{\sigma \in S_6/(S_4 \times S_2)} \pi(\sigma) \, \Delta_{j_{\sigma(1)} j_{\sigma(2)}} x_{j_{\sigma(3)}} \wedge x_{j_{\sigma(4)}} \wedge x_{j_{\sigma(5)}} \wedge x_{j_{\sigma(6)}} \\ &+ \sum_{\sigma \in S_6/(S_4 \times S_4)} \pi(\sigma) \, \mathrm{Pf}\,(\boldsymbol{\Delta})_{j_{\sigma(1)} j_{\sigma(2)} j_{\sigma(3)} j_{\sigma(4)}} x_{j_{\sigma(5)}} \wedge x_{j_{\sigma(6)}} \\ &\left. + \mathrm{Pf}\,(\boldsymbol{\Delta})_{j_1 j_2 j_3 j_4 j_5 j_6} I_{\mathcal{F}} \right\}. \end{aligned} \tag{234}$$

Note that

$$\mathrm{Pf}\,(\boldsymbol{\Delta})_{j_1 j_2 j_3 j_4} = \Delta_{j_1 j_2} \Delta_{j_3 j_4} - \Delta_{j_1 j_3} \Delta_{j_2 j_4} + \Delta_{j_1 j_4} \Delta_{j_2 j_3} \tag{235}$$

and

$$\begin{aligned} \mathrm{Pf}\,(\boldsymbol{\Delta})_{j_1 j_2 j_3 j_4 j_5 j_6} &= \mathrm{Pf}\,(\boldsymbol{\Delta})_{j_1 j_2 j_3 j_4} \Delta_{j_5 j_6} - \mathrm{Pf}\,(\boldsymbol{\Delta})_{j_1 j_2 j_3 j_5} \Delta_{j_4 j_6} \\ &+ \mathrm{Pf}\,(\boldsymbol{\Delta})_{j_1 j_2 j_4 j_5} \Delta_{j_3 j_6} - \mathrm{Pf}\,(\boldsymbol{\Delta})_{j_1 j_3 j_4 j_5} \Delta_{j_2 j_6} + \mathrm{Pf}\,(\boldsymbol{\Delta})_{j_2 j_3 j_4 j_5} \Delta_{j_1 j_6}. \end{aligned} \tag{236}$$

Let $r = 3$ then

$$\sum_{\substack{1\leq j_1<j_2<j_3<j_4\leq 6\\ 1\leq j_5<j_6\leq 6}} A_{4j_1j_2j_3j_4}A_{2j_5j_6}\left\{\sum_{\sigma\in S_6/(S_4\times S_2)}\pi(\sigma)\,\Delta_{j_{\sigma(1)}j_{\sigma(2)}}x_{j_{\sigma(3)}}\wedge x_{j_{\sigma(4)}}\wedge x_{j_{\sigma(5)}}\wedge x_{j_{\sigma(6)}}\right.$$
$$+\sum_{\sigma\in S_6/(S_4\times S_2)}\pi(\sigma)\,\Delta_{j_{\sigma(1)}j_{\sigma(2)}}x_{j_{\sigma(3)}}\wedge x_{j_{\sigma(4)}}\wedge x_{j_{\sigma(5)}}\wedge x_{j_{\sigma(6)}}$$
$$+\sum_{\sigma\in S_6/(S_4\times S_4)}\pi(\sigma)\,\mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_{\sigma(1)}j_{\sigma(2)}j_{\sigma(3)}j_{\sigma(4)}}x_{j_{\sigma(5)}}\wedge x_{j_{\sigma(6)}}$$
$$\left.+\,\mathrm{Pf}\left(\boldsymbol{\Delta}\right)_{j_1j_2j_3j_4j_5j_6}I_{\mathcal{F}}\right\}. \tag{237}$$

Two indices in common gives the second order wedge product term

$$\begin{gathered}
123412\,(34)\,,123512\,(35)\,,123612\,(36)\,,124512\,(45)\,,124612\,(46)\,,125612\,(56)\,,\\
123413\,(24)\,,123513\,(25)\,,123613\,(26)\,,134513\,(45)\,,134613\,(46)\,,135613\,(56)\,,\\
123414\,(23)\,,124514\,(25)\,,124614\,(26)\,,134514\,(35)\,,134614\,(36)\,,145614\,(56)\,,\\
123515\,(23)\,,124515\,(24)\,,125615\,(26)\,,134515\,(34)\,,135615\,(36)\,,145615\,(46)\,,\\
123616\,(23)\,,124616\,(24)\,,125616\,(26)\,,134616\,(34)\,,135616\,(35)\,,145616\,(45)\,,
\end{gathered}\tag{238}$$

$$\begin{gathered}
123423\,(14)\,,123523\,(15)\,,123623\,(16)\,,234523\,(43)\,,234623\,(46)\,,235623\,(56)\,,\\
123424\,(13)\,,124524\,(15)\,,124624\,(16)\,,234524\,(35)\,,234624\,(36)\,,245624\,(56)\,,\\
123525\,(13)\,,124525\,(14)\,,125625\,(16)\,,234525\,(34)\,,235625\,(36)\,,245625\,(46)\,,\\
123626\,(13)\,,124626\,(14)\,,125626\,(15)\,,234626\,(34)\,,235626\,(35)\,,245626\,(45)\,,
\end{gathered}$$

$$\begin{gathered}
123434\,(12)\,,134534\,(15)\,,134634\,(16)\,,234534\,(25)\,,234634\,(26)\,,345634\,(56)\,,\\
123535\,(12)\,,143535\,(14)\,,135635\,(16)\,,234535\,(24)\,,235635\,(26)\,,345635\,(46)\,,\\
123636\,(12)\,,134636\,(16)\,,135636\,(15)\,,234636\,(24)\,,235636\,(25)\,,345636\,(45)\,,
\end{gathered}$$

$$\begin{gathered}
124545\,(12)\,,145645\,(16)\,,134545\,(13)\,,234545\,(23)\,,345645\,(34)\,,245645\,(26)\,,\\
145646\,(15)\,,124646\,(12)\,,134646\,(13)\,,234646\,(23)\,,245646\,(25)\,,345646\,(35)\,,
\end{gathered}$$

$$125656\,(12)\,,135656\,(13)\,,145656\,(14)\,,235656\,(23)\,,245656\,(24)\,,345656\,(34)\,. \tag{239}$$

Hence the coefficient of $x_1 \wedge x_2$ is

$$-\frac{1}{4}\left(A_{1256}A_{56}+A_{1246}A_{46}+A_{1236}A_{36}+A_{1245}A_{45}+A_{1235}A_{35}+A_{1234}A_{34}\right) \tag{240}$$

One index in common gives the fourth order wedge product term

134512 (3452) , 134612 (3462) , 135612 (3562) , 145612 (4562) , 234512 (3451) , 234612 (3461) , 235612 (3561) ,
245612 (4561) ,

124513 (2453) , 124613 (2463) , 135613 (3561) , 145613 (4561) , 234513 (2451) , 234613 (2461) , 235613 (2561) ,
345613 (4561) ,

123514 (2354) , 123614 (2364) , 145614 (3561) , 135614 (3561) , 234514 (2351) , 234614 (2361) , 245614 (2561) ,
345614 (3561) ,

123415 (2345) , 123615 (2365) , 145615 (3461) , 134615 (3461) , 234515 (2341) $*$, 235615 (2361) , 245615 (2461) ,
345615 (3461) ,

123416 (2346) , 123516 (2356) , 145616 (3451) , 134516 (3451) , 234616 (2341) $*$, 235616 (2351) , 245616 (2451) ,
345616 (3451) ,

124523 (1453) , 125623 (1563) , 124623 (1463) , 245623 (4563) , 134523 (1453) , 134623 (1462) , 135623 (1562) ,
345623 (4562) ,

123524 (1354) , 123624 (1364) , 125624 (1564) , 235624 (3564) , 134524 (1453) , 134624 (1462) , 145624 (1562) ,
345624 (4562) ,

123425 (1345) , 123625 (1365) , 124625 (1465) , 234625 (3465) , 135425 (1342) , 135625 (1562) , 145625 (1462) ,
354625 (3462) ,

123426 (1346) , 123526 (1356) , 124526 (1456) , 234526 (3456) , 136426 (1342) , 136526 (1352) , 146526 (1452) ,
364526 (3452) ,

123534 (1254) , 123634 (1264) , 235634 (2564) , 135634 (1564) , 124534 (1253) , 124634 (1263) , 145634 (1563) ,
245634 (2563) ,

123435 (1245) , 123635 (1265) , 234635 (2465) , 134635 (1465) , 124535 (1243) $*$, 125635 (1263) , 145635 (1463) ,
245635 (2463) ,

123536 (1256) , 123436 (1246) , 234536 (2456) , 134536 (1436) , 124636 (1243) $*$, 125636 (1253) , 145636 (1453) ,
245636 (2463) ,

$123545\,(1234)*, 125645\,(1264)\,, 235645\,(2364)\,, 135645\,(1364)\,, 124645\,(1265)\,, 134645\,(1365)\,, 234645\,(2365)\,,$

$123445\,(1235)\,,$

$123446\,(1236)\,, 124546\,(1256)\,, 234546\,(2356)\,, 134546\,(1356)\,, 123646\,(1234)*, 125646\,(1254)\,, 135646\,(1354)\,,$

$235646\,(2354)\,,$

$123556\,(1236)\,, 124556\,(1246)\,, 234556\,(2346)\,, 134556\,(1346)\,, 124566\,(1245)\,, 134656\,(1345)\,, 124656\,(1245)\,,$

$$234656\,(2345)\,, \tag{241}$$

Hence the coefficient of $x_1 \wedge x_2 \wedge x_3 \wedge x_4$ is

$$-\frac{1}{2}\left(A_{1236}A_{46} + A_{1235}A_{45} - A_{1246}A_{36} - A_{1245}A_{35} - A_{2345}A_{15} - A_{2346}A_{16} + A_{1345}A_{25} + A_{1346}A_{26}\right) \tag{242}$$

No indices in common give sixth order wedge product terms

$$\begin{aligned} &345612, 245613, 235614, 234615, 234516, 145623, 135624, 134625, \\ &134526, 125634, 124635, 124536, 123645, 123546, 123456. \end{aligned} \tag{243}$$

Hence the coefficient of $x_1 \wedge x_2 \wedge x_3 \wedge x_4 \wedge x_5 \wedge x_6$ is

$$\begin{aligned} &A_{3456}A_{12} - A_{2456}A_{13} + A_{2356}A_{14} - A_{2346}A_{15} + A_{2345}A_{16} + A_{1456}A_{23} - A_{1356}A_{24} + A_{1346}A_{25} \\ &- A_{1345}A_{26} + A_{1256}A_{34} - A_{1246}A_{35} + A_{1245}A_{36} + A_{1236}A_{45} - A_{1235}A_{46} + A_{1234}A_{56}, \end{aligned} \tag{244}$$

and the sixth order term is

$$A_4 \wedge A_2 = \left(A_4 \wedge A_2\right)_{123456} x_1 \wedge x_2 \wedge x_3 \wedge x_4 \wedge x_5 \wedge x_6, \tag{245}$$

where $\left(A_4 \wedge A_2\right)_{123456}$ is given by the preceding expression.

Next we consider the homogeneous fourth degree operator

$$A_4 = \sum_{1 \le j_1 < j_2 < j_3 < j_4 \le 2r} A_{4 j_1 j_2 j_3 j_4} x_{j_1} x_{j_2} x_{j_3} x_{j_4} \tag{246}$$

and its square

$$\left(A_4\right)^2 = \sum_{\substack{1 \le j_1 < j_2 \le 2r \\ 1 \le j_3 < j_4 \le 2r}} A_{4 j_1 j_2 j_3 j_4} A_{4 j_5 j_6 j_7 j_8} x_{j_1} x_{j_2} x_{j_3} x_{j_4} x_{j_5} x_{j_6} x_{j_7} x_{j_8}. \tag{247}$$

Utilizing the relationship

$$\begin{aligned}
&x_{j_1}x_{j_2}x_{j_3}x_{j_4}x_{j_5}x_{j_6}x_{j_7}x_{j_8} = x_{j_1}\wedge x_{j_2}\wedge x_{j_3}\wedge x_{j_4}\wedge x_{j_5}\wedge x_{j_6}\wedge x_{j_7}\wedge x_{j_8}\\
&+\sum_{\sigma\in S_8/(S_2\times S_6)}\pi(\sigma)\,\mathrm{Pf}(\mathbf{\Delta})_{j_{\sigma(1)}j_{\sigma(2)}}\,x_{j_{\sigma(3)}}\wedge x_{j_{\sigma(4)}}\wedge x_{j_{\sigma(5)}}\wedge x_{j_{\sigma(6)}}\wedge x_{j_{\sigma(7)}}\wedge x_{j_{\sigma(8)}}\\
&+\sum_{\sigma\in S_8/(S_4\times S_4)}\pi(\sigma)\,\mathrm{Pf}(\mathbf{\Delta})_{j_{\sigma(1)}j_{\sigma(2)}j_{\sigma(3)}j_{\sigma(4)}}\,x_{j_{\sigma(5)}}\wedge x_{j_{\sigma(6)}}\wedge x_{j_{\sigma(7)}}\wedge x_{j_{\sigma(8)}}\\
&+\sum_{\sigma\in S_8/(S_6\times S_2)}\pi(\sigma)\,\mathrm{Pf}(\mathbf{\Delta})_{j_{\sigma(1)}j_{\sigma(2)}j_{\sigma(3)}j_{\sigma(4)}j_{\sigma(5)}j_{\sigma(6)}}\,x_{j_{\sigma(7)}}\wedge x_{j_{\sigma(8)}}+\mathrm{Pf}(\mathbf{\Delta})_{j_1j_2j_3j_4j_5j_6j_7j_8}\,I_{\mathcal{F}}
\end{aligned} \tag{248}$$

one obtains

$$(A_4)^2 = \sum_{\substack{1\le j_1<j_2<j_3<j_4\le 2r\\ 1\le j_5<j_6<j_7<j_8\le 2r}} A_{4j_1j_2j_3j_4}A_{4j_5j_6j_7j_8}\,\{x_{j_1}\wedge x_{j_2}\wedge x_{j_3}\wedge x_{j_4}\wedge x_{j_5}\wedge x_{j_6}\wedge x_{j_7}\wedge x_{j_8} \tag{249}$$

$$+\sum_{\sigma\in S_8/(S_2\times S_6)}\pi(\sigma)\,\Delta_{j_{\sigma(1)}j_{\sigma(2)}}x_{j_{\sigma(3)}}\wedge x_{j_{\sigma(4)}}\wedge x_{j_{\sigma(5)}}\wedge x_{j_{\sigma(6)}}\wedge x_{j_{\sigma(7)}}\wedge x_{j_{\sigma(8)}} \tag{250}$$

$$+\sum_{\sigma\in S_8/(S_4\times S_4)}\pi(\sigma)\,\mathrm{Pf}(\mathbf{\Delta})_{j_{\sigma(1)}j_{\sigma(2)}j_{\sigma(3)}j_{\sigma(4)}}\,x_{j_{\sigma(5)}}\wedge x_{j_{\sigma(6)}}\wedge x_{j_{\sigma(7)}}\wedge x_{j_{\sigma(8)}} \tag{251}$$

$$\left.+\sum_{\sigma\in S_8/(S_6\times S_2)}\pi(\sigma)\,\mathrm{Pf}(\mathbf{\Delta})_{j_{\sigma(1)}j_{\sigma(2)}j_{\sigma(3)}j_{\sigma(4)}j_{\sigma(5)}j_{\sigma(6)}}\,x_{j_{\sigma(7)}}\wedge x_{j_{\sigma(8)}}+\mathrm{Pf}(\mathbf{\Delta})_{j_1j_2j_3j_4j_5j_6j_7j_8}\,I_{\mathcal{F}}\right\}. \tag{252}$$

One can consider the special case $r = 4$ the subscripts that have no values in common are

$$\binom{8}{4}\binom{8-4}{4} = \binom{8}{4} = \frac{8*7*6*5}{1*2*3*4} = 70 \tag{253}$$

e.g.

$$12345678, 12354678, 12364578, 12374568, 12384567, \tag{254}$$

which gives the terms

$$\begin{aligned}
&A_{41234}A_{45678}x_1\wedge x_2\wedge x_3\wedge x_4\wedge x_5\wedge x_6\wedge x_7\wedge x_8\\
&A_{41235}A_{44678}x_1\wedge x_2\wedge x_3\wedge x_5\wedge x_4\wedge x_6\wedge x_7\wedge x_8\\
&A_{41236}A_{44578}x_1\wedge x_2\wedge x_3\wedge x_6\wedge x_4\wedge x_5\wedge x_7\wedge x_8\\
&A_{41237}A_{44568}x_1\wedge x_2\wedge x_3\wedge x_7\wedge x_4\wedge x_5\wedge x_6\wedge x_8\\
&A_{41238}A_{44567}x_1\wedge x_2\wedge x_3\wedge x_8\wedge x_4\wedge x_5\wedge x_6\wedge x_7
\end{aligned} \tag{255}$$

and the final coefficient is

$$\left(A_{41234}A_{45678}-A_{41235}A_{44678}+A_{41236}A_{44578}-A_{41237}A_{44568}+A_{41238}A_{44567}+\cdots\right)x_1\wedge x_2\wedge x_3\wedge x_4\wedge x_5\wedge x_6\wedge x_7\wedge \tag{256}$$

The ordered pairs (j,k) can also be expressed as a single index as $(J(j,k))$,where the transformation is

$$J(j,k)=2r(j-1)-j(j+1)/2+k;1\leq j<k\leq 2r. \tag{257}$$

One can check this formula for $r=3$ by:

$$\begin{array}{ccc}
j & k & J(j,k) \\
1 & 2 & 6(1-1)-1(1+1)/2+2=1 \\
1 & 3 & 6(1-1)-1(1+1)/2+3=2 \\
1 & 4 & 6(1-1)-1(1+1)/2+4=3 \\
1 & 5 & 6(1-1)-1(1+1)/2+5=4 \\
1 & 6 & 6(1-1)-1(1+1)/2+6=5 \\
2 & 3 & 6(2-1)-2(2+1)/2+3=6 \\
2 & 4 & 6(2-1)-2(2+1)/2+4=7 \\
2 & 5 & 6(2-1)-2(2+1)/2+5=8 \\
2 & 6 & 6(2-1)-2(2+1)/2+6=9 \\
3 & 4 & 6(3-1)-3(3+1)/2+4=10 \\
3 & 5 & 6(3-1)-3(3+1)/2+5=11 \\
3 & 6 & 6(3-1)-3(3+1)/2+6=12 \\
4 & 5 & 6(4-1)-4(4+1)/2+5=13 \\
4 & 6 & 6(4-1)-4(4+1)/2+6=14 \\
5 & 6 & 6(5-1)-5(5+1)/2+6=15
\end{array} \tag{258}$$

One can classify this term into $0-$substition, $1-$substition, $2-$substitions, $3-$substitions, and $4-$substitions:

1. $0-$substition :

$$12345678 \tag{259}$$

2. 1−substition :

$$\binom{4}{1}^2 = 16 \tag{260}$$

$$\begin{aligned}
&1235, 1236, 1237, 1238, \;\; 4678, 5478, 5648, 5674,\\
&1254, 1264, 1274, 1284, \;\; 3678, 5378, 5638, 5673,\\
&1534, 1634, 1734, 1834, \;\; 2678, 5278, 5628, 5672,\\
&5234, 6234, 7234, 8234, \;\; 1678, 5178, 5618, 5671,
\end{aligned} \tag{261}$$

$$A_{1235}A_{4678}x_1x_2x_3x_5x_4x_6x_7x_8 = -A_{1235}A_{4678}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{262}$$

$$A_{1236}A_{5478}x_1x_2x_3x_6x_5x_4x_7x_8 = -A_{1236}A_{4578}x_1x_2x_3x_6x_5x_4x_7x_8 = A_{1236}A_{4578}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{263}$$

$$A_{1237}A_{5648}x_1x_2x_3x_7x_5x_6x_4x_8 = -A_{1237}A_{5648}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{264}$$

$$A_{1238}A_{5674}x_1x_2x_3x_8x_5x_6x_7x_4 = -A_{1238}A_{5674}x_1x_2x_3x_5x_4x_6x_7x_8 = A_{1238}A_{5647}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{265}$$

$$A_{1254}A_{3678}x_1x_2x_3x_5x_4x_6x_7x_8 = -A_{1235}A_{4678}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{266}$$

$$A_{1264}A_{5378}x_1x_2x_3x_6x_5x_4x_7x_8 = -A_{1236}A_{4578}x_1x_2x_3x_6x_5x_4x_7x_8 = A_{1236}A_{4578}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{267}$$

$$A_{1274}A_{5638}x_1x_2x_3x_7x_5x_6x_4x_8 = -A_{1237}A_{5648}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{268}$$

$$A_{1284}A_{5673}x_1x_2x_3x_8x_5x_6x_7x_4 = -A_{1238}A_{5674}x_1x_2x_3x_5x_4x_6x_7x_8 = A_{1238}A_{5647}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{269}$$

$$A_{1534}A_{2678}x_1x_2x_3x_5x_4x_6x_7x_8 = -A_{1235}A_{4678}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{270}$$

$$A_{1634}A_{5278}x_1x_2x_3x_6x_5x_4x_7x_8 = -A_{1236}A_{4578}x_1x_2x_3x_6x_5x_4x_7x_8 = A_{1236}A_{4578}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{271}$$

$$A_{1734}A_{5628}x_1x_2x_3x_7x_5x_6x_4x_8 = -A_{1237}A_{5648}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{272}$$

$$A_{1834}A_{5672}x_1x_2x_3x_8x_5x_6x_7x_4 = -A_{1238}A_{5674}x_1x_2x_3x_5x_4x_6x_7x_8 = A_{1238}A_{5647}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{273}$$

$$A_{5234}A_{1678}x_1x_2x_3x_5x_4x_6x_7x_8 = -A_{1235}A_{4678}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{274}$$

$$A_{6234}A_{5178}x_1x_2x_3x_6x_5x_4x_7x_8 = -A_{1236}A_{4578}x_1x_2x_3x_6x_5x_4x_7x_8 = A_{1236}A_{4578}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{275}$$

$$A_{7234}A_{5618}x_1x_2x_3x_7x_5x_6x_4x_8 = -A_{1237}A_{5648}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{276}$$

$$A_{8234}A_{5671}x_1x_2x_3x_8x_5x_6x_7x_4 = -A_{1238}A_{5674}x_1x_2x_3x_5x_4x_6x_7x_8 = A_{1238}A_{5647}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{277}$$

3. 2−substitions:

$$\binom{4}{2}^2 = 36 \tag{278}$$

$$\begin{array}{ll}
1256, 1356, 1456, 2356, 2456, 3456, & 3478, 2478, 2378, 1478, 1378, 1278 \\
1257, 1357, 1457, 2357, 2457, 3457, & 3468, 2468, 2368, 1468, 1368, 1268 \\
1258, 1358, 1458, 2358, 2458, 3458, & 3467, 2467, 2367, 1467, 1367, 1267 \\
1267, 1367, 1467, 2367, 2467, 3467, & 3458, 2458, 2358, 1458, 1358, 1258 \\
1268, 1368, 1468, 2368, 2468, 3468, & 3457, 2457, 2357, 1457, 1357, 1257 \\
1278, 1378, 1478, 2378, 2478, 3478, & 3456, 2456, 2356, 1456, 1356, 1256
\end{array} \tag{279}$$

$$J(j,k) = 2r(j-1) - j(j+1)/2 + k; 1 \leq j < k \leq 2r. \tag{280}$$

$$J(5,6) = 8(5-1) - 5(5+1)/2 + 6 = 32 - 15 + 6 = 23 \tag{281}$$

$$J(3,4) = 8(3-1) - 3(3+1)/2 + 4 = 16 - 6 + 4 = 14 \tag{282}$$

$$J(7,8) = 8(7-1) - 7(7+1)/2 + 8 = 54 - 28 + 8 = 34 \tag{283}$$

$$A_{1256}A_{3478}x_1x_2x_5x_6x_3x_4x_7x_8 = A_{1256}A_{3478}x_1x_2x_3x_4x_5x_6x_7x_8 = \mathcal{A}_{1(23)}\mathcal{A}_{(14)(34)}y_{12}y_{34}y_{56}y_{78} = \mathcal{A}_{1(23)}\mathcal{A}_{(1} \tag{284}$$

$$A_{1356}A_{2478}x_1x_3x_5x_6x_2x_4x_7x_8 = -A_{1356}A_{2478}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{285}$$

$$A_{1456}A_{2378}x_1x_4x_5x_6x_2x_3x_7x_8 = A_{1456}A_{2378}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{286}$$

$$A_{2356}A_{1478}x_2x_3x_5x_6x_1x_4x_7x_8 = -A_{2356}A_{1478}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{287}$$

$$A_{2456}A_{1378}x_2x_4x_5x_6x_1x_3x_7x_8 = -A_{2456}A_{1378}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{288}$$

$$A_{3456}A_{1278}x_3x_4x_5x_6x_1x_2x_7x_8 = A_{3456}A_{1278}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{289}$$

$$A_{1257}A_{3468}x_1x_2x_5x_7x_3x_4x_6x_8 = -A_{1257}A_{3468}x_1x_2x_3x_5x_4x_6x_7x_8 \tag{290}$$

$$A_{1357}A_{2468}x_1x_3x_5x_7x_2x_4x_6x_8 = A_{1357}A_{2468}x_1x_2x_3x_5x_4x_6x_7x_8 \tag{291}$$

$$A_{1457}A_{2368}x_1x_4x_5x_7x_2x_3x_6x_8 = -A_{1457}A_{2368}x_1x_2x_3x_5x_4x_6x_7x_8 \tag{292}$$

$$A_{2357}A_{1468}x_2x_3x_5x_7x_1x_4x_6x_8 = A_{2357}A_{1468}x_1x_2x_3x_5x_4x_6x_7x_8 \tag{293}$$

$$A_{2457}A_{1368}x_2x_4x_5x_7x_1x_3x_6x_8 = A_{2457}A_{1368}x_1x_2x_3x_5x_4x_6x_7x_8 \tag{294}$$

$$A_{3457}A_{1268}x_3x_4x_5x_7x_1x_2x_6x_8 = A_{3457}A_{1268}x_1x_2x_3x_5x_4x_6x_7x_8 \tag{295}$$

$$A_{1256}A_{3478}x_1x_2x_5x_6x_3x_4x_7x_8 = \tag{296}$$

$$A_{1356}A_{2478}x_1x_3x_5x_6x_2x_4x_7x_8 = \tag{297}$$

$$A_{1456}A_{2378}x_1x_4x_5x_6x_2x_3x_7x_8 = \tag{298}$$

$$A_{2356}A_{1478}x_2x_3x_5x_6x_1x_4x_7x_8 = \tag{299}$$

$$A_{2456}A_{1378}x_2x_4x_5x_6x_1x_3x_7x_8 = \tag{300}$$

$$A_{3456}A_{1278}x_3x_4x_5x_6x_1x_2x_7x_8 = \tag{301}$$

$$A_{1256}A_{3478}x_1x_2x_5x_6x_3x_4x_7x_8 = \tag{302}$$

$$A_{1356}A_{2478}x_1x_3x_5x_6x_2x_4x_7x_8 = \tag{303}$$

$$A_{1456}A_{2378}x_1x_4x_5x_6x_2x_3x_7x_8 = \tag{304}$$

$$A_{2356}A_{1478}x_2x_3x_5x_6x_1x_4x_7x_8 = \tag{305}$$

$$A_{2456}A_{1378}x_2x_4x_5x_6x_1x_3x_7x_8 = \tag{306}$$

$$A_{3456}A_{1278}x_3x_4x_5x_6x_1x_2x_7x_8 = \tag{307}$$

$$A_{1256}A_{3478}x_1x_2x_5x_6x_3x_4x_7x_8 = \tag{308}$$

$$A_{1356}A_{2478}x_1x_3x_5x_6x_2x_4x_7x_8 = \tag{309}$$

$$A_{1456}A_{2378}x_1x_4x_5x_6x_2x_3x_7x_8 = \tag{310}$$

$$A_{2356}A_{1478}x_2x_3x_5x_6x_1x_4x_7x_8 = \tag{311}$$

$$A_{2456}A_{1378}x_2x_4x_5x_6x_1x_3x_7x_8 = \tag{312}$$

$$A_{3456}A_{1278}x_3x_4x_5x_6x_1x_2x_7x_8 = \tag{313}$$

$$A_{1256}A_{3478}x_1x_2x_5x_6x_3x_4x_7x_8 = \tag{314}$$

$$A_{1356}A_{2478}x_1x_3x_5x_6x_2x_4x_7x_8 = \tag{315}$$

$$A_{1456}A_{2378}x_1x_4x_5x_6x_2x_3x_7x_8 = \tag{316}$$

$$A_{2356}A_{1478}x_2x_3x_5x_6x_1x_4x_7x_8 = \tag{317}$$

$$A_{2456}A_{1378}x_2x_4x_5x_6x_1x_3x_7x_8 = \tag{318}$$

$$A_{3456}A_{1278}x_3x_4x_5x_6x_1x_2x_7x_8 = \tag{319}$$

4. 3−substitions:

$$\binom{4}{3}^2 = 16 \tag{320}$$

$$\begin{aligned} &4678, 5478, 5648, 5674, \;\; 1235, 1236, 1237, 1238, \\ &3678, 5378, 5638, 5673, \;\; 1254, 1264, 1274, 1284, \\ &2678, 5278, 5628, 5672, \;\; 1534, 1634, 1734, 1834, \\ &1678, 5178, 5618, 5671, \;\; 5234, 6234, 7234, 8234, \end{aligned} \tag{321}$$

$$A_{1235}A_{4678}x_1x_2x_3x_5x_4x_6x_7x_8 = -A_{1235}A_{4678}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{322}$$

$$A_{1236}A_{5478}x_1x_2x_3x_6x_5x_4x_7x_8 = -A_{1236}A_{4578}x_1x_2x_3x_6x_5x_4x_7x_8 = A_{1236}A_{4578}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{323}$$

$$A_{1237}A_{5648}x_1x_2x_3x_7x_5x_6x_4x_8 = -A_{1237}A_{5648}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{324}$$

$$A_{1238}A_{5674}x_1x_2x_3x_8x_5x_6x_7x_4 = -A_{1238}A_{5674}x_1x_2x_3x_5x_4x_6x_7x_8 = A_{1238}A_{5647}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{325}$$

$$A_{1254}A_{3678}x_1x_2x_3x_5x_4x_6x_7x_8 = -A_{1235}A_{4678}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{326}$$

$$A_{1264}A_{5378}x_1x_2x_3x_6x_5x_4x_7x_8 = -A_{1236}A_{4578}x_1x_2x_3x_6x_5x_4x_7x_8 = A_{1236}A_{4578}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{327}$$

$$A_{1274}A_{5638}x_1x_2x_3x_7x_5x_6x_4x_8 = -A_{1237}A_{5648}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{328}$$

$$A_{1284}A_{5673}x_1x_2x_3x_8x_5x_6x_7x_4 = -A_{1238}A_{5674}x_1x_2x_3x_5x_4x_6x_7x_8 = A_{1238}A_{5647}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{329}$$

$$A_{1534}A_{2678}x_1x_2x_3x_5x_4x_6x_7x_8 = -A_{1235}A_{4678}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{330}$$

$$A_{1634}A_{5278}x_1x_2x_3x_6x_5x_4x_7x_8 = -A_{1236}A_{4578}x_1x_2x_3x_6x_5x_4x_7x_8 = A_{1236}A_{4578}x_1x_2x_3x_5x_4x_6x_7x_8 \tag{331}$$

$$A_{1734}A_{5628}x_1x_2x_3x_7x_5x_6x_4x_8 = -A_{1237}A_{5648}x_1x_2x_3x_4x_5x_6x_7x_8 \tag{332}$$

$$A_{1834}A_{5672}x_1x_2x_3x_8x_5x_6x_7x_4 = -A_{1238}A_{5674}x_1x_2x_3x_5x_4x_6x_7x_8 = A_{1238}A_{5647}x_1x_2x_3x_5x_4x_6x_7x_8 \tag{333}$$

$$A_{5234}A_{1678}x_1x_2x_3x_5x_4x_6x_7x_8 = -A_{1235}A_{4678}x_1x_2x_3x_5x_4x_6x_7x_8 \tag{334}$$

$$A_{6234}A_{5178}x_1x_2x_3x_6x_5x_4x_7x_8 = -A_{1236}A_{4578}x_1x_2x_3x_6x_5x_4x_7x_8 = A_{1236}A_{4578}x_1x_2x_3x_5x_4x_6x_7x_8 \tag{335}$$

$$A_{7234}A_{5618}x_1x_2x_3x_7x_5x_6x_4x_8 = -A_{1237}A_{5648}x_1x_2x_3x_5x_4x_6x_7x_8 \tag{336}$$

$$A_{8234}A_{5671}x_1x_2x_3x_8x_5x_6x_7x_4 = -A_{1238}A_{5674}x_1x_2x_3x_5x_4x_6x_7x_8 = A_{1238}A_{5647}x_1x_2x_3x_5x_4x_6x_7x_8 \tag{337}$$

5. 4−substitions:

$$\binom{4}{4}^2 = 1 \tag{338}$$

$$56781234 \tag{339}$$

One can consider the special case $r = 4$ the subscripts that have one values in common are

$$\binom{8}{4}\binom{8-4}{3} = 70 * 4 = 280 \tag{340}$$

e.g.

12341567, 12351467, 12361457, 12371456, 12451367, 12461357, 12471356, 12561347, 12571346, 12671356,
13451267, 13461257, 13471256, 13561247, 13571246, 13671245,
14561237, 14571236, 14671235,
15671234. (341)

the coefficients are

$$(A_{1234}A_{1567} - A_{1235}A_{1467} + A_{1236}A_{1457} - A_{1237}A_{1456} + A_{1245}A_{1367} - A_{1246}A_{1357} + A_{1247}A_{1356} - A_{1256}A_{1347}$$
$$- A_{1257}A_{1346} + A_{1267}A_{1345} - A_{1345}A_{1267} + A_{1346}A_{1257} - A_{1347}A_{1256} - A_{1356}A_{1247} + A_{1357}A_{1246} - A_{1367}A_{1245}$$
$$+A_{1456}A_{1237} - A_{1457}A_{1236} + A_{1467}A_{1235}, -A_{1567}A_{1234})\, x_2 \wedge x_3 \wedge x_4 \wedge x_5 \wedge x_6 \wedge x_7. \tag{342}$$

So it seems that this term cancels.

One can consider the special case $r = 4$ the subscripts that have two values in common are

$$\binom{8}{4}\binom{8-4}{2} = 70 * 12 = 840 \tag{343}$$

e.g.

$$12341256, 12351246, 12361245, 12451236, 12461235, 12561234. \tag{344}$$

the coefficients are given by

$$A_{1234}A_{1256}x_3 \wedge x_4 \wedge x_5 \wedge x_6 \tag{345}$$
$$A_{1235}A_{1246}x_3 \wedge x_5 \wedge x_4 \wedge x_6 \tag{346}$$
$$A_{1236}A_{1245}x_3 \wedge x_6 \wedge x_4 \wedge x_5 \tag{347}$$
$$A_{1245}A_{1236}x_4 \wedge x_5 \wedge x_3 \wedge x_6 \tag{348}$$
$$A_{1246}A_{1235}x_4 \wedge x_6 \wedge x_3 \wedge x_5 \tag{349}$$
$$A_{1256}A_{1234}x_5 \wedge x_6 \wedge x_3 \wedge x_4 \tag{350}$$

which gives on reordering

$$A_{1234}A_{1256}x_3 \wedge x_4 \wedge x_5 \wedge x_6 - A_{1235}A_{1246}x_3 \wedge x_4 \wedge x_5 \wedge x_6 + A_{1236}A_{1245}x_3 \wedge x_4 \wedge x_5 \wedge x_6 \tag{351}$$

$$+ A_{1245}A_{1236}x_3 \wedge x_4 \wedge x_5 \wedge x_6 - A_{1246}A_{1235}x_3 \wedge x_4 \wedge x_5 \wedge x_6 + A_{1256}A_{1234}x_3 \wedge x_4 \wedge x_5 \wedge x_6 \tag{352}$$

$$= 2\left(A_{1234}A_{1256} - A_{1235}A_{1246} + A_{1236}A_{1245}\right) x_3 \wedge x_4 \wedge x_5 \wedge x_6 \tag{353}$$

which in general is non zero.

One can consider the special case $r = 4$ the subscripts that have three values in common are

$$\binom{8}{4}\binom{8-4}{1} = 70 * 4 = 280 \tag{354}$$

e.g.

$$12341235, 12341236, 12341237, 12341238, 12351234, 12351236, 12351237, 12351238, \tag{355}$$

$$12361234, 12361235, 12361237, 12361238, 12371238, \tag{356}$$

the coefficients are

$$A_{1234}A_{1235} - A_{1235}A_{1234} + A_{1234}A_{1236} - A_{1236}A_{1234} + A_{1234}A_{1235} - A_{1235}A_{1234} \cdots . \tag{357}$$

So it seems that this term cancels.

One can consider the special case $r = 4$ the subscripts that have four values in common are

$$\binom{8}{4}\binom{8-4}{0} = 70 * 1 = 70 \tag{358}$$

$$12341234 + 12351235 + \cdots . \tag{359}$$

III.

IV. GEOMETRIC ALGEBRA

The operator and Clifford products in $\mathcal{F} = \mathcal{B}\left(\mathcal{H}_{\mathcal{F}}\right)$ are the same and are also known as the *geometric product.* Thus $\mathcal{F}$ can be thought of as an operator algebra, Clifford algebra or Geometric algebra. The product of two grade 1 operators (vectors)

$$a = \sum_{1 \le j \le 2r} \alpha_j x_j \text{ and } b = \sum_{1 \le j \le 2r} \beta_j x_j \tag{360}$$

is given by

$$\begin{aligned} ab &= \sum_{1 \le j,k \le 2r} \alpha_j \beta_k x_j x_k = \sum_{1 \le j \le 2r} \alpha_j \beta_j x_j^2 + \sum_{1 \le j \ne k \le 2r} \alpha_j \beta_k x_j x_k \\ &= \frac{1}{2} \sum_{1 \le j \le 2r} \alpha_j \beta_j + \sum_{1 \le j < k \le 2r} \left(\alpha_j \beta_k - \alpha_k \beta_j\right) x_j x_k \\ &= \frac{1}{2}\left(a \middle| b\right) + a \wedge b \end{aligned} \tag{361}$$

and the product ab expressed in this fashion is called the *geometric product* of the *vectors* a and b. The products $(a|\, b)$ and $a \wedge b$ can be expressed or defined by

$$(a|\, b) = \frac{1}{2}\{ab + ba\}$$

and

$$a \wedge b = \frac{1}{2}\{ab - ba\}.$$

A. Degree of Operators

N^{th} degree operators are defined to be N^{th} degree polynomials in the generators of the Clifford algebra and are of the form

$$A_N = \alpha_0 I_{\mathcal{F}} + \sum_{1 \le M \le N} \sum_{1 \le j_1 < \cdots < j_M \le 2r} \alpha_{M j_1 \cdots j_M} x_{j_1} \cdots x_{j_M} \tag{362}$$

and grade N operators are homogeneous of degree N i.e.

$$B_N = \sum_{1 \le j_1 < \cdots < j_N \le 2r} \alpha_{N j_1 \cdots j_N} x_{j_1} \cdots x_{j_N}. \tag{363}$$

Blades are outer $\equiv$ wedge products of *orthogonal* vectors, while versors are outer $\equiv$ wedge of vectors.

1. *First Degree Operators*

First degree operators are first degree polynomials in the generators of the Clifford algebra

$$A_1 = \alpha_0 I_{\mathcal{F}} + \sum_{1 \le j \le 2r} \alpha_{1j} x_j.$$

Grade 1 operators are homogeneous of degree 1 and can be expressed in terms of a non a self adjoint basis as

$$\begin{aligned} A_1 &= \sum_{1 \le j \le 2r} \alpha_j x_j = \sum_{1 \le j \le r} \{\alpha_j x_j + \alpha_{j+r} x_{j+r}\} = \sum_{1 \le j \le r} \left\{\alpha_j \left(a_j^\dagger + a_j\right) + \alpha_{j+r} i \left(a_j^\dagger - a_j\right)\right\} \\ &= \sum_{1 \le j \le r} \left\{\alpha_j a_j^\dagger + \alpha_j a_j + i\alpha_{j+r} a_j^\dagger - i\alpha_{j+r} a_j\right\} = \sum_{1 \le j \le r} \left\{(\alpha_j + i\alpha_{j+r})\, a_j^\dagger + (\alpha_j - i\alpha_{j+r})\, a_j\right\}. \end{aligned} \tag{364}$$

2. Second Degree Operators

Second degree operators are second degree polynomials in the generators of the Clifford algebra

$$A_2 = \alpha_0 I_{\mathcal{F}} + \sum_{1\leq j\leq 2r} \alpha_{1j} x_j + \sum_{1\leq j<k\leq 2r} \alpha_{2jk} x_j x_k.$$

Homogeneous second degree operators of degree 2 can be expressed in terms of a non self adjoint basis as

$$\begin{aligned}
A_2 &= \sum_{1\leq j<k\leq 2r} \alpha_{jk} x_j x_k \\
&= 2 \sum_{1\leq j<k\leq r} (\alpha_{jk} - \alpha_{kj}) \left(a_j^\dagger + a_j\right) \left(a_k^\dagger + a_k\right) + \\
&+ i \sum_{1\leq j\leq r, r+1\leq k\leq 2r} (\alpha_{jk} - \alpha_{kj}) \left(a_j^\dagger + a_j\right) \left(a_k^\dagger - a_k\right) - 2 \sum_{r+1\leq j<k\leq 2r} (\alpha_{jk} - \alpha_{kj}) \left(a_j^\dagger - a_j\right) \left(a_k^\dagger - a_k\right) \\
&= 2 \sum_{1\leq j<k\leq r} (\alpha_{jk} - \alpha_{kj}) \left(a_j^\dagger a_k^\dagger + a_j^\dagger a_k + a_j a_k^\dagger + a_j a_k\right) \\
&+ i \sum_{1\leq j\leq r, r+1\leq k\leq 2r} (\alpha_{jk} - \alpha_{kj}) \left(a_j^\dagger a_k^\dagger - a_j^\dagger a_k + a_j a_k^\dagger - a_j a_k\right) \\
&- 2 \sum_{r+1\leq j<k\leq 2r} (\alpha_{jk} - \alpha_{kj}) \left(a_j^\dagger a_k^\dagger - a_j^\dagger a_k - a_j a_k^\dagger + a_j a_k\right). \qquad (365)
\end{aligned}$$

Grade 2 operators are homogeneous of degree 2 i.e.

$$B_2 = \sum_{1\leq j_1<j_2\leq 2r} \alpha_{2j_1j_2} x_{j_1} x_{j_2}. \qquad (366)$$

V. ONE PARTICLE OPERATORS

One can first consider one particle operators of

3. Diagonal form

Diagonal one particle operators expressed in terms of the generators of the Clifford algebra are

$$\lambda_1 = \sum_{1\leq j\leq r} \beta_{1j} b_j^\dagger b_j = \frac{1}{2} \sum_{1\leq j,k\leq r} \beta_{1j} \left(y_j - i y_{j+r}\right)\left(y_j + i y_{j+r}\right) \tag{367}$$

$$= \frac{1}{2} \sum_{1\leq j\leq r} \beta_{1j} \left\{y_j y_j + y_{j+r} y_{j+r} + i y_j y_{j+r} - i y_{j+r} y_j\right\} \tag{368}$$

$$= \frac{1}{2} \left(\operatorname{Tr} \lambda_1\right) I_{\mathcal{F}} + \frac{i}{2} \sum_{1\leq j\leq r} \beta_{1j} \left\{\left(y_j y_{j+r} - y_{j+r} y_j\right)\right\} \tag{369}$$

$$= \frac{1}{2} \left(\operatorname{Tr} \lambda_1\right) I_{\mathcal{F}} + i \sum_{1\leq j\leq r} \beta_{1j} y_j y_{j+r}, \tag{370}$$

which is quite general as any normal one particle operator can be diagonalized. If the operator is self adjoint $\left\{\beta_{1j} \in \mathbb{R} \middle| 1 \leq j \leq r\right\}$. One can observe from the preceding that a one particle operator has both grade zero and one components and is a first degree polynomial.

We can find the square of the diagonal form of a normal one particle operator when $r = 2$ as

$$\begin{aligned}
&\left(\frac{1}{2} \left(\operatorname{Tr} \lambda_1\right) I_{\mathcal{F}} + i \left(\beta_{11} y_1 y_3 + \beta_{12} y_2 y_4\right)\right)^2 \\
&= \frac{1}{4} \left(\operatorname{Tr} \lambda_1\right)^2 I_{\mathcal{F}} + i \left(\operatorname{Tr} \lambda_1\right)\left(\beta_{11} y_1 y_3 + \beta_{12} y_2 y_4\right) - \left(\beta_{11} y_1 y_3 + \beta_{12} y_2 y_4\right)\left(\beta_{11} y_1 y_3 + \beta_{12} y_2 y_4\right) \\
&= \frac{1}{4} \left(\operatorname{Tr} \lambda_1\right)^2 I_{\mathcal{F}} + i \left(\operatorname{Tr} \lambda_1\right)\left(\beta_{11} y_1 y_3 + \beta_{12} y_2 y_4\right) + \frac{1}{4}\beta_{11}^2 I_{\mathcal{F}} + \frac{1}{4}\beta_{12}^2 I_{\mathcal{F}} \\
&\qquad\qquad + \beta_{11}\beta_{12} y_1 y_3 y_2 y_4 + \beta_{11}\beta_{12} y_2 y_4 y_1 y_3 \\
&= \frac{1}{4} \left(\operatorname{Tr} \lambda_1\right)^2 I_{\mathcal{F}} + i \left(\operatorname{Tr} \lambda_1\right)\left(\beta_{11} y_1 y_3 + \beta_{12} y_2 y_4\right) + \frac{1}{4}\beta_{11}^2 I_{\mathcal{F}} + \frac{1}{4}\beta_{12}^2 I_{\mathcal{F}} - 2\beta_{11}\beta_{12} y_1 y_2 y_3 y_4 \\
&= \frac{1}{4} \left\{\left(\operatorname{Tr} \lambda_1\right)^2 + \beta_{11}^2 + \beta_{12}^2\right\} I_{\mathcal{F}} + i \left(\operatorname{Tr} \lambda_1\right)\left(\beta_{11} y_1 y_3 + \beta_{12} y_2 y_4\right) - 2\beta_{11}\beta_{12} y_1 y_2 y_3 y_4 \\
&= \frac{1}{4} \left\{\left(\operatorname{Tr} \lambda_1\right)^2 + \operatorname{Tr} \lambda_1^2\right\} I_{\mathcal{F}} + i \left(\operatorname{Tr} \lambda_1\right)\left(\beta_{11} y_1 y_3 + \beta_{12} y_2 y_4\right) - 2\beta_{11}\beta_{12} y_1 y_2 y_3 y_4
\end{aligned} \tag{371}$$

Thus the idempotency condition gives

$$\begin{aligned}
&\frac{1}{2}\left(\operatorname{Tr} \lambda_1 \left(\frac{1}{2} \operatorname{Tr} \lambda_1 - 1\right) + \frac{1}{2}\left(\beta_{11}^2 + \beta_{12}^2\right)\right) I_{\mathcal{F}} + i \left(\operatorname{Tr} \lambda_1 - 1\right)\left(\beta_{11} y_1 y_3 + \beta_{12} y_2 y_4\right) \\
&- 2\beta_{11}\beta_{12} y_1 y_2 y_3 y_4 = 0 \Longrightarrow \\
&\frac{1}{2}\left(\operatorname{Tr} \lambda_1 \left(\frac{1}{2} \operatorname{Tr} \lambda_1 - 1\right) + \frac{1}{2} \operatorname{Tr} \lambda_1^2\right) I_{\mathcal{F}} + i \left(\operatorname{Tr} \lambda_1 - 1\right)\left(\beta_{11} y_1 y_3 + \beta_{12} y_2 y_4\right) \\
&- 2\beta_{11}\beta_{12} y_1 y_2 y_3 y_4 = 0.
\end{aligned} \tag{372}$$

One can express this more transparently as

$$\left(\mathrm{Tr}\,\lambda_1^2\right) I_{\mathcal{F}} = 2\left(\mathrm{Tr}\,\lambda_1\right)\left(\left(\mathrm{Tr}\,\lambda_1\right) - \frac{1}{2}\right) I_{\mathcal{F}} \tag{373}$$

$$i\left(\mathrm{Tr}\,\lambda_1\right)\left(\beta_{11}y_1y_3 + \beta_{12}y_2y_4\right) = i\left(\beta_{11}y_1y_3 + \beta_{12}y_2y_4\right) \tag{374}$$

$$2\beta_{11}\beta_{12}y_1y_2y_3y_4 = 0 \tag{375}$$

$$\mathrm{Tr}\,\lambda_1 = \beta_{11} + \beta_{12} \tag{376}$$

Noting

$$\begin{aligned} y_1y_2y_3y_4 &= -\frac{1}{4}\left(a_1^\dagger + a_1\right)\left(a_2^\dagger + a_2\right)\left(a_3^\dagger - a_3\right)\left(a_4^\dagger - a_4\right) \\ &= -\frac{1}{4}\left(a_1^\dagger a_2^\dagger + a_1^\dagger a_2 + a_1 a_2^\dagger + a_1 a_2\right)\left(a_3^\dagger a_4^\dagger - a_3^\dagger a_4 - a_3 a_4^\dagger + a_3 a_4\right) \neq 0 \end{aligned} \tag{377}$$

one can deduce that either or both $\beta_{11}, \beta_{12} = 0$ thus

$$\begin{aligned} \left(\mathrm{Tr}\,\lambda_1\right)\left(\beta_{11}y_1y_3\right) &= \beta_{11}y_1y_3 \\ &\Rightarrow \mathrm{Tr}\,\lambda_1 = \beta_{11} = 1. \end{aligned} \tag{378}$$

For any r one obtains

$$\begin{aligned} &\left(\frac{1}{2}\left(\mathrm{Tr}\,\lambda_1\right) I_{\mathcal{F}} + i\sum_{1\leq j\leq r}\beta_{1j}y_jy_{j+r}\right)^2 \\ &= \frac{1}{4}\left(\mathrm{Tr}\,\lambda_1\right)^2 I_{\mathcal{F}} + i\left(\mathrm{Tr}\,\lambda_1\right)\sum_{1\leq j\leq r}\beta_{1j}y_jy_{j+r} - \sum_{1\leq j\leq r}\beta_{1j}y_jy_{j+r}\sum_{1\leq j\leq r}\beta_{1j}y_jy_{j+r} \\ &= \frac{1}{4}\left(\mathrm{Tr}\,\lambda_1\right)^2 I_{\mathcal{F}} + i\left(\mathrm{Tr}\,\lambda_1\right)\sum_{1\leq j\leq r}\beta_{1j}y_jy_{j+r} + \frac{1}{4}\sum_{1\leq j\leq r}\beta_{1j}^2 I_{\mathcal{F}} + 2\sum_{1\leq j_1<j_2\leq r}\beta_{1j_1}\beta_{1j_2}y_{j_1}y_{j_1+r}y_{j_2}y_{j_2+r} \\ &= \frac{1}{4}\left(\mathrm{Tr}\,\lambda_1\right)^2 I_{\mathcal{F}} + i\left(\mathrm{Tr}\,\lambda_1\right)\sum_{1\leq j\leq r}\beta_{1j}y_jy_{j+r} + \frac{1}{4}\sum_{1\leq j\leq r}\beta_{1j}^2 I_{\mathcal{F}} - 2\sum_{1\leq j_1<j_2\leq r}\beta_{1j_1}\beta_{1j_2}y_{j_1}y_{j_2}y_{j_1+r}y_{j_2+r} \\ &= \frac{1}{4}\left\{\left(\mathrm{Tr}\,\lambda_1\right)^2 + \sum_{1\leq j\leq r}\beta_{1j}^2 I_{\mathcal{F}}\right\} I_{\mathcal{F}} + i\left(\mathrm{Tr}\,\lambda_1\right)\sum_{1\leq j\leq r}\beta_{1j}y_jy_{j+r} - 2\sum_{1\leq j_1<j_2\leq r}\beta_{1j_1}\beta_{1j_2}y_{j_1}y_{j_2}y_{j_1+r}y_{j_2+r} \\ &= \frac{1}{4}\left\{\left(\mathrm{Tr}\,\lambda_1\right)^2 + \left(\mathrm{Tr}\,\lambda_1^2\right) I_{\mathcal{F}}\right\} I_{\mathcal{F}} + i\left(\mathrm{Tr}\,\lambda_1\right)\sum_{1\leq j\leq r}\beta_{1j}y_jy_{j+r} - 2\sum_{1\leq j_1<j_2\leq r}\beta_{1j_1}\beta_{1j_2}y_{j_1}y_{j_2}y_{j_1+r}y_{j_2+r} \end{aligned} \tag{379}$$

The idempotency condition

$$\left(\frac{1}{2}\left(\mathrm{Tr}\,\lambda_1\right) I_{\mathcal{F}} + i\sum_{1\leq j\leq r}\beta_{1j}y_jy_{j+r}\right)^2 - \left(\frac{1}{2}\left(\mathrm{Tr}\,\lambda_1\right) I_{\mathcal{F}} + i\sum_{1\leq j\leq r}\beta_{1j}y_jy_{j+r}\right) = 0 \tag{380}$$

gives

$$\frac{1}{4}\left\{(\operatorname{Tr}\lambda_1)\left((\operatorname{Tr}\lambda_1)-2\right)+\left(\operatorname{Tr}\lambda_1^2\right)\right\} I_{\mathcal{F}}+i\left(\operatorname{Tr}\lambda_1-1\right)\sum_{1\leq j\leq r}\beta_{1j}y_jy_{j+r}$$
$$-2\sum_{1\leq j_1<j_2\leq r}\beta_{1j_1}\beta_{1j_2}y_{j_1}y_{j_2}y_{j_1+r}y_{j_2+r}=0. \tag{381}$$

One can apply the same reasoning as above to deduce that $\operatorname{Tr}\lambda_1=\beta_{11}=1$ and all other β_{1j} 's are zero

4. General form

Any one particle operator λ_1 can be expressed as

$$\lambda_1=\sum_{1\leq j,k\leq r}\lambda_{1jk}a_j^\dagger a_k=\frac{1}{2}\sum_{1\leq j,k\leq r}\lambda_{1jk}\left(x_j-ix_{j+r}\right)\left(x_k+ix_{k+r}\right)$$
$$=\frac{1}{2}\sum_{1\leq j,k\leq r}\lambda_{1jk}\left\{x_jx_k+x_{j+r}x_{k+r}+ix_jx_{k+r}-ix_{j+r}x_k\right\}, \tag{382}$$

which can be further developed as

$$\lambda_1=\frac{1}{2}\left(\operatorname{Tr}\lambda_1\right)I_{\mathcal{F}}+\frac{1}{2}\sum_{1\leq j<k\leq r}\left(\lambda_{1jk}-\lambda_{1kj}\right)\left\{x_jx_k+x_{j+r}x_{k+r}+i\left(x_jx_{k+r}-x_{j+r}x_k\right)\right\}$$
$$=\frac{1}{2}\left(\operatorname{Tr}\lambda_1\right)I_{\mathcal{F}}+\sum_{1\leq j<k\leq 2r}\eta_{jk}x_jx_k, \tag{383}$$

where the coefficients η_{jk} are antisymmetric and given by

$$\eta_{jk}=\frac{1}{2}\left(\lambda_{1jk}-\lambda_{1kj}\right); \quad 1\leq j<k\leq r; r+1\leq j<k\leq 2r \tag{384}$$
$$\eta_{jk}=-\frac{i}{2}\left(\lambda_{1jk}-\lambda_{1kj}\right); \quad 1\leq j\leq r; r+1\leq k\leq 2r \tag{385}$$
$$\eta_{jj}=0; \quad 1\leq j\leq 2r. \tag{386}$$

One can compute the square of λ_1 when $r=2$ i.e.

$$
\begin{aligned}
(\lambda_1)^2 &= \left(\frac{1}{2}(\operatorname{Tr}\lambda_1)\, I_{\mathcal{F}} + \eta_{12}x_1x_2 + \eta_{13}x_1x_3 + \eta_{14}x_1x_4 + \eta_{23}x_2x_3 + \eta_{24}x_2x_4 + \eta_{34}x_3x_4\right)^2 \\
&= \frac{1}{4}(\operatorname{Tr}\lambda_1)^2 I_{\mathcal{F}} + (\operatorname{Tr}\lambda_1)(\eta_{12}x_1x_2 + \eta_{13}x_1x_3 + \eta_{14}x_1x_4 + \eta_{23}x_2x_3 + \eta_{24}x_2x_4 + \eta_{34}x_3x_4) \\
&- \frac{1}{4}\left(\eta_{12}^2 + \eta_{13}^2 + \eta_{14}^2 + \eta_{23}^2 + \eta_{24}^2 + \eta_{34}^2\right) I_{\mathcal{F}} \\
&+ \eta_{12}\eta_{13}x_1x_2x_1x_3 + \eta_{12}\eta_{14}x_1x_2x_1x_4 + \eta_{12}\eta_{23}x_1x_2x_2x_3 + \eta_{12}\eta_{24}x_1x_2x_2x_4 \\
&+ \eta_{13}\eta_{12}x_1x_3x_1x_2 + \eta_{13}\eta_{14}x_1x_3x_1x_4 + \eta_{13}x_1x_3\eta_{23}x_2x_3 + \eta_{13}\eta_{34}x_1x_3x_3x_4 \\
&+ \eta_{14}\eta_{12}x_1x_4x_1x_2 + \eta_{14}\eta_{13}x_1x_4x_1x_3 + \eta_{14}x_1x_4\eta_{24}x_2x_4 + \eta_{14}\eta_{34}x_1x_4x_3x_4 \\
&+ \eta_{23}\eta_{12}x_2x_3x_1x_2 + \eta_{23}\eta_{13}x_2x_3x_1x_3 + \eta_{23}\eta_{24}x_2x_3x_2x_4 + \eta_{23}\eta_{34}x_2x_3x_3x_4 \\
&+ \eta_{24}\eta_{12}x_2x_4x_1x_2 + \eta_{24}\eta_{14}x_2x_4x_1x_4 + \eta_{24}\eta_{23}x_2x_4x_2x_3 + \eta_{24}\eta_{34}x_2x_4x_3x_4 \\
&+ \eta_{34}\eta_{13}x_3x_4x_1x_3 + \eta_{34}\eta_{14}x_3x_4x_1x_4 + \eta_{34}\eta_{23}x_3x_4x_2x_3 + \eta_{34}\eta_{24}x_3x_4x_2x_4 \\
&+ \eta_{12}\eta_{34}x_1x_2x_3x_4 + \eta_{13}\eta_{24}x_1x_3x_2x_4 + \eta_{14}\eta_{23}x_1x_4x_2x_3 + \eta_{23}\eta_{14}x_2x_3x_1x_4 \\
&+ \eta_{24}x\eta_{132}x_4x_1x_3 + \eta_{34}\eta_{12}x_3x_4x_1x_2
\end{aligned}
$$

$$
\begin{aligned}
&= \left((\operatorname{Tr}\lambda_1)^2 - \frac{1}{4}\left(\eta_{12}^2 + \eta_{13}^2 + \eta_{14}^2 + \eta_{23}^2 + \eta_{24}^2 + \eta_{34}^2\right)\right) I_{\mathcal{F}} \\
&+ ((\operatorname{Tr}\lambda_1)\,\eta_{23} + \eta_{34}\eta_{24} - \eta_{12}\eta_{13} + \eta_{13}\eta_{12} - \eta_{24}\eta_{34})\, x_2x_3 \\
&+ ((\operatorname{Tr}\lambda_1)\,\eta_{24} + \eta_{23}\eta_{34} - \eta_{12}\eta_{14} + \eta_{14}\eta_{12} - \eta_{34}\eta_{23})\, x_2x_4 \\
&+ ((\operatorname{Tr}\lambda_1)\,\eta_{13} + \eta_{12}\eta_{23} - \eta_{14}\eta_{34} - \eta_{23}\eta_{12} + \eta_{34}\eta_{143})\, x_1x_3 \\
&+ ((\operatorname{Tr}\lambda_1)\,\eta_{14} + \eta_{12}\eta_{24} - \eta_{24}\eta_{12} + \eta_{13}\eta_{34} - \eta_{34}\eta_{13})\, x_1x_4 \\
&+ ((\operatorname{Tr}\lambda_1)\,\eta_{34} + \eta_{14}\eta_{13} - \eta_{13}\eta_{14} - \eta_{23}\eta_{24} + \eta_{24}\eta_{23})\, x_3x_4 \\
&+ ((\operatorname{Tr}\lambda_1)\,\eta_{12} + \eta_{23}\eta_{13} - \eta_{13}\eta_{23} - \eta_{14}\eta_{24} + \eta_{24}\eta_{14})\, x_1x_2 \\
&+ 2(\eta_{12}\eta_{34} - \eta_{13}\eta_{24} + \eta_{14}\eta_{23})\, x_1x_2x_3x_4
\end{aligned}
$$

$$
\begin{aligned}
&= \left((\operatorname{Tr}\lambda_1)^2 - \frac{1}{4}\left(\eta_{12}^2 + \eta_{13}^2 + \eta_{14}^2 + \eta_{23}^2 + \eta_{24}^2 + \eta_{34}^2\right)\right) I_{\mathcal{F}} \\
&+ (\operatorname{Tr}\lambda_1)(\eta_{23}x_2x_3 + \eta_{24}x_2x_4 + \eta_{13}x_1x_3 + \eta_{14}x_1x_4 + \eta_{34}x_3x_4 + \eta_{12}x_1x_2) \\
&+ 2(\eta_{12}\eta_{34} - \eta_{13}\eta_{24} + \eta_{14}\eta_{23})\, x_1x_2x_3x_4
\end{aligned}
$$

$$
\begin{aligned}
&= \left(\operatorname{Tr}\lambda_1\left(\operatorname{Tr}\lambda_1 - 1\right) - \left(\eta_{12}^2 + \eta_{13}^2 + \eta_{14}^2 + \eta_{23}^2 + \eta_{24}^2 + \eta_{34}^2\right)\right) I_{\mathcal{F}} \\
&+ \left(\operatorname{Tr}\lambda_1 - 1\right)\left(\eta_{23}x_2x_3 + \eta_{24}x_2x_4 + \eta_{13}x_1x_3 + \eta_{14}x_1x_4 + \eta_{34}x_3x_4 + \eta_{12}x_1x_2\right) \\
&+ 2\left(\eta_{12}\eta_{34} - \eta_{13}\eta_{24} + \eta_{14}\eta_{23}\right)x_1x_2x_3x_4 = 0. \qquad (387)
\end{aligned}
$$

recall

$$\eta_{jk} = \frac{1}{2}\left(\lambda_{1jk} - \lambda_{1kj}\right); \quad 1 \le j < k \le r; r+1 \le j < k \le 2r \qquad (388)$$

$$\eta_{jk} = -\frac{i}{2}\left(\lambda_{1jk} - \lambda_{1kj}\right); \quad 1 \le j \le r; r+1 \le k \le 2r. \qquad (389)$$

This gives in general for any r

$$
\begin{aligned}
&\left(\operatorname{Tr}\lambda_1\left(\operatorname{Tr}\lambda_1 - 1\right) - \sum_{1\le j<k\le r}\eta_{jk}^2\right) I_{\mathcal{F}} + \left(\operatorname{Tr}\lambda_1 - 1\right)\sum_{1\le j<k\le r}\eta_{jk}x_jx_k \\
&+ 2\sum_{1\le j_1<j_2<j_3<j_4\le r}\operatorname{Pf}\left(\eta\right)_{j_1j_2j_3j_4} x_{j_1}x_{j_2}x_{j_3}x_{j_4} = 0. \qquad (390)
\end{aligned}
$$

5.

6. One Particle and One Hole Idempotents

The creators and annihilators

$$a_j^\dagger = \frac{1}{\sqrt{2}}\left(x_j - ix_{j+r}\right) \qquad (391)$$

$$a_j = \frac{1}{\sqrt{2}}\left(x_j + ix_{j+r}\right) \qquad (392)$$

produce the one particle operators

$$a_j^\dagger a_k = \frac{1}{2}\left(x_j - ix_{j+r}\right)\left(x_k + ix_{k+r}\right) = \frac{1}{2}\left(x_jx_k + ix_jx_{k+r} - ix_{j+r}x_k + x_{j+r}x_{k+r}\right) \qquad (393)$$

$$= \frac{1}{2}\left\{\left(x_jx_k + x_{j+r}x_{k+r}\right) + i\left(x_jx_{k+r} - x_{j+r}x_k\right)\right\}. \qquad (394)$$

One particle idempotents are given by

$$n_j = a_j^\dagger a_j = \frac{1}{2}(x_j - ix_{j+r})(x_j + ix_{j+r}) = \frac{1}{2}\left(x_j^2 + ix_jx_{j+r} - ix_{j+r}x_j + x_{j+r}^2\right) \tag{395}$$

$$= \frac{1}{2}I_{\mathcal{F}} + ix_jx_{j+r} \tag{396}$$

$$(n_j)^2 = \left(\frac{1}{2}I_{\mathcal{F}} + ix_jx_{j+r}\right)^2 = \left(\frac{1}{2}I_{\mathcal{F}} + ix_jx_{j+r}\right)\left(\frac{1}{2}I_{\mathcal{F}} + ix_jx_{j+r}\right) \tag{397}$$

$$= \frac{1}{4}I_{\mathcal{F}} + \frac{1}{2}ix_jx_{j+r} + \frac{1}{2}ix_jx_{j+r} - x_jx_{j+r}x_jx_{j+r} \tag{398}$$

$$= \frac{1}{4}I_{\mathcal{F}} + ix_jx_{j+r} + \frac{1}{4}I_{\mathcal{F}} = \frac{1}{2}I_{\mathcal{F}} + ix_jx_{j+r} = n_j. \tag{399}$$

and one hole idempotents are given by

$$h_j = a_ja_j^\dagger = \frac{1}{2}(x_j + ix_{j+r})(x_j - ix_{j+r}) = \frac{1}{2}\left(x_j^2 - ix_jx_{j+r} + ix_{j+r}x_j + x_{j+r}^2\right) \tag{400}$$

$$= \frac{1}{2}I_{\mathcal{F}} - ix_jx_{j+r}. \tag{401}$$

The projectors $\{n_j, h_j\}$ are infinite dimensional when $r = \infty$ showing that in this case there are *no* finite one particle operator projections!

Consider

$$p = \frac{1}{2}I_{\mathcal{F}} + ixy, \tag{402}$$

where

$$x^2 = y^2 = \frac{1}{2}I_{\mathcal{F}}, \tag{403}$$

then

$$p^2 = \left(\frac{1}{2}I_{\mathcal{F}} + ixy\right)\left(\frac{1}{2}I_{\mathcal{F}} + ixy\right) = \frac{1}{4}I_{\mathcal{F}} + \frac{1}{2}ixy + \frac{1}{2}ixy - xyxy = \frac{1}{2}I_{\mathcal{F}} + ixy = p. \tag{404}$$

Noting that

$$x_j = \frac{1}{\sqrt{2}}\left(a_j^\dagger + a_j\right) = x_j^\dagger \tag{405}$$

$$x_{j+r} = \frac{i}{\sqrt{2}}\left(a_j^\dagger - a_j\right) = x_{j+r}^\dagger \tag{406}$$

one can see that in particular

$$p_{12} = \left(\frac{1}{2}I_{\mathcal{F}} + ix_1x_2\right) = \frac{1}{2}\left(I_{\mathcal{F}} + i\left(a_1^\dagger a_2^\dagger + a_1a_2^\dagger + a_1^\dagger a_2 + a_1a_2\right)\right). \tag{407}$$

One can also note that

$$p = \frac{1}{\sqrt{2}}\left(I_{\mathcal{F}} + \sqrt{2}x\right) \tag{408}$$

are also idempotents when

$$x^2 = \frac{1}{2}I_{\mathcal{F}}. \tag{409}$$

VI. TWO PARTICLE OPERATORS

One can first consider two particle operators of

7. Decomposable Diagonal form

This type of particle operators can be expanded as

$$\begin{aligned}
\lambda_2 &= \frac{1}{2} \sum_{1 \leq j_1, j_2 \leq r} \lambda_{2 j_1 j_2 j_1 j_2} a_{j_1}^{\dagger} a_{j_2}^{\dagger} a_{j_2} a_{j_1} \\
&= \frac{1}{8} \sum_{1 \leq j_1, j_2 \leq r} \lambda_{2 j_1 j_2 j_1 j_2} \left(x_{j_1} - i x_{j_1+r}\right)\left(x_{j_2} - i x_{j_2+r}\right)\left(x_{j_2} + i x_{j_2+r}\right)\left(x_{j_1} + i x_{j_1+r}\right) \\
&= \frac{1}{8} \sum_{1 \leq j_1, j_2 \leq r} \lambda_{2 j_1 j_2 j_1 j_2} \left(x_{j_1} x_{j_2} - i x_{j_1+r} x_{j_2} - i x_{j_1} x_{j_2+r} - x_{j_1+r} x_{j_2+r}\right) \times \\
&\left(x_{j_2} x_{j_1} + i x_{j_2+r} x_{j_1} + i x_{j_2} x_{j_1+r} - x_{j_2+r} x_{j_1+r}\right) \\
&= \frac{1}{8} \sum_{1 \leq j_1, j_2 \leq r} \lambda_{2 j_1 j_2 j_1 j_2} \{ \\
&+ x_{j_1} x_{j_2} x_{j_2} x_{j_1} - i x_{j_1+r} x_{j_2} x_{j_2} x_{j_1} - i x_{j_1} x_{j_2+r} x_{j_2} x_{j_1} - x_{j_1+r} x_{j_2+r} x_{j_2} x_{j_1} \\
&+ i x_{j_1} x_{j_2} x_{j_2+r} x_{j_1} + x_{j_1+r} x_{j_2} x_{j_2+r} x_{j_1} + x_{j_1} x_{j_2+r} x_{j_2+r} x_{j_1} - i x_{j_1+r} x_{j_2+r} x_{j_2+r} x_{j_1} \\
&+ i x_{j_1} x_{j_2} x_{j_2} x_{j_1+r} + x_{j_1+r} x_{j_2} x_{j_2} x_{j_1+r} + x_{j_1} x_{j_2+r} x_{j_2} x_{j_1+r} - i x_{j_1+r} x_{j_2+r} x_{j_2} x_{j_1+r} \\
&- x_{j_1} x_{j_2} x_{j_2+r} x_{j_1+r} + i x_{j_1+r} x_{j_2} x_{j_2+r} x_{j_1+r} + i x_{j_1} x_{j_2+r} x_{j_2+r} x_{j_1+r} + x_{j_1+r} x_{j_2+r} x_{j_2+r} x_{j_1+r} \}
\end{aligned} \tag{410}$$

The zeroth degree terms are

$$\begin{aligned}
&\frac{1}{8} \sum_{1 \leq j_1, j_2 \leq r} \lambda_{2 j_1 j_2 j_1 j_2} \left\{x_{j_1} x_{j_2} x_{j_2} x_{j_1} + x_{j_1} x_{j_2+r} x_{j_2+r} x_{j_1} + x_{j_1+r} x_{j_2} x_{j_2} x_{j_1+r} + x_{j_1+r} x_{j_2+r} x_{j_2+r} x_{j_1+r}\right\} \\
&= \frac{1}{4} \sum_{1 \leq j_1 < j_2 \leq r} \lambda_{2 j_1 j_2 j_1 j_2} I_{\mathcal{F}} = \frac{1}{4} \operatorname{Tr}\left\{\lambda_2\right\} I_{\mathcal{F}}
\end{aligned} \tag{411}$$

The second degree terms are

$$\begin{aligned}
&= \frac{1}{8} \sum_{1 \leq j_1, j_2 \leq r} \lambda_{2 j_1 j_2 j_1 j_2} \{ - i x_{j_1+r} x_{j_2} x_{j_2} x_{j_1} - i x_{j_1} x_{j_2+r} x_{j_2} x_{j_1} \\
&+ i x_{j_1} x_{j_2} x_{j_2+r} x_{j_1} - i x_{j_1+r} x_{j_2+r} x_{j_2+r} x_{j_1} + i x_{j_1} x_{j_2} x_{j_2} x_{j_1+r} \\
&- i x_{j_1+r} x_{j_2} x_{j_2+r} x_{j_1+r} - i x_{j_1+r} x_{j_2+r} x_{j_2} x_{j_1+r} + i x_{j_1} x_{j_2+r} x_{j_2+r} x_{j_1+r} \}
\end{aligned}$$

$$= \frac{1}{16} \sum_{1\le j_1,j_2\le r} \lambda_{2j_1j_2j_1j_2} \{-ix_{j_1+r}x_{j_1} - ix_{j_2+r}x_{j_2}$$

$$+ ix_{j_2}x_{j_2+r} - ix_{j_1+r}x_{j_1} + ix_{j_1}x_{j_1+r} - ix_{j_2+r}x_{j_2} - ix_{j_2+r}x_{j_2} + ix_{j_1}x_{j_1+r}\}$$

$$= \frac{i}{2} \sum_{1\le j_1,j_2\le r} \lambda_{2j_1j_2j_1j_2} x_{j_1}x_{j_1+r} = \frac{i}{2} \sum_{1\le j_1\le r} L_2^1(\lambda_2)_{j_1j_1} x_{j_1}x_{j_1+r}. \tag{412}$$

The fourth degree terms are

$$\frac{1}{8} \sum_{1\le j_1,j_2\le r} \lambda_{2j_1j_2j_1j_2} \{x_{j_1}x_{j_2}x_{j_1+r}x_{j_2+r} + x_{j_1}x_{j_2}x_{j_1+r}x_{j_2+r} + x_{j_1}x_{j_2}x_{j_1+r}x_{j_2+r} + x_{j_1}x_{j_2}x_{j_1+r}x_{j_2+r}\}$$

$$= \frac{1}{2} \sum_{1\le j_1,j_2\le r} \lambda_{2j_1j_2j_1j_2} x_{j_1}x_{j_2}x_{j_1+r}x_{j_2+r} = \sum_{1\le j_1<j_2\le r} \lambda_{2j_1j_2j_1j_2} x_{j_1}x_{j_2}x_{j_1+r}x_{j_2+r} \tag{413}$$

Hence

$$\lambda_2 = \frac{1}{4} \operatorname{Tr}\{\lambda_2\} I_{\mathcal{F}} + \frac{i}{2} \sum_{1\le j_1\le r} L_2^1(\lambda_2)_{j_1j_1} x_{j_1}x_{j_1+r} + \sum_{1\le j_1<j_2\le r} \lambda_{2j_1j_2j_1j_2} x_{j_1}x_{j_2}x_{j_1+r}x_{j_2+r}$$

$$= \frac{1}{4} L_2^0(\lambda_2) I_{\mathcal{F}} + \frac{i}{2} \sum_{1\le j_1\le r} L_2^1(\lambda_2)_{j_1j_1} x_{j_1}x_{j_1+r} + \sum_{1\le j_1<j_2\le r} L_2^2(\lambda_2)_{j_1j_2j_1j_2} x_{j_1}x_{j_2}x_{j_1+r}x_{j_2+r} \tag{414}$$

8. *General form*

General two particle operators can be expanded as

$$\lambda_2 = \frac{1}{4} \sum_{1\le j_1,j_2,j_3,j_4\le r} \lambda_{2j_1j_2j_3j_4} a_{j_1}^\dagger a_{j_2}^\dagger a_{j_4} a_{j_3}$$

$$= \frac{1}{16} \sum_{1\le j_1,j_2,j_3,j_4\le r} \lambda_{2j_1j_2j_3j_4} (x_{j_1} - ix_{j_1+r})(x_{j_2} - ix_{j_2+r})(x_{j_4} + ix_{j_4+r})(x_{j_3} + ix_{j_3+r})$$

$$= \frac{1}{16} \sum_{1\le j_1,j_2,j_3,j_4\le r} \lambda_{2j_1j_2j_3j_4} (x_{j_1}x_{j_2} - ix_{j_1+r}x_{j_2} - ix_{j_1}x_{j_2+r} - x_{j_1+r}x_{j_2+r})$$

$$\times (x_{j_4}x_{j_3} + ix_{j_4+r}x_{j_3} + ix_{j_4}x_{j_3+r} - x_{j_4+r}x_{j_3+r}) \tag{415}$$

$$= \frac{1}{16} \sum_{1\le j_1,j_2,j_3,j_4\le r} \lambda_{2j_1j_2j_3j_4} \{x_{j_1}x_{j_2}x_{j_4}x_{j_3} - ix_{j_1+r}x_{j_2}x_{j_4}x_{j_3} - ix_{j_1}x_{j_2+r}x_{j_4}x_{j_3} - x_{j_1+r}x_{j_2+r}x_{j_4}x_{j_3}$$

$$+ ix_{j_1}x_{j_2}x_{j_4+r}x_{j_3} + x_{j_1+r}x_{j_2}x_{j_4+r}x_{j_3} + x_{j_1}x_{j_2+r}x_{j_4+r}x_{j_3} - ix_{j_1+r}x_{j_2+r}x_{j_4+r}x_{j_3}$$

$$+ ix_{j_1}x_{j_2}x_{j_4}x_{j_3+r} + x_{j_1+r}x_{j_2}x_{j_4}x_{j_3+r} + x_{j_1}x_{j_2+r}x_{j_4}x_{j_3+r} - ix_{j_1+r}x_{j_2+r}x_{j_4}x_{j_3+r}$$

$$-x_{j_1}x_{j_2}x_{j_4+r}x_{j_3+r} + ix_{j_1+r}x_{j_2}x_{j_4+r}x_{j_3+r} + ix_{j_1}x_{j_2+r}x_{j_4+r}x_{j_3+r} + x_{j_1+r}x_{j_2+r}x_{j_4+r}x_{j_3+r}\}. \tag{416}$$

Let $r = 4$ then

$$\lambda_2 = \sum_{1\le j_1<j_2\le 4;1\le,j_3<j_4\le 4} \lambda_{2j_1j_2j_3j_4} a^\dagger_{j_1} a^\dagger_{j_2} a_{j_4} a_{j_3}$$
$$= \frac{1}{4} \sum_{1\le j_1<j_2\le 4;1\le,j_3<j_4\le 4} \lambda_{2j_1j_2j_3j_4} (x_{j_1} - ix_{j_1+r})(x_{j_2} - ix_{j_2+r})(x_{j_4} + ix_{j_4+r})(x_{j_3} + ix_{j_3+r})$$

$$= \frac{1}{4} \sum_{1\le j_1<j_2\le 4;1\le,j_3<j_4\le 4} \lambda_{2j_1j_2j_3j_4} \{x_{j_1}x_{j_2}x_{j_4}x_{j_3} - ix_{j_1+r}x_{j_2}x_{j_4}x_{j_3} - ix_{j_1}x_{j_2+r}x_{j_4}x_{j_3} - x_{j_1+r}x_{j_2+r}x_{j_4}x_{j_3}$$
$$+ ix_{j_1}x_{j_2}x_{j_4+r}x_{j_3} + x_{j_1+r}x_{j_2}x_{j_4+r}x_{j_3} + x_{j_1}x_{j_2+r}x_{j_4+r}x_{j_3} - ix_{j_1+r}x_{j_2+r}x_{j_4+r}x_{j_3}$$
$$+ ix_{j_1}x_{j_2}x_{j_4}x_{j_3+r} + x_{j_1+r}x_{j_2}x_{j_4}x_{j_3+r} + x_{j_1}x_{j_2+r}x_{j_4}x_{j_3+r} - ix_{j_1+r}x_{j_2+r}x_{j_4}x_{j_3+r}$$
$$-x_{j_1}x_{j_2}x_{j_4+r}x_{j_3+r} + ix_{j_1+r}x_{j_2}x_{j_4+r}x_{j_3+r} + ix_{j_1}x_{j_2+r}x_{j_4+r}x_{j_3+r} + x_{j_1+r}x_{j_2+r}x_{j_4+r}x_{j_3+r}\}. \tag{417}$$

⌈Zero differences

$$= \frac{1}{4}\lambda_{21212} \{x_1x_2x_2x_1 - ix_5x_2x_2x_1 - ix_1x_6x_2x_1 - x_5x_6x_2x_1$$
$$+ ix_1x_2x_6x_1 + x_5x_2x_6x_1 + x_1x_6x_6x_1 - ix_5x_6x_6x_1$$
$$+ ix_1x_2x_2x_5 + x_5x_2x_2x_5 + x_1x_6x_2x_5 - ix_5x_6x_2x_5$$
$$-x_1x_2x_6x_5 + ix_5x_2x_6x_5 + ix_1x_6x_6x_5 + x_5x_6x_6x_5\}. \tag{418}$$

$$= \frac{1}{4}\lambda_{21212} \left\{ \frac{1}{4}I_{\mathcal{F}} + i\frac{1}{2}x_1x_5 + i\frac{1}{2}x_2x_6 + x_5x_2x_6x_1 \right.$$
$$+ i\frac{1}{2}x_2x_6 + x_5x_2x_6x_1 + \frac{1}{4}I_{\mathcal{F}} + i\frac{1}{2}x_1x_5$$
$$+ i\frac{1}{2}x_1x_5 + \frac{1}{4}I_{\mathcal{F}} + x_5x_2x_6x_1 + i\frac{1}{2}x_2x_6$$
$$\left. + x_5x_2x_1x_6 + i\frac{1}{2}x_2x_6 + i\frac{1}{2}x_1x_5 + \frac{1}{4}I_{\mathcal{F}} \right\}. \tag{419}$$

$$= \lambda_{21212} \left\{ \frac{1}{4}I_{\mathcal{F}} + \frac{i}{2}x_1x_5 + \frac{i}{2}ix_2x_6 + x_1x_2x_5x_6 \right\} \tag{420}$$

One difference

$$
\begin{aligned}
&= \frac{1}{4}\lambda_{21213}\left\{x_1x_2x_3x_1 - ix_5x_2x_3x_1 - ix_1x_6x_3x_1 - x_5x_6x_3x_1\right.\\
&+ ix_1x_2x_7x_1 + x_5x_2x_7x_1 + x_1x_6x_7x_1 - ix_5x_6x_7x_1\\
&+ ix_1x_2x_3x_5 + x_5x_2x_3x_5 + x_1x_6x_3x_5 - ix_5x_6x_3x_5\\
&\left.-x_1x_2x_7x_5 + ix_5x_2x_7x_5 + ix_1x_6x_7x_5 + x_5x_6x_7x_5\right\}.
\end{aligned} \tag{421}
$$

$$
\begin{aligned}
&= \frac{1}{4}\lambda_{21213}\left\{\frac{1}{2}x_2x_3 - ix_5x_2x_3x_1 - i\frac{1}{2}x_6x_3 - x_5x_6x_3x_1\right.\\
&+ i\frac{1}{2}x_2x_7 + x_5x_2x_7x_1 + \frac{1}{2}x_6x_7 - ix_5x_6x_7x_1\\
&+ ix_1x_2x_3x_5 + \frac{1}{2}x_2x_3 + x_1x_6x_3x_5 - i\frac{1}{2}x_6x_3\\
&\left.-x_1x_2x_7x_5 + i\frac{1}{2}x_2x_7 + ix_1x_6x_7x_5 + \frac{1}{2}x_6x_7\right\}.
\end{aligned} \tag{422}
$$

$$
\begin{aligned}
&= \frac{1}{4}\lambda_{21213}\left\{x_2x_3 + x_6x_7 + 2x_1x_3x_5x_6 + 2x_1x_2x_5x_7 + ix_3x_6\right.\\
&\left.+ ix_2x_7 - ix_1x_2x_3x_5 - ix_1x_5x_6x_7 + ix_1x_2x_3x_5 + ix_1x_5x_6x_7\right\}.
\end{aligned}
$$

Two differences

$$
\begin{aligned}
&= \frac{1}{4}\lambda_{21243}\left\{x_1x_2x_3x_4 - ix_5x_2x_3x_4 - ix_1x_6x_3x_4 - x_5x_6x_3x_4\right.\\
&+ ix_1x_2x_8x_3 + x_5x_2x_8x_3 + x_1x_6x_8x_3 - ix_5x_6x_8x_3\\
&+ ix_1x_2x_4x_8 + x_5x_2x_4x_8 + x_1x_6x_4x_8 - ix_5x_6x_4x_8\\
&\left.-x_1x_2x_7x_8 + ix_5x_2x_7x_8 + ix_1x_6x_7x_8 + x_5x_6x_7x_8\right\}.
\end{aligned} \tag{423}
$$

$$
\begin{aligned}
&= \frac{1}{4}\lambda_{21243}\left\{x_1x_2x_3x_4 - x_3x_4x_5x_6 - x_2x_3x_5x_8 + x_1x_3x_6x_8\right.\\
&+ x_2x_4x_5x_8 - x_1x_4x_6x_8 - x_1x_2x_7x_8 + x_5x_6x_7x_8\\
&+ ix_2x_3x_4x_5 - ix_1x_3x_4x_6 - ix_1x_2x_3x_8 + ix_3x_5x_6x_8\\
&\left.+ ix_1x_2x_4x_8 - ix_4x_5x_6x_8 - ix_2x_5x_7x_8 + ix_1x_6x_7x_8\right\}.
\end{aligned} \tag{424}
$$

$\rfloor$

The zeroth degree terms are

$$\frac{1}{16}\sum_{1\leq j_1,j_2\leq r}\lambda_{2j_1j_2j_1j_2}\left\{x_{j_1}x_{j_2}x_{j_2}x_{j_1}+x_{j_1}x_{j_2+r}x_{j_2+r}x_{j_1}+x_{j_1+r}x_{j_2}x_{j_2}x_{j_1+r}+x_{j_1+r}x_{j_2+r}x_{j_2+r}x_{j_1+r}\right\}$$
$$=\frac{1}{4}\sum_{1\leq j_1<j_2\leq r}\lambda_{2j_1j_2j_1j_2}I_{\mathcal{F}}=\frac{1}{4}\operatorname{Tr}\left\{\lambda_2\right\}I_{\mathcal{F}}. \tag{425}$$

The second degree terms are after setting $j_3=j_1, j_3=j_2, j_4=j_1$, and $j_4=j_2$, then $j_4=j_3$ and interchanging j_1 and j_2 (in the following sums we can consider (j_2, j_2+r) but not (j_2, j_2))

$$\frac{1}{16}\sum_{1\leq j_1,j_2\neq j_3\leq r}\lambda_{2j_1j_2j_1j_3}\left\{x_{j_1}x_{j_2}x_{j_3}x_{j_1}+x_{j_1}x_{j_2}x_{j_3}x_{j_1}-ix_{j_1}x_{j_2+r}x_{j_3}x_{j_1}\right.$$
$$+ix_{j_1}x_{j_2}x_{j_3+r}x_{j_1}+x_{j_1}x_{j_2+r}x_{j_3+r}x_{j_1}-ix_{j_1+r}x_{j_2+r}x_{j_3}x_{j_1+r}$$
$$\left.+ix_{j_1+r}x_{j_2}x_{j_3+r}x_{j_1+r}+x_{j_1+r}x_{j_2+r}x_{j_3+r}x_{j_1+r}\right\}$$

$$-\frac{1}{16}\sum_{1\leq j_1,j_2\neq j_3\leq r}\lambda_{2j_1j_2j_1j_3}\left\{x_{j_1}x_{j_2}x_{j_1}x_{j_3}-ix_{j_1}x_{j_2+r}x_{j_1}x_{j_3}\right.$$
$$+x_{j_1+r}x_{j_2}x_{j_1+r}x_{j_3}-ix_{j_1+r}x_{j_2+r}x_{j_1+r}x_{j_3}$$
$$+ix_{j_1}x_{j_2}x_{j_1}x_{j_3+r}+x_{j_1}x_{j_2+r}x_{j_1}x_{j_3+r}$$
$$\left.+ix_{j_1+r}x_{j_2}x_{j_1+r}x_{j_3+r}+x_{j_1+r}x_{j_2+r}x_{j_1+r}x_{j_3+r}\right\}$$

$$-\frac{1}{16}\sum_{1\leq j_1,j_2\neq j_3\leq r}\lambda_{2j_1j_2j_1j_3}\left\{x_{j_2}x_{j_1}x_{j_3}x_{j_2}-ix_{j_2+r}x_{j_1}x_{j_3}x_{j_1}\right.$$
$$+ix_{j_1}x_{j_1}x_{j_3+r}x_{j_1}+x_{j_2+r}x_{j_1}x_{j_3+r}x_{j_1}$$
$$+x_{j_2}x_{j_1+r}x_{j_3}x_{j_1+r}-ix_{j_2+r}x_{j_1+r}x_{j_3}x_{j_1+r}$$
$$\left.+ix_{j_2}x_{j_1+r}x_{j_3+r}x_{j_1+r}+x_{j_2+r}x_{j_1+r}x_{j_3+r}x_{j_1+r}\right\}$$

$$+\frac{1}{16}\sum_{1\leq j_1,j_2\neq j_3\leq r}\lambda_{2j_1j_2j_1j_3}\left\{x_{j_2}x_{j_1}x_{j_1}x_{j_3}-ix_{j_2+r}x_{j_1}x_{j_1}x_{j_3}\right.$$
$$+x_{j_2}x_{j_1+r}x_{j_1+r}x_{j_3}-ix_{j_2+r}x_{j_1+r}x_{j_1+r}x_{j_3}$$
$$+ix_{j_2}x_{j_1}x_{j_1}x_{j_3+r}+x_{j_2+r}x_{j_1}x_{j_1}x_{j_3+r}$$
$$\left.+ix_{j_2}x_{j_1+r}x_{j_1+r}x_{j_3+r}+x_{j_2+r}x_{j_1+r}x_{j_1+r}x_{j_3+r}\right\}$$

$$=\frac{1}{32}\sum_{1\le j_1,j_2\neq j_3\le r}\lambda_{2j_1j_2j_1j_3}\left\{x_{j_2}x_{j_3}-ix_{j_2+r}x_{j_3}\right.$$

$$\left.+ix_{j_2}x_{j_3+r}+x_{j_2+r}x_{j_3+r}-ix_{j_2+r}x_{j_3}+ix_{j_2}x_{j_3+r}+x_{j_2+r}x_{j_3+r}\right\}$$

$$+\frac{1}{32}\sum_{1\le j_1,j_2\neq j_3\le r}\lambda_{2j_1j_2j_1j_3}\left\{x_{j_2}x_{j_3}-ix_{j_2+r}x_{j_3}\right.$$

$$\left.+x_{j_2}x_{j_3}-ix_{j_2+r}x_{j_3}+ix_{j_2}x_{j_3+r}+x_{j_2+r}x_{j_3+r}+ix_{j_2}x_{j_3+r}+x_{j_2+r}x_{j_3+r}\right\}$$

$$+\frac{1}{32}\sum_{1\le j_1,j_2\neq j_3\le r}\left\{x_{j_2}x_{j_3}-ix_{j_2+r}x_{j_3}\right.$$

$$\left.+ix_{j_1}x_{j_3+r}+x_{j_2+r}x_{j_3+r}+x_{j_2}x_{j_3}-ix_{j_2+r}x_{j_3}+ix_{j_2}x_{j_3+r}+x_{j_2+r}x_{j_3+r}\right\}$$

$$+\frac{1}{32}\sum_{1\le j_1,j_2\neq j_3\le r}\left\{x_{j_2}x_{j_3}-ix_{j_2+r}x_{j_3}\right.$$

$$\left.+x_{j_2}x_{j_3}-ix_{j_2+r}x_{j_3}+ix_{j_2}x_{j_3+r}+x_{j_2+r}x_{j_3+r}+ix_{j_2}x_{j_3+r}+x_{j_2+r}x_{j_3+r}\right\}$$

$$=\frac{1}{4}\sum_{1\le j_2\neq j_3\le r}L_2^1\left(\lambda_2\right)_{j_2j_3}\left\{x_{j_2}x_{j_3}+x_{j_2+r}x_{j_3+r}+ix_{j_3}x_{j_2+r}+ix_{j_2}x_{j_3+r}\right\}$$

as the products are antisymmetric this gives that the second degree terms are

$$\frac{1}{2}\sum_{1\le j_2<j_3\le r}\left(L_2^1\left(\lambda_2\right)_{j_2j_3}-L_2^1\left(\lambda_2\right)_{j_3j_2}\right)\left\{x_{j_2}x_{j_3}+x_{j_2+r}x_{j_3+r}+i\left(x_{j_3}x_{j_2+r}+x_{j_2}x_{j_3+r}\right)\right\}. \quad (426)$$

$\lceil$

$$\frac{1}{2}\left(L_2^1\left(\lambda_2\right)_{12}-L_2^1\left(\lambda_2\right)_{21}\right)\left\{x_1x_2+x_3x_4+i\left(x_2x_3+x_2x_4\right)\right\} \quad (427)$$

$\rfloor$

The fourth degree terms are

$$\frac{1}{16}\sum_{1\le j_1\neq j_2\neq j_3\neq j_4\le r}\lambda_{2j_1j_2j_3j_4}\left\{x_{j_1}x_{j_2}x_{j_4}x_{j_3}-x_{j_1+r}x_{j_2+r}x_{j_4}x_{j_3}+x_{j_1+r}x_{j_2}x_{j_4+r}x_{j_3}+x_{j_1}x_{j_2+r}x_{j_4+r}x_{j_3}\right.$$

$$-x_{j_1}x_{j_2}x_{j_4+r}x_{j_3+r}+x_{j_1+r}x_{j_2}x_{j_4}x_{j_3+r}+x_{j_1}x_{j_2+r}x_{j_4}x_{j_3+r}+x_{j_1+r}x_{j_2+r}x_{j_4+r}x_{j_3+r}$$

$$+ix_{j_1}x_{j_2}x_{j_4+r}x_{j_3}-ix_{j_1+r}x_{j_2}x_{j_4}x_{j_3}-ix_{j_1}x_{j_2+r}x_{j_4}x_{j_3}$$

$$\left.+ix_{j_1}x_{j_2}x_{j_4}x_{j_3+r}-ix_{j_1+r}x_{j_2+r}x_{j_4}x_{j_3+r}+ix_{j_1}x_{j_2+r}x_{j_4+r}x_{j_3+r}-ix_{j_1+r}x_{j_2+r}x_{j_4+r}x_{j_3}+ix_{j_1+r}x_{j_2}x_{j_4+r}x_{j_3+r}\right\}.$$

Two particle operators are a special form of *even* fourth degree operators, which are fourth degree polynomials in the generators of the Clifford algebra

$$A_2 = \alpha_0 I_{\mathcal{F}} + \sum_{1 \le j_1 < j_2 \le 2r} \alpha_{2 j_1 j_2} x_{j_1} x_{j_2} + \sum_{1 \le j_1 < j_2 < j_3 < j_4 \le 2r} \alpha_{4 j_1 j_2 j_3 j_4} x_{j_1} x_{j_2} x_{j_3} x_{j_4}. \tag{428}$$

The square of such operators is

$$\begin{aligned}
(A_2)^2 &= (A_{20} + A_{22} + A_{24})^2 = (A_{20} + A_{22} + A_{24})(A_{20} + A_{22} + A_{24}) \\
&= A_{20}A_{20} + A_{20}A_{22} + A_{20}A_{24} + A_{22}A_{20} + A_{22}A_{22} + A_{22}A_{24} + A_{24}A_{20} + A_{24}A_{22} + A_{24}A_{24} \\
&= \alpha_0^2 I_{\mathcal{F}} + 2\alpha_0 A_{22} + 2\alpha_0 A_{24} + (A_{22})^2 + A_{22}A_{24} + A_{24}A_{22} + (A_{24})^2 .
\end{aligned} \tag{429}$$

$$(A_{20})^2 = \alpha_0^2 I_{\mathcal{F}} \tag{430}$$

$$A_{20}A_{22} = \alpha_0 A_{22} \tag{431}$$

$$A_{20}A_{24} = \alpha_0 A_{24} \tag{432}$$

$$A_{22}A_{22} = \sum_{\substack{1 \le j_1 < j_2 \le 2r \\ 1 \le j_3 < j_4 \le 2r}} \alpha_{2 j_1 j_2} \alpha_{2 j_3 j_4} x_{j_1} x_{j_2} x_{j_3} x_{j_4} \tag{433}$$

⌈In the case $r=2$ one has

$$\begin{aligned}
&\alpha_{212}\alpha_{212}x_1x_2x_1x_2+\alpha_{212}\alpha_{213}x_1x_2x_1x_3+\alpha_{212}\alpha_{214}x_1x_2x_1x_4\\
&+\alpha_{212}\alpha_{223}x_1x_2x_2x_3+\alpha_{212}\alpha_{224}x_1x_2x_2x_4+\alpha_{212}\alpha_{234}x_1x_2x_3x_4\\
&+\alpha_{213}\alpha_{212}x_1x_3x_1x_2+\alpha_{213}\alpha_{213}x_1x_3x_1x_3+\alpha_{213}\alpha_{214}x_1x_3x_1x_4\\
&+\alpha_{213}\alpha_{223}x_1x_3x_2x_3+\alpha_{213}\alpha_{224}x_1x_3x_2x_4+\alpha_{213}\alpha_{234}x_1x_3x_3x_4\\
&+\alpha_{214}\alpha_{212}x_1x_4x_1x_2+\alpha_{214}\alpha_{213}x_1x_4x_1x_3+\alpha_{214}\alpha_{214}x_1x_4x_1x_4\\
&+\alpha_{214}\alpha_{223}x_1x_4x_2x_3+\alpha_{214}\alpha_{224}x_1x_4x_2x_4+\alpha_{214}\alpha_{234}x_1x_4x_3x_4\\
&+\alpha_{223}\alpha_{212}x_2x_3x_1x_2+\alpha_{223}\alpha_{213}x_2x_3x_1x_3+\alpha_{223}\alpha_{214}x_2x_3x_1x_4\\
&+\alpha_{223}\alpha_{223}x_2x_3x_2x_3+\alpha_{223}\alpha_{224}x_2x_3x_2x_4+\alpha_{223}\alpha_{234}x_2x_3x_3x_4\\
&+\alpha_{224}\alpha_{212}x_2x_4x_1x_2+\alpha_{224}\alpha_{213}x_2x_4x_1x_3+\alpha_{224}\alpha_{214}x_2x_4x_1x_4\\
&+\alpha_{224}\alpha_{223}x_2x_4x_2x_3+\alpha_{224}\alpha_{224}x_2x_4x_2x_4+\alpha_{224}\alpha_{234}x_2x_4x_3x_4\\
&+\alpha_{234}\alpha_{212}x_3x_4x_1x_2+\alpha_{234}\alpha_{213}x_3x_4x_1x_3+\alpha_{234}\alpha_{214}x_3x_4x_1x_4\\
&+\alpha_{234}\alpha_{223}x_3x_4x_2x_3+\alpha_{234}\alpha_{224}x_3x_4x_2x_4+\alpha_{234}\alpha_{234}x_3x_4x_3x_4
\end{aligned}\tag{434}$$

$$\begin{aligned}
=&-\frac{1}{4}\alpha_{212}\alpha_{212}I_{\mathcal{F}}-\frac{1}{2}\alpha_{212}\alpha_{213}x_2x_3-\frac{1}{2}\alpha_{212}\alpha_{214}x_2x_4+\frac{1}{2}\alpha_{212}\alpha_{223}x_1x_3+\frac{1}{2}\alpha_{212}\alpha_{224}x_1x_4+\alpha_{212}\alpha_{234}x_1x_2x_3x_4\\
&-\frac{1}{2}\alpha_{213}\alpha_{212}x_3x_2-\alpha_{213}\alpha_{213}\frac{1}{4}I_{\mathcal{F}}-\frac{1}{2}\alpha_{213}\alpha_{214}x_3x_4-\frac{1}{2}\alpha_{213}\alpha_{223}x_1x_2-\alpha_{213}\alpha_{224}x_1x_2x_3x_4+\frac{1}{2}\alpha_{213}\alpha_{234}x_1x_4\\
&-\frac{1}{2}\alpha_{214}\alpha_{212}x_4x_2-\frac{1}{2}\alpha_{214}\alpha_{213}x_4x_3-\alpha_{214}\alpha_{214}\frac{1}{4}I_{\mathcal{F}}+\alpha_{214}\alpha_{223}x_1x_2x_3x_4-\frac{1}{2}\alpha_{214}\alpha_{224}x_1x_2-\frac{1}{2}\alpha_{214}\alpha_{234}x_1x_3\\
&+\frac{1}{2}\alpha_{223}\alpha_{212}x_3x_1-\frac{1}{2}\alpha_{223}\alpha_{213}x_2x_1+\alpha_{223}\alpha_{214}x_1x_2x_3x_4-\alpha_{223}\alpha_{223}\frac{1}{4}I_{\mathcal{F}}-\frac{1}{2}\alpha_{223}\alpha_{224}x_3x_4+\frac{1}{2}\alpha_{223}\alpha_{234}x_2x_4\\
&+\frac{1}{2}\alpha_{224}\alpha_{212}x_4x_1-\alpha_{224}\alpha_{213}x_1x_2x_3x_4-\frac{1}{2}\alpha_{224}\alpha_{214}x_2x_1-\frac{1}{2}\alpha_{224}\alpha_{223}x_4x_3-\alpha_{224}\alpha_{224}\frac{1}{4}I_{\mathcal{F}}-\alpha_{224}\alpha_{234}\frac{1}{2}x_2x_3\\
&+\alpha_{234}\alpha_{212}x_1x_2x_3x_4+\frac{1}{2}\alpha_{234}\alpha_{213}x_4x_1-\frac{1}{2}\alpha_{234}\alpha_{214}x_3x_1+\frac{1}{2}\alpha_{234}\alpha_{223}x_4x_2-\frac{1}{2}\alpha_{234}\alpha_{224}x_3x_2-\alpha_{234}\alpha_{234}\frac{1}{4}I_{\mathcal{F}}
\end{aligned}\tag{435}$$

$$
\begin{aligned}
& -\frac{1}{4} I_{\mathcal{F}} \left\{ \alpha_{212}\alpha_{212} + \alpha_{213}\alpha_{213} + \alpha_{214}\alpha_{214} + \alpha_{223}\alpha_{223} + \alpha_{224}\alpha_{224} + \alpha_{234}\alpha_{234} \right\} \\
& + \frac{1}{2} \left\{ -\alpha_{212}\alpha_{213} + \alpha_{213}\alpha_{212} - \alpha_{224}\alpha_{234} + \alpha_{234}\alpha_{224} \right\} x_2 x_3 \\
& + \frac{1}{2} \left\{ -\alpha_{212}\alpha_{214} + \alpha_{212}\alpha_{214} + \alpha_{223}\alpha_{234} - \alpha_{234}\alpha_{223} \right\} x_2 x_4 \\
& + \frac{1}{2} \left\{ \alpha_{212}\alpha_{223} - \alpha_{214}\alpha_{234} - \alpha_{223}\alpha_{212} + \alpha_{234}\alpha_{214} \right\} x_1 x_3 \\
& + \frac{1}{2} \left\{ \alpha_{212}\alpha_{224} - \alpha_{224}\alpha_{212} - \alpha_{234}\alpha_{213} + \alpha_{213}\alpha_{234} \right\} x_1 x_4 \\
& + \frac{1}{2} \left\{ -\alpha_{213}\alpha_{214} + \alpha_{214}\alpha_{213} - \alpha_{223}\alpha_{224} + \alpha_{224}\alpha_{223} \right\} x_3 x_4 \\
& + \frac{1}{2} \left\{ -\alpha_{213}\alpha_{223} - \alpha_{214}\alpha_{224} + \alpha_{223}\alpha_{213} + \alpha_{224}\alpha_{214} \right\} x_1 x_2 \\
& + 2 \left\{ \alpha_{212}\alpha_{234} - \alpha_{213}\alpha_{224} + \alpha_{214}\alpha_{223} \right\} x_1 x_2 x_3 x_4 \qquad (436)
\end{aligned}
$$

$$
\begin{aligned}
&= -\frac{1}{4} \left(\sum_{1 \le j_1 < j_2 \le 2r} \alpha_{2 j_1 j_2}^2 \right) I_{\mathcal{F}} + 2 \operatorname{Pf}\left\{ \alpha_2 \right\} x_1 x_2 x_3 x_4 \\
&= \frac{1}{8} \operatorname{Tr}\left\{ \alpha_2^t \alpha_2 \right\} I_{\mathcal{F}} + 2 \operatorname{Pf}\left\{ \alpha_2 \right\} x_1 x_2 x_3 x_4. \qquad (437)
\end{aligned}
$$

One should observe

$$
2 \operatorname{Pf}\left\{ \alpha_2 \right\} x_1 x_2 x_3 x_4 = \alpha_2 \wedge \alpha_2, \qquad (438)
$$

however

$$
\alpha_2 \wedge \alpha_2 \neq \alpha_2 \wedge \alpha_2 = \mathcal{C}_2\left(\alpha_2 \right). \qquad (439)
$$

$\rfloor$

$$
\begin{aligned}
A_{22} A_{22} &= \frac{1}{8} \operatorname{Tr}\left\{ \alpha_2^t \alpha_2 \right\} I_{\mathcal{F}} + 2 \sum_{\substack{1 \le j_1 < j_2 \le 2r \\ 1 \le j_3 < j_4 \le 2r}} \operatorname{Pf}\left\{ \alpha_2 \right\}_{j_1 j_2 j_3 j_4} x_{j_1} x_{j_2} x_{j_3} x_{j_4} \qquad (440) \\
&= \frac{1}{8} \operatorname{Tr}\left\{ \alpha_2^t \alpha_2 \right\} I_{\mathcal{F}} + \alpha_2 \wedge \alpha_2. \qquad (441)
\end{aligned}
$$

$$
A_{22} A_{24} = A_{24} A_{22} = \sum_{\substack{1 \le j_1 < j_2 \le 2r \\ 1 \le j_3 < j_4 < j_5 < j_6 \le 2r}} \alpha_{2 j_1 j_2} \alpha_{4 j_5 j_6 j_7 j_8} x_{j_1} x_{j_2} x_{j_3} x_{j_4} x_{j_5} x_{j_6} \qquad (442)
$$

$$121234 : -\frac{1}{4}\alpha_{212}\alpha_{41234}x_3x_4 \tag{443}$$
$$121235 : -\frac{1}{4}\alpha_{212}\alpha_{41235}x_3x_5 \tag{444}$$
$$121236 : -\frac{1}{4}\alpha_{212}\alpha_{41236}x_3x_6 \tag{445}$$
$$121245 : -\frac{1}{4}\alpha_{212}\alpha_{41234}x_4x_5 \tag{446}$$
$$121246 : -\frac{1}{4}\alpha_{212}\alpha_{41246}x_4x_6 \tag{447}$$
$$121256 : -\frac{1}{4}\alpha_{212}\alpha_{41256}x_5x_6; 131356 : -\frac{1}{4}\alpha_{213}\alpha_{41356}x_5x_6; 141456 : -\frac{1}{4}\alpha_{214}\alpha_{41456}x_5x_6; 232356 : -\frac{1}{4}\alpha_{223}\alpha_{42356}x \tag{448}$$
$$122456 : -\frac{1}{4}\alpha_{224}\alpha_{42456}x_5x_6; 343456 : -\frac{1}{4}\alpha_{234}\alpha_{43456}x_5x_6 \tag{449}$$

$$121345 : -\frac{1}{2}\alpha_{212}\alpha_{41345}x_2x_3x_4x_5 \tag{450}$$
$$121346 : \tag{451}$$
$$121356 : \tag{452}$$
$$121456 : \tag{453}$$
$$122345 : \tag{454}$$
$$122346 : \tag{455}$$
$$122356 : \tag{456}$$
$$122456 : \tag{457}$$
$$123456 : \tag{458}$$

$$A_{22}A_{24} = A_{24}A_{22} = \left(\alpha_{212}\alpha_{43456} - \alpha_{213}\alpha_{42456} + \alpha_{214}\alpha_{43256}\right)x_1x_2x_3x_4x_5x_6 \tag{459}$$

$$A_{24}A_{24} = \sum_{\substack{1\le j_1<j_2<j_3<j_4\le 2r \\ 1\le j_5<j_6<j_7<j_8\le 2r}} \alpha_{4j_1j_2j_3j_4}\alpha_{4j_5j_6j_7j_8}x_{j_1}x_{j_2}x_{j_3}x_{j_4}x_{j_5}x_{j_6}x_{j_7}x_{j_8} \tag{460}$$

$$A_{24}A_{24} = \alpha_4\wedge\alpha_4 + \frac{1}{2}L_3^4\left(\alpha_4\wedge\alpha_4\right) - \frac{1}{4}L_2^4\left(\alpha_4\wedge\alpha_4\right) + \frac{1}{8}L_1^4\left(\alpha_4\wedge\alpha_4\right) - \frac{1}{16}L_0^4\left(\alpha_4\wedge\alpha_4\right)I_{\mathcal{F}} \tag{461}$$

A. Grassmann Powers

The identity

$$A \wedge B = (-1)^{pq}\, B \wedge A, \tag{462}$$

where

$$A = \langle A \rangle_p \,, B = \langle B \rangle_q \tag{463}$$

gives

$$A \wedge A = (-1)^{p^2}\, A \wedge A \tag{464}$$

thus if p^2 is odd then

$$A \wedge A = 0, \tag{465}$$

i.e.

$$A \wedge A = 0; p = 1, 3, 5, \cdots \tag{466}$$

1. Let $p = 2$ then

$$A \wedge A = \sum_{1 \le j_1 < j_2 < j_3 < j_4 \le 2r} 2\left(A_{j_1 j_2} A_{j_3 j_4} - A_{j_1 j_3} A_{j_2 j_4} + A_{j_1 j_4} A_{j_2 j_3}\right) x_{j_1} x_{j_2} x_{j_3} x_{j_4} \tag{467}$$

$$= \sum_{1 \le j_1 < j_2 < j_3 < j_4 \le 2r} 2\,\mathrm{Pf}\left(\mathbb{A}_2\left(j_1 j_2 j_3 j_4\right)\right) x_{j_1} x_{j_2} x_{j_3} x_{j_4}, \tag{468}$$

where $\mathbb{A}_2\left(j_1 j_2 j_3 j_4\right)$ is the antisymmetric matrix

$$\begin{pmatrix} 0 & A_{j_1 j_2} & A_{j_1 j_3} & A_{j_1 j_4} \\ -A_{j_1 j_2} & 0 & A_{j_2 j_3} & A_{j_2 j_4} \\ -A_{j_1 j_3} & -A_{j_2 j_3} & 0 & A_{j_3 j_4} \\ -A_{j_1 j_4} & -A_{j_2 j_4} & -A_{j_3 j_4} & 0 \end{pmatrix} \tag{469}$$

and

$$\mathrm{Pf}\left(\mathbb{A}_2\left(j_1 j_2 j_3 j_4\right)\right) = A_{j_1 j_2} A_{j_3 j_4} - A_{j_1 j_3} A_{j_2 j_4} + A_{j_1 j_4} A_{j_2 j_3}. \tag{470}$$

2. Let $p = 4$ then

$$A \wedge A = \sum_{1 \le j_1 < j_2 < j_3 < j_4 < j_5 < j_6 < j_7 < j_8 \le 2r} 2\left(A_{j_1 j_2} A_{j_3 j_4} - A_{j_1 j_3} A_{j_2 j_4} + A_{j_1 j_4} A_{j_2 j_3}\right) x_{j_1} x_{j_2} x_{j_3} x_{j_4} x_{j_5} x_{j_6} x_{j_7} x_{j_8} \tag{471}$$

$$= \sum_{1 \le j_1 < j_2 < j_3 < j_4 < j_5 < j_6 < j_7 < j_8 \le 2r} 2\,\mathrm{Pf}\left(\mathbb{A}_2\left(j_1 j_2 j_3 j_4\right)\right) x_{j_1} x_{j_2} x_{j_3} x_{j_4} x_{j_5} x_{j_6} x_{j_7} x_{j_8}, \tag{472}$$

$$\operatorname{Pf}(a_1,\cdots,a_{2n}) = \sum_{j=2}^{2n} \operatorname{Pf}(a_1,a_j)(-1)^j \operatorname{Pf}(a_2,\cdots,\hat{a}_j,\cdots,a_{2n}) \tag{473}$$

One can consider the special case $r=4$ the subscripts that have no values in common are

$$\binom{8}{4}\binom{8-4}{4} = \binom{8}{4} = \frac{8*7*6*5}{1*2*3*4} = 70 \tag{474}$$

e.g.

$$12345678, 12354678, 12364578, 12374568, 12384567, \tag{475}$$

which gives the terms

$$\begin{gathered}
A_{1234}A_{5678}x_1 \wedge x_2 \wedge x_3 \wedge x_4 \wedge x_5 \wedge x_6 \wedge x_7 \wedge x_8 \\
A_{1235}A_{4678}x_1 \wedge x_2 \wedge x_3 \wedge x_5 \wedge x_4 \wedge x_6 \wedge x_7 \wedge x_8 \\
A_{1236}A_{4578}x_1 \wedge x_2 \wedge x_3 \wedge x_6 \wedge x_4 \wedge x_5 \wedge x_7 \wedge x_8 \\
A_{1237}A_{4568}x_1 \wedge x_2 \wedge x_3 \wedge x_7 \wedge x_4 \wedge x_5 \wedge x_6 \wedge x_8 \\
A_{1238}A_{4567}x_1 \wedge x_2 \wedge x_3 \wedge x_8 \wedge x_4 \wedge x_5 \wedge x_6 \wedge x_7
\end{gathered} \tag{476}$$

and the final coefficient is

$$\left(A_{1234}A_{5678} - A_{1235}A_{4678} + A_{1236}A_{4578} - A_{1237}A_{4568} + A_{1238}A_{4567}\right) x_1 \wedge x_2 \wedge x_3 \wedge x_4 \wedge x_5 \wedge x_6 \wedge x_7 \wedge x_8 \tag{477}$$

$$\begin{gathered}
A_{1234}A_{5678}x_1 \wedge x_2 \wedge x_3 \wedge x_4 \wedge x_5 \wedge x_6 \wedge x_7 \wedge x_8 \\
A_{1254}A_{3678}x_1 \wedge x_2 \wedge x_3 \wedge x_4 \wedge x_5 \wedge x_6 \wedge x_7 \wedge x_8 \\
A_{1264}A_{5378}x_1 \wedge x_2 \wedge x_3 \wedge x_4 \wedge x_5 \wedge x_6 \wedge x_7 \wedge x_8 \\
A_{1274}A_{5638}x_1 \wedge x_2 \wedge x_3 \wedge x_4 \wedge x_5 \wedge x_6 \wedge x_7 \wedge x_8 \\
A_{1284}A_{5673}x_1 \wedge x_2 \wedge x_3 \wedge x_4 \wedge x_5 \wedge x_6 \wedge x_7 \wedge x_8
\end{gathered} \tag{478}$$

e

$$\operatorname{Pf}(a_1,\cdots,a_4) = \sum_{j=2}^{4} \operatorname{Pf}(a_1,a_j)(-1)^j \operatorname{Pf}(a_2,\cdots,\hat{a}_j,\cdots,a_4) \tag{479}$$

$$= \operatorname{Pf}(a_1,a_2)(-1)^2 \operatorname{Pf}(a_3,a_4) + \operatorname{Pf}(a_1,a_3)(-1)^3 \operatorname{Pf}(a_2,a_4) \tag{480}$$

$$+ \operatorname{Pf}(a_1,a_4)(-1)^4 \operatorname{Pf}(a_2,a_3) \tag{481}$$

$$= \frac{1}{4}(a_1a_2 - a_2a_1)(a_3a_4 - a_4a_3) - \frac{1}{4}(a_1a_3 - a_3a_1)(a_2a_4 - a_4a_2) \tag{482}$$

$$+ \frac{1}{4}(a_1a_4 - a_4a_1)(a_2a_3 - a_3a_2) \tag{483}$$

VII. NOTES FROM CHISOLM ON GEOMETRIC ALGEBRA

$$AB = \sum_{0\le m,n\le 2r} A_m B_n \tag{484}$$

$$A_m B_n = \sum_{j=0}^{\min(m,n)} \langle A_m B_n \rangle_{|m-n|+2j} \tag{485}$$

So in particular

$$A^2 = \sum_{0\le m,n\le 2r} A_m A_n \tag{486}$$

$$A_m A_n = \sum_{j=0}^{\min(m,n)} \langle A_m A_n \rangle_{|m-n|+2j} \tag{487}$$

and for fourth order operators

$$\begin{aligned} A^2 &= A_0^2 + A_1^2 + A_2^2 + A_3^2 + A_4^2 \\ &\quad + A_0A_1 + A_0A_2 + A_0A_3 + A_0A_4 + A_1A_0 + A_2A_0 + A_3A_0 + A_4A_0 \\ &\quad + A_1A_2 + A_1A_3 + A_1A_4 + A_2A_1 + A_3A_1 + A_4A_1 \\ &\quad + A_2A_3 + A_2A_4 + A_3A_2 + A_4A_2 \\ &\quad + A_3A_4 + A_4A_3, \end{aligned} \tag{488}$$

where

1.

$$A_4^2 = \left\langle A_4^2 \right\rangle_0 + \left\langle A_4^2 \right\rangle_2 + \left\langle A_4^2 \right\rangle_4 + \left\langle A_4^2 \right\rangle_6 + \left\langle A_4^2 \right\rangle_8 \tag{489a}$$

$$A_3^2 = \left\langle A_3^2 \right\rangle_0 + \left\langle A_3^2 \right\rangle_2 + \left\langle A_3^2 \right\rangle_4 + \left\langle A_3^2 \right\rangle_6 \tag{489b}$$

$$A_2^2 = \left\langle A_2^2 \right\rangle_0 + \left\langle A_2^2 \right\rangle_2 + \left\langle A_2^2 \right\rangle_4 \tag{489c}$$

$$A_1^2 = \left\langle A_1^2 \right\rangle_0 + \left\langle A_1^2 \right\rangle_2 \tag{489d}$$

$$A_0^2 = \left\langle A_0^2 \right\rangle_0, \tag{489e}$$

thus

$$\begin{aligned} (A|\,A) &= \left\langle A_4^2 \right\rangle_0 + \left\langle A_3^2 \right\rangle_0 + \left\langle A_2^2 \right\rangle_0 + \left\langle A_1^2 \right\rangle_0 + \left\langle A_0^2 \right\rangle_0 \\ &= (A_0|\,A_0) + (A_1|\,A_1) + (A_2|\,A_2) + (A_3|\,A_3) + (A_4|\,A_4) \\ &= \Bigg\{ \alpha_0^2 + \frac{1}{2} \sum_{1\le j_1\le 2r} (\alpha_{1j_1})^2 - \frac{1}{4} \sum_{1\le j_1<j_2\le 2r} (\alpha_{2j_1j_2})^2 - \frac{1}{8} \sum_{1\le j_1<j_2<j_3\le 2r} (\alpha_{3j_1j_2j_3})^2 + \frac{1}{16} \sum_{1\le j_1<j_2<j_3<j_4\le 2r} (\alpha_4 \end{aligned}$$

One can also express the inner products of the even number components as

$$
\begin{aligned}
(A_0|\, A_0) &= \operatorname{Tr}\left\{\mathbb{A}_0^\dagger \mathbb{A}_0\right\} \\
(A_2|\, A_2) &= \frac{1}{4}\operatorname{Tr}\left\{\mathbb{A}_2^\dagger \mathbb{A}_2\right\} \\
(A_4|\, A_4) &= \frac{1}{16}\operatorname{Tr}\left\{\mathbb{A}_4^\dagger \mathbb{A}_4\right\},
\end{aligned} \tag{490}
$$

where we have used the ordered form of the basis.

2.

$$A_3 A_4 = \langle A_3 A_4\rangle_1 + \langle A_3 A_4\rangle_3 + \langle A_3 A_4\rangle_5 + \langle A_3 A_4\rangle_7 \tag{491}$$

$$A_4 A_3 = \langle A_4 A_3\rangle_1 + \langle A_4 A_3\rangle_3 + \langle A_4 A_3\rangle_5 + \langle A_4 A_3\rangle_7\,, \tag{492}$$

3.

$$A_2 A_3 = \langle A_2 A_3\rangle_1 + \langle A_2 A_3\rangle_3 + \langle A_2 A_3\rangle_5 \tag{493}$$

$$A_3 A_2 = \langle A_3 A_2\rangle_1 + \langle A_3 A_2\rangle_3 + \langle A_3 A_2\rangle_5\,, \tag{494}$$

4.

$$A_2 A_4 = \langle A_2 A_4\rangle_2 + \langle A_2 A_4\rangle_4 + \langle A_2 A_4\rangle_6 \tag{495}$$

$$A_4 A_2 = \langle A_4 A_2\rangle_2 + \langle A_4 A_2\rangle_4 + \langle A_4 A_2\rangle_6\,. \tag{496}$$

5.

$$A_1 A_2 = \langle A_1 A_2\rangle_1 + \langle A_1 A_2\rangle_3 \tag{497}$$

$$A_2 A_1 = \langle A_2 A_1\rangle_1 + \langle A_2 A_1\rangle_3 \tag{498}$$

6.

$$A_1 A_3 = \langle A_1 A_3\rangle_2 + \langle A_1 A_3\rangle_4 \tag{499}$$

$$A_3 A_1 = \langle A_3 A_1\rangle_2 + \langle A_3 A_1\rangle_4 \tag{500}$$

7.

$$A_1 A_4 = \langle A_1 A_4\rangle_3 + \langle A_1 A_4\rangle_5 \tag{501}$$

$$A_4 A_1 = \langle A_4 A_1\rangle_3 + \langle A_4 A_1\rangle_5\,. \tag{502}$$

In our case

$$A = A_0 + A_2 + A_4, \tag{503}$$

hence

$$\begin{aligned} A^2 = {} & \left\langle A_4^2 \right\rangle_0 + \left\langle A_4^2 \right\rangle_2 + \left\langle A_4^2 \right\rangle_4 + \left\langle A_4^2 \right\rangle_6 + \left\langle A_4^2 \right\rangle_8 + \left\langle A_2^2 \right\rangle_0 + \left\langle A_2^2 \right\rangle_2 + \left\langle A_2^2 \right\rangle_4 \\ & + \left\langle A_2 A_4 \right\rangle_2 + \left\langle A_2 A_4 \right\rangle_4 + \left\langle A_2 A_4 \right\rangle_6 + \left\langle A_4 A_2 \right\rangle_2 + \left\langle A_4 A_2 \right\rangle_4 + \left\langle A_4 A_2 \right\rangle_6 + 2A_0 A_2 + 2A_0 A_4, \end{aligned} \tag{504}$$

which can be rearranged as

$$\begin{aligned} A^2 = {} & \left\langle A_4^2 \right\rangle_0 + \left\langle A_2^2 \right\rangle_0 \\ & + \left\langle A_4^2 \right\rangle_2 + \left\langle A_2^2 \right\rangle_2 + \left\langle A_2 A_4 \right\rangle_2 + \left\langle A_4 A_2 \right\rangle_2 + 2A_0 A_2 \\ & + 2A_0 A_4 + \left\langle A_4^2 \right\rangle_4 + \left\langle A_2 A_4 \right\rangle_4 + \left\langle A_4 A_2 \right\rangle_4 + \left\langle A_2^2 \right\rangle_4 \\ & + \left\langle A_4^2 \right\rangle_6 + \left\langle A_2 A_4 \right\rangle_6 + \left\langle A_4 A_2 \right\rangle_6 \\ & + \left\langle A_4^2 \right\rangle_8, \end{aligned} \tag{505}$$

As

$$\left\langle A_r B_s \right\rangle_{r+s-2j} = (-1)^{rs-j} \left\langle B_s A_r \right\rangle_{r+s-2j} \tag{506}$$

this gives

$$\begin{aligned} A^2 = {} & A_0^2 + \left\langle A_4^2 \right\rangle_0 + \left\langle A_2^2 \right\rangle_0 \\ & + 2 \left\langle A_2 A_4 \right\rangle_2 + 2A_0 A_2 \\ & + \left\langle A_4^2 \right\rangle_4 + \left\langle A_2^2 \right\rangle_4 + 2A_0 A_4 \\ & + 2 \left\langle A_2 A_4 \right\rangle_6 \\ & + \left\langle A_4^2 \right\rangle_8, \end{aligned} \tag{507}$$

as

$$\langle A_4 A_4\rangle_{8-6} = (-1)^{16-3}\langle A_4 A_4\rangle_{8-6} \Rightarrow \langle A_4^2\rangle_2 = 0 \tag{508a}$$

$$\langle A_4 A_4\rangle_{8-4} = (-1)^{16-2}\langle A_4 A_4\rangle_{8-4} = \langle A_4 A_4\rangle_4 \tag{508b}$$

$$\langle A_4 A_4\rangle_{8-2} = (-1)^{16-1}\langle A_4 A_4\rangle_{8-2} \Rightarrow \langle A_4^2\rangle_6 = 0 \tag{508c}$$

$$\langle A_4 A_4\rangle_{8-0} = (-1)^{16-0}\langle A_4 A_4\rangle_8 = \langle A_4^2\rangle_8 \tag{508d}$$

$$\langle A_4 A_2\rangle_{6-4} = (-1)^{8-2}\langle A_2 A_4\rangle_{6-4} = \langle A_2 A_4\rangle_2 \tag{508e}$$

$$\langle A_2 A_4\rangle_{6-2} = (-1)^{8-1}\langle A_4 A_2\rangle_{6-2} \Rightarrow \langle A_2 A_4\rangle_4 = -\langle A_4 A_2\rangle_4 \tag{508f}$$

$$\langle A_4 A_2\rangle_{6-0} = (-1)^{8-0}\langle A_2 A_4\rangle_{6-0} = \langle A_2 A_4\rangle_6 \tag{508g}$$

$$\langle A_2 A_2\rangle_{4-2} = (-1)^{4-1}\langle A_2 A_2\rangle_{4-2} \Rightarrow \langle A_2^2\rangle_2 = 0 \tag{508h}$$

$$\langle A_2 A_2\rangle_{4-0} = (-1)^{4-0}\langle A_2 A_2\rangle_{4-0} = \langle A_2^2\rangle_4 . \tag{508i}$$

It is easy to show that

$$\langle A_4^2\rangle_0 = (A_4|\, A_4)\, I_{\mathcal{F}} \tag{509a}$$

$$\langle A_2^2\rangle_0 = \frac{1}{8}\operatorname{Tr}\left\{\mathbb{A}_2^t \mathbb{A}_2\right\} I_{\mathcal{F}} = (A_2|\, A_2)\, I_{\mathcal{F}} \tag{509b}$$

and

$$\langle A_4^2\rangle_8 = A_4 \wedge A_4 \tag{510a}$$

$$\langle A_2^2\rangle_4 = A_2 \wedge A_2 \tag{510b}$$

$$\langle A_2 A_4\rangle_6 = A_2 \wedge A_4. \tag{510c}$$

One can check this by considering representative tensor elements for eq.($508a$)

$$\langle A_4 A_2\rangle_{(2)12} = -\left(A_{1256}A_{56} + A_{1246}A_{46} + A_{1236}A_{36} + A_{1245}A_{45} + A_{1235}A_{35} + A_{1234}A_{34}\right) \neq 0 \tag{511}$$

$$\langle A_2 A_4\rangle_{(2)12} = -\left(A_{56}A_{1256} + A_{46}A_{1246} + A_{36}A_{1236} + A_{45}A_{1245} + A_{35}A_{1235} + A_{34}A_{1234}\right) \neq 0 \tag{512}$$

for eq.($508c$)

$$\begin{aligned}\langle A_4^2\rangle_{(6)234567} =& A_{1234}A_{1567} - A_{1235}A_{1467} + A_{1236}A_{1457} - A_{1237}A_{1456} + A_{1245}A_{1367} - A_{1246}A_{1357} + A_{1247}A_{1356} - \\ & - A_{1257}A_{1346} + A_{1267}A_{1345} - A_{1345}A_{1267} + A_{1346}A_{1257} - A_{1347}A_{1256} - A_{1356}A_{1247} + A_{1357}A_{1246} \\ & + A_{1456}A_{1237} - A_{1457}A_{1236} + A_{1467}A_{1235}, -A_{1567}A_{1234} = 0\end{aligned} \tag{513}$$

for eq. (508b)

$$\left\langle A_4^2\right\rangle_{(4)3456} = -A_{1234}A_{1256} + A_{1235}A_{1246} - A_{1236}A_{1245} - A_{1245}A_{1236} + A_{1246}A_{1235} - A_{1256}A_{1234} \tag{514}$$

$$= 2\left(A_{1235}A_{1246} - A_{1234}A_{1256} - A_{1236}A_{1245}\right) \tag{515}$$

for eq. (**??**)

$$\left\langle A_4^2\right\rangle_{(2)45} = A_{1234}A_{1235} - A_{1235}A_{1234} + A_{1234}A_{1236} - A_{1236}A_{1234} + A_{1234}A_{1235} - A_{1235}A_{1234} \cdots = 0 \tag{516}$$

The *reflection or involution* condition $A^2 = I_{\mathcal{F}}$ gives

$$A^2 = A_0^2 + \left\langle A_4^2\right\rangle_0 + \left\langle A_2^2\right\rangle_0 + 2\left\langle A_2A_4\right\rangle_2 + 2A_0A_2 + \left\langle A_4^2\right\rangle_4 + \left\langle A_2^2\right\rangle_4 + 2A_0A_4 + 2\left\langle A_2A_4\right\rangle_6 + \left\langle A_4^2\right\rangle_8 = I_{\mathcal{F}}, \tag{517}$$

leading to

$$A_0^2 + \left\langle A_4^2\right\rangle_0 + \left\langle A_2^2\right\rangle_0 = I_{\mathcal{F}} \tag{518}$$

$$\left\langle A_2A_4\right\rangle_2 + A_0A_2 = 0 \tag{519}$$

$$\left\langle A_4^2\right\rangle_4 + \left\langle A_2^2\right\rangle_4 + 2A_0A_4 = 0 \tag{520}$$

$$\left\langle A_2A_4\right\rangle_6 = A_2 \wedge A_4 = 0 \tag{521}$$

$$\left\langle A_4^2\right\rangle_8 = A_4 \wedge A_4 = 0 \tag{522}$$

If A is a reflection then

$$\mathcal{A}_+ = \frac{1}{\sqrt{2}}\left(I_{\mathcal{F}} + A\right)$$
$$\mathcal{A}_- = \frac{1}{\sqrt{2}}\left(I_{\mathcal{F}} - A\right)$$

are *orthogonal idempotents*, as

$$\mathcal{A}_+^2 = \left(\frac{1}{\sqrt{2}}\left(I_{\mathcal{F}} + A\right)\right)^2 = \frac{1}{2}\left(I_{\mathcal{F}} + 2A + I_{\mathcal{F}}\right) = \mathcal{A}_+,$$

$$\mathcal{A}_-^2 = \left(\frac{1}{\sqrt{2}}\left(I_{\mathcal{F}} - A\right)\right)^2 = \frac{1}{2}\left(I_{\mathcal{F}} - 2A + I_{\mathcal{F}}\right) = \mathcal{A}_1$$

and

$$\mathcal{A}_+\mathcal{A}_- = \mathcal{A}_-\mathcal{A}_+ = 0 = \frac{1}{\sqrt{2}}\left(I_{\mathcal{F}} + A\right)\frac{1}{\sqrt{2}}\left(I_{\mathcal{F}} - A\right) = \frac{1}{2}\left(I_{\mathcal{F}} + A\right)\left(I_{\mathcal{F}} - A\right).$$

The *idempotency* condition $A^2 = A$ gives

$$A^2 - A = \left\{A_0\left(A_0 - 1\right) + \left\langle A_4^2 \right\rangle_0 + \left\langle A_2^2 \right\rangle_0\right\} + \left\{2\left\langle A_2 A_4 \right\rangle_2 + \left(2A_0 - 1\right) A_2\right\} + \left\{\left\langle A_4^2 \right\rangle_4 + \left\langle A_2^2 \right\rangle_4 + \left(2A_0 - 1\right) A_4\right.$$
$$+ 2\left\langle A_2 A_4 \right\rangle_6 + \left\langle A_4^2 \right\rangle_8 = 0 \tag{523}$$

which gives

$$A_0\left(A_0 - I_{\mathcal{F}}\right) + \left\langle A_4^2 \right\rangle_0 + \left\langle A_2^2 \right\rangle_0 = 0 \tag{524a}$$
$$2\left\langle A_2 A_4 \right\rangle_2 + \left(2A_0 - I_{\mathcal{F}}\right) A_2 = 0 \tag{524b}$$
$$\left\langle A_4^2 \right\rangle_4 + \left\langle A_2^2 \right\rangle_4 + \left(2A_0 - I_{\mathcal{F}}\right) A_4 = 0 \tag{524c}$$
$$\left\langle A_2 A_4 \right\rangle_6 = 0 \tag{524d}$$
$$\left\langle A_4^2 \right\rangle_8 = 0 \tag{524e}$$

and thus

$$A_0\left(A_0 - 1\right) + \left(A_4 \middle| A_4\right) I_{\mathcal{F}} + \left(A_2 \middle| A_2\right) I_{\mathcal{F}} = 0 \tag{525}$$
$$2\left\langle A_2 A_4 \right\rangle_2 + \left(2A_0 - 1\right) A_2 = 0 \tag{526}$$
$$\left\langle A_4^2 \right\rangle_4 + A_2 \wedge A_2 + \left(2A_0 - 1\right) A_4 = 0 \tag{527}$$
$$A_2 \wedge A_4 = 0 \tag{528}$$
$$A_4 \wedge A_4 = 0 \tag{529}$$

and in more detail

$$A_0\left(A_0 - 1\right) + \sum_{1 \leq j_1 < j_2 < j_3 < j_4 \leq 2r} A_{4 j_1 j_2 j_3 j_4}^2 + \sum_{1 \leq j_1 < j_2 \leq 2r} A_{2 j_1 j_2}^2 = 0 \tag{530}$$
$$2\left\langle A_2 A_4 \right\rangle_2 + \left(2\alpha - 1\right) A_2 = 0 \tag{531}$$
$$\left\langle A_4^2 \right\rangle_4 + A_2 \wedge A_2 + \left(2\alpha - 1\right) A_4 = 0 \tag{532}$$
$$A_2 \wedge A_4 = 0 \tag{533}$$
$$A_4 \wedge A_4 = 0. \tag{534}$$

A. Canonical form of 4-forms

The operator A_4 can be expanded as

$$A_4 = \frac{1}{4!} \sum_{1 \leq j_1, j_2, j_3, j_4 \leq 2r} A_{j_1 j_2 j_3 j_4} x_{j_1} x_{j_2} x_{j_3} x_{j_4} = \frac{1}{4!} \sum_{1 \leq j_1, j_2, j_3, j_4 \leq 2r} A_{j_1 j_2 j_3 j_4} x_{j_1} \wedge x_{j_2} \wedge x_{j_3} \wedge x_{j_4} \tag{535}$$

$$= \sum_{1 \leq j_1 < j_2 < j_3 < j_4 \leq 2r} A_{j_1 j_2 j_3 j_4} x_{j_1} x_{j_2} x_{j_3} x_{j_4} = \frac{1}{4} \sum_{\substack{1 \leq j_1 < j_2 \leq 2r \\ 1 < j_3 < j_4 \leq 2r}} A_{j_1 j_2 j_3 j_4} x_{j_1} x_{j_2} x_{j_3} x_{j_4}, \tag{536}$$

where the coefficients $\{A_{j_1 j_2 j_3 j_4}\}$ are antisymmetric in the subscripts and

$$A_{j_1 j_2 j_3 j_4} = 0 \text{ if } j_1 = j_3 \text{ or } j_4,\ j_2 = j_3 \text{ or } j_4,\ j_1 = j_2 \text{ or } \ j_3 = j_4.$$

$\lceil$ consider the example for $r = 2$ arranged as 12×12 matrix

$$\begin{array}{ccccccccccccc}
 & 12 & 13 & 14 & 21 & 23 & 24 & 31 & 32 & 34 & 41 & 42 & 43 \\
12 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & A_{1234} & 0 & 0 & A_{1243} \\
13 & 0 & 0 & 0 & 0 & 0 & A_{1324} & 0 & 0 & 0 & 0 & * & 0 \\
14 & 0 & 0 & 0 & 0 & A_{1423} & 0 & 0 & A_{1432} & 0 & 0 & 0 & 0 \\
21 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & * & 0 & 0 & * \\
23 & 0 & 0 & * & 0 & 0 & 0 & 0 & 0 & 0 & * & 0 & 0 \\
24 & 0 & * & 0 & 0 & 0 & 0 & * & 0 & 0 & 0 & 0 & 0 \\
31 & 0 & 0 & 0 & 0 & 0 & * & 0 & 0 & 0 & 0 & * & 0 \\
32 & 0 & 0 & * & 0 & 0 & 0 & 0 & 0 & 0 & * & 0 & 0 \\
34 & * & 0 & 0 & * & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
41 & 0 & 0 & 0 & 0 & * & 0 & 0 & * & 0 & 0 & 0 & 0 \\
42 & 0 & * & 0 & 0 & 0 & 0 & * & 0 & 0 & 0 & 0 & 0 \\
43 & * & 0 & 0 & * & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{array}$$

or as a 6×6 *symmetric* matrix

$$\begin{array}{c} \\ 12 \\ 13 \\ 14 \\ 23 \\ 24 \\ 34 \end{array} \begin{array}{c} \begin{array}{cccccc} 12 & 13 & 14 & 23 & 24 & 34 \end{array} \\ \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & A_{1234} \\ 0 & 0 & 0 & 0 & A_{1324} & 0 \\ 0 & 0 & 0 & A_{1423} & 0 & 0 \\ 0 & 0 & A_{2314} & 0 & 0 & 0 \\ 0 & A_{2413} & 0 & 0 & 0 & 0 \\ A_{3412} & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array}$$

$\rfloor$

As

$$x_{j_1} x_{j_2} x_{j_3} x_{j_4} = x_{j_3} x_{j_4} x_{j_1} x_{j_2}$$

one has

$$A_{j_1 j_2 j_3 j_4} = A_{j_3 j_4 j_1 j_2}$$

and thus the matrix $\mathbb{A} = \{ A_{J_{12} J_{34}} | \, 1 \leq J_{12}, J_{34} \leq r\,(2r-1)\}$ is symmetric i.e.

$$A_{J_{12} J_{34}} = A_{J_{34} J_{12}}; 1 \leq J_{12}, J_{34} \leq r\,(2r-1)$$

and the operator A_4 can be expanded as

$$A_4 = \frac{1}{4} \sum_{1 \leq J_{12}, J_{34} \leq r(2r-1)} A_{4 J_{12} J_{34}} y_{J_{12}} y_{J_{34}} = \frac{1}{4} \sum_{1 \leq J_{12}, J_{34} \leq r(2r-1)} A_{4 J_{12} J_{34}} y_{J_{12}} \wedge y_{J_{34}},$$

where

$$y_{J_{12}} = x_{j_1} x_{j_2}$$
$$y_{J_{34}} = x_{j_3} x_{j_4}.$$

If the matrix $\mathbb{A} = \{ A_{J_{12} J_{34}} | \, 1 \leq J_{12}, J_{34} \leq r\,(2r-1)\}$ is real symmetric, (which it must be if A_4 is self adjoint), one can diagonalize it with real orthogonal transformations i.e. $\mathbb{O} \in O(r\,(2r-1)\,, \mathbb{R})$, to obtain

$$\mathbb{O}^t \mathbb{A} \mathbb{O} = \begin{pmatrix} \alpha_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \alpha_{r(2r-1)} \end{pmatrix} \Longleftrightarrow \mathbb{A} = \mathbb{O} \begin{pmatrix} \alpha_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \alpha_{r(2r-1)} \end{pmatrix} \mathbb{O}^t$$

giving

$$A_{J_{12}J_{34}} = \sum_{1 \le K \le r(2r-1)} O_{J_{12}K} \alpha_K O_{J_{34}K}$$

and thus

$$A_4 = \frac{1}{4} \sum_{1 \le J_{12}, J_{34} \le r(2r-1)} A_{4J_{12}J_{34}} y_{J_{12}} y_{J_{34}} = \frac{1}{4} \sum_{1 \le J_{12}, J_{34} \le r(2r-1)} \sum_{1 \le K \le r(2r-1)} O_{J_{12}K} \alpha_K O_{J_{34}K} y_{J_{12}} y_{J_{34}}.$$

One can define the generalized geminal operator

$$\mathcal{O}_K = \sum_{1 \le J_{12} \le r(2r-1)} O_{J_{12}K} y_{J_{12}} = \sum_{1 \le j_1 \ne j_2 \le 2r} O_{kj_1j_2} x_{j_1} x_{j_2}; \quad O_{kj_1j_2} = -O_{kj_2j_1},$$

which has the canonical form

$$\mathcal{O}_K = \sum_{1 \le l \le r} c_{Kl} \varphi_{Kl} \varphi_{Kl+r}, \quad c_{Kl} \in \mathbb{R}^+$$

Note that $\{\varphi_{Kl}\}$ *are self adjoint* operators belonging to $\mathcal{F}$ that are linear combinations of single creation and annihilation operators. The operator A_4 can thus be expressed as

$$\begin{aligned} A_4 &= \frac{1}{4} \sum_{1 \le J_{12}, J_{34} \le r(2r-1)} A_{4J_{12}J_{34}} y_{J_{12}} y_{J_{34}} = \frac{1}{4} \sum_{1 \le K \le r(2r-1)} \alpha_K \sum_{1 \le J_{12}, J_{34} \le r(2r-1)} O_{J_{12r(2r+1)2}K} O_{J_{34}K} y_{J_{12}} y_{J_{34}} \\ &= \frac{1}{4} \sum_{1 \le K \le r(2r-1)} \alpha_K \sum_{1 \le J_{12} \le r(2r-1)} O_{J_{12}K} y_{J_{12}} \sum_{1 \le J_{34} \le r(2r-1)} O_{J_{34}K} y_{J_{34}} = \frac{1}{4} \sum_{1 \le K \le r(2r-1)} \alpha_K \mathcal{O}_K \wedge \mathcal{O}_K \\ &= \frac{1}{4} \sum_{1 \le K \le 2r(2r-1)} \alpha_K \mathcal{O}_K^{2\wedge}. \end{aligned}$$

Using the canonical form for a second degree antisymmetric polynomial this gives the canonical form for a fourth degree antisymmetric polynomial as

$$\begin{aligned} A_4 &= \frac{1}{4} \sum_{1 \le K \le r(2r-1)} \alpha_K \sum_{1 \le l_1, l_2 \le r} c_{Kl_1} c_{Kl_2} \varphi_{Kl_1} \varphi_{Kl_1+r} \varphi_{Kl_2} \varphi_{Kl_2+r} = \frac{1}{4} \sum_{1 \le K \le r(2r-1)} \alpha_K \sum_{1 \le l_1 \ne l_2 \le r} c_{Kl_1} c_{Kl_2} \varphi_{Kl_1} \varphi_{Kl_1+} \\ &= \frac{1}{2} \sum_{1 \le K \le r(2r-1)} \alpha_K \sum_{1 \le l_1 < l_2 \le r} c_{Kl_1} c_{Kl_2} \varphi_{Kl_1} \varphi_{Kl_1+r} \varphi_{Kl_2} \varphi_{Kl_2+r} = \sum_{1 \le K \le r(2r-1)} \alpha_K \mathcal{O}_K \wedge \mathcal{O}_K. \end{aligned}$$

One should note that $\{\alpha_K\}$ and $\{c_{Kl} \ge 0\}$ are real and $\{\mathcal{O}_K\}$ are self adjoint. The coefficient α_K could be negative

⌈ If

$$\mathcal{O}_K \wedge \mathcal{O}_K = \mathcal{O}_K^{2\wedge} = 0$$

then

$$\operatorname{rank}\{\mathcal{O}_K\} = 1$$

as

$$\mathcal{O}_K^{2\wedge} = 2 \sum_{1 \le l < m \le r} c_{Kl} c_{Km} \varphi_{Kl} \varphi_{Kl+r} \varphi_{Km} \varphi_{Km+r}$$

and

$$\left(\varphi_{Kp} \right| \left. \varphi_{Kq} \right) = \delta_{pq} I_{\mathcal{F}}$$

so that

$$\varphi_{Kl} \varphi_{Kl+r} \varphi_{Km} \varphi_{Km+r} \neq 0 \text{ if } l \neq m.$$

$\rfloor$

The Grassmann square of A_4 is

$$\begin{aligned}
A_4^{2\wedge} = A_4 \wedge A_4 &= \left(\frac{1}{4}\right)^2 \sum_{1 \le K \le r(2r-1)} \alpha_K \mathcal{O}_K \wedge \mathcal{O}_K \wedge \sum_{1 \le L \le r(2r-1)} \alpha_L \mathcal{O}_L \wedge \mathcal{O}_L \\
&= \left(\frac{1}{4}\right)^2 \sum_{\substack{1 \le K \le r(2r-1) \\ 1 \le L \le r(2r-1)}} \alpha_K \alpha_L \mathcal{O}_K \wedge \mathcal{O}_L \wedge \mathcal{O}_K \wedge \mathcal{O}_L \\
&= \left(\frac{1}{4}\right)^2 \sum_{\substack{1 \le K \le r(2r-1) \\ 1 \le L \le r(2r-1)}} \alpha_K \alpha_L \mathcal{O}_{KL} \wedge \mathcal{O}_{KL} \\
&= \left(\frac{1}{4}\right)^2 \sum_{1 \le K, L \le r(2r-1)} \alpha_K \alpha_L \sum_{1 \le m,n,p,q \le r} c_{Km} c_{Kn} c_{Lp} c_{Lq} \varphi_{Km} \varphi_{Km+r} \varphi_{Kn} \varphi_{Kn+r} \varphi_{Lp} \varphi_{Lp+r} \varphi_{Lq} \varphi_{Lq+r}.
\end{aligned}$$

If A is a reflection then

$$\begin{aligned}
\mathcal{A}_+ &= \frac{1}{\sqrt{2}} \left(I_{\mathcal{F}} + A \right) \\
\mathcal{A}_- &= \frac{1}{\sqrt{2}} \left(I_{\mathcal{F}} - A \right)
\end{aligned}$$

are idempotents, as

$$\mathcal{A}_+^2 = \left(\frac{1}{\sqrt{2}} \left(I_{\mathcal{F}} + A \right) \right)^2 = \frac{1}{2} \left(I_{\mathcal{F}} + 2A + I_{\mathcal{F}} \right) = \mathcal{A}_+$$

and

$$\mathcal{A}_-^2 = \left(\frac{1}{\sqrt{2}} \left(I_{\mathcal{F}} - A \right) \right)^2 = \frac{1}{2} \left(I_{\mathcal{F}} - 2A + I_{\mathcal{F}} \right) = \mathcal{A}_1.$$

Also

$$\mathcal{A}_+ \mathcal{A}_- = \mathcal{A}_- \mathcal{A}_+ = 0 = \frac{1}{\sqrt{2}} \left(I_{\mathcal{F}} + A \right) \frac{1}{\sqrt{2}} \left(I_{\mathcal{F}} - A \right) = \frac{1}{2} \left(I_{\mathcal{F}} + A \right) \left(I_{\mathcal{F}} - A \right).$$

$$\left\langle A_4^2 \right\rangle_4 = \frac{1}{4} \left(A_{1234} A_{1256} - A_{1235} A_{1246} + A_{1236} A_{1245} \right) x_3 \wedge x_4 \wedge x_5 \wedge x_6 \tag{537}$$

$$\left\langle A_2 A_4 \right\rangle_{(2)12} = -\frac{1}{4} \left(A_{1256} A_{56} + A_{1246} A_{46} + A_{1236} A_{36} + A_{1245} A_{45} + A_{1235} A_{35} + A_{1234} A_{34} \right) x_1 \wedge x_2 \tag{538}$$